SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
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Somaiya TRUST

Module 3 Unsupervised Learning – (PCA)

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What is the curse of dimensionality?

Try building several machine learning models to analyze the performance of a Formula One (F1) driver.

Consider the following cases:

Model_1 - 2 features (circuit name, country name)

Model_2 - 4 features (circuit name, country name, weather, max speed of the car)

Model_3 - 8 features (circuit name, country name, weather, max speed of the car , say driver's experience, number of wins, car condition, driver's physical fitness)

Model_4 - 16 features (circuit name, country name, weather, max speed of the car , say driver's experience, number of wins, car condition, driver's physical fitness, age, latitude, longitude, driver's height, hair color, car color, the car company, driver's marital status)

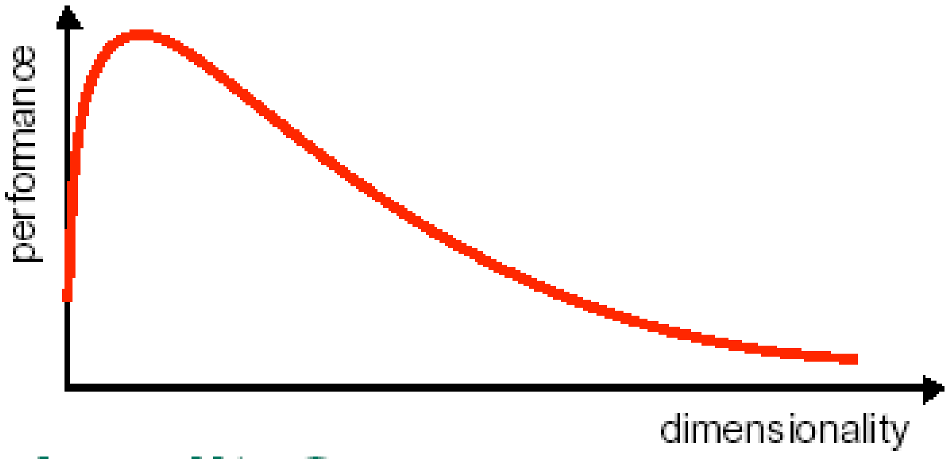
Model_5 - 32 features, **Model_6** - 64 features, **Model_7** - 128 features, **Model_8** - 256 features, **Model_9** - 512 features, **Model_10** - 1024 features.

What is the curse of dimensionality?

- Assuming the training data remains constant, it is observed that on increasing the number of features the accuracy tends to increase until a certain threshold value and after that, it starts to decrease.
- The accuracy of Model_1 < accuracy of Model_2 < accuracy of Model_3 but if we try to extrapolate this trend it doesn't hold true for all the models having more than 8 features.
- Now you might wonder if we are providing some extra information for the model to learn why is it so that the performance starts to degrade.

What is the curse of dimensionality?

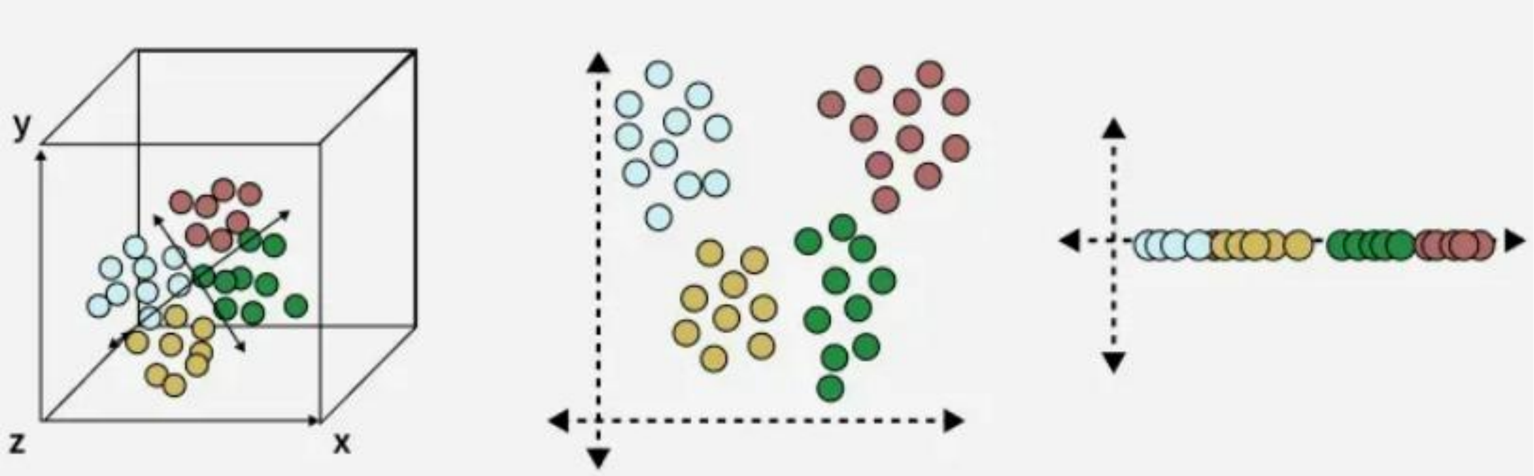
- For a given sample size, there is a maximum number of features above which the performance of classifier will degrade rather than improve.



- The curse of dimensionality was first termed by Richard E. Bellman when considering problems in dynamic programming.

What is Dimensionality Reduction

- Dimensionality Reduction is the process of reducing the number of input variables (features) in a dataset while preserving as much important information as possible.



Why Dimensionality Reduction?

- Prevent – Curse of Dimensionality
- Improve the performance of the model
- Visualize the data – understand the data

Dimensionality Reduction

Advantages

- Simplify the pattern representation and the classifiers
- Faster classifier with less memory consumption
- Alleviate curse of dimensionality with limited data sample

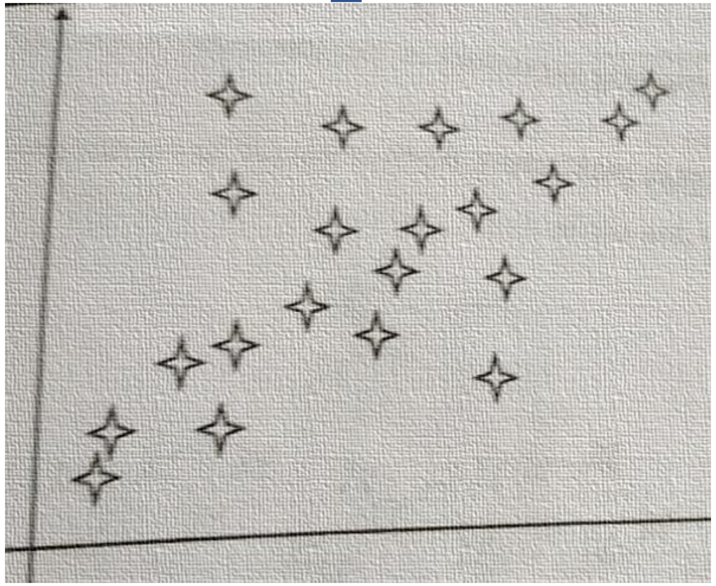
Disadvantages

- Loss information
- Increased error in the resulting recognition system

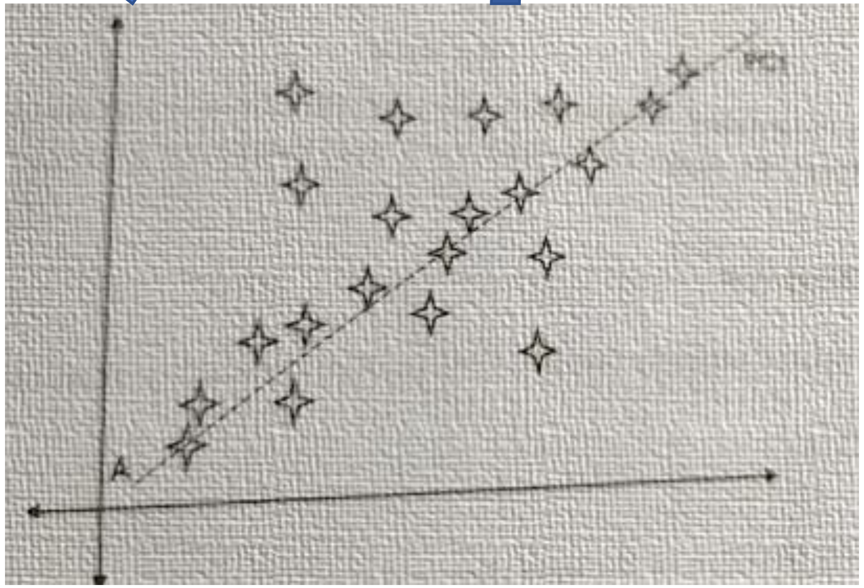
Feature Reduction: PCA

- PCA (Principal Component Analysis) is an unsupervised dimension reduction method.
- It aims at finding the most important dimensions (called principal components) for the given dataset.
- A principal component is a direction in which the data has the largest variation.

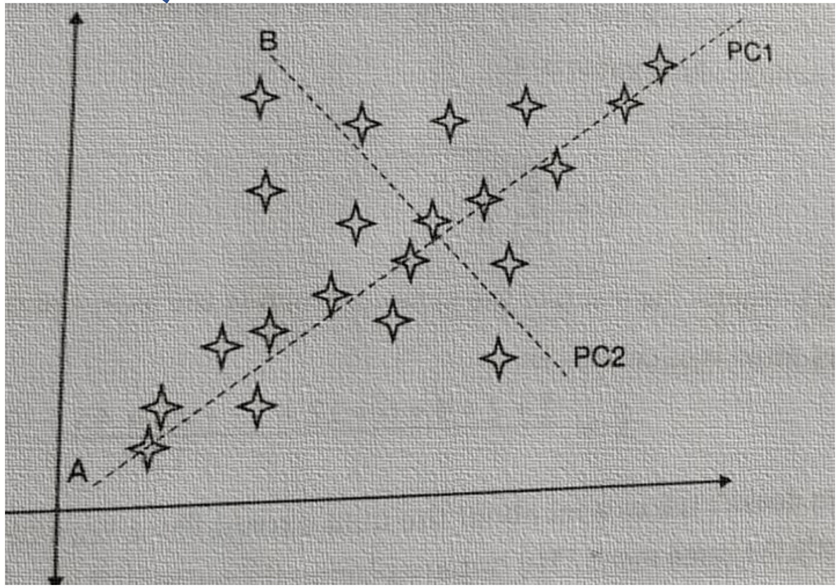
PCA Conceputal Steps:



Centre the data by subtracting off the mean



Choose the direction with largest variation, place an axis in that direction



Look at the remaining variations, find another axis that is that is orthogonal to the first axis and covers maximum variations

Statistical Concepts Required for PCA

1. Mean – the mean of a dataset is:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Where

xi = data points

n = number of samples

Statistical Concepts Required for PCA

2. Standard deviation measures the spread of data around the mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n-1}} \qquad s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}}$$

Large value → data widely spread
Small value → data tightly clustered.

Statistical Concepts Required for PCA

3. Variance is square of the standard deviation.

$$Var(X) = \sigma^2$$

$$var(X) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(X_i - \bar{X})}{(n-1)}$$

It measure how far data points are from the mean.

Statistical Concepts Required for PCA

- Covariance measures how two variables vary together.

$$Cov(X,Y) = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{n-1}$$

Covariance Value	Meaning
Positive	Variables increase together
Negative	One increases while other decreases
Zero	No relationship

Covariance Matrix

Representing Covariance between dimensions as a matrix e.g. for 3 dimensions:

$$C = \begin{bmatrix} cov(x,x) & cov(x,y) & cov(x,z) \\ cov(y,x) & cov(y,y) & cov(y,z) \\ cov(z,x) & cov(z,y) & cov(z,z) \end{bmatrix}$$

- Diagonal is the variances of x, y and z
- cov(x,y) = cov(y,x) hence matrix is symmetrical about the diagonal
- N-dimensional data will result in NxN covariance matrix

Covariance

- Exact value is not as important as it's sign.
- (+) A positive value of covariance indicates both dimensions increase or decrease together e.g. as the number of hours studied increases, the marks in that subject increase.
- (-) A negative value indicates while one increases the other decreases, or vice-versa e.g. time spent on social media vs performance in exam.
- (zero) If covariance is zero: the two dimensions are independent of each other e.g. heights of students vs the marks obtained in a subject

Eigen vectors

- $M = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 5 \end{bmatrix}$ The resulting vector is not integer multiple of the original vector
- $N = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ The resulting vector is exactly 4 times the original vector. This is called eigenvector.

- *We can transform and change matrices into new vectors by multiplying a matrix with a vector. The multiplication of the matrix by a vector computes a new vector. This is the transformed vector.*

Eigenvector

- If the new transformed vector is just a scaled form of the original vector then the original vector is known to be an **eigenvector** of the original matrix
- An eigenvector is a vector that does not change when a transformation is applied to it, except that it becomes a scaled version of the original vector.
- **Eigenvalue**—The scalar that is used to transform (stretch) an Eigenvector.

Eigenvector and Eigenvalue

- An eigenvector **does not change direction** in a transformation:
- In this shear mapping the red arrow changes direction, but the blue arrow does not. The blue arrow is an eigenvector of this shear mapping because it does not change direction, and since its length is unchanged, its eigenvalue is 1.
- **1** means no change,
- **2** means doubling in length,
- **−1** means pointing backwards along the eigenvalue's direction

The Mathematics

- For a square matrix **A**, an Eigenvector and Eigenvalue make this equation true:

$$Av = \lambda v$$

Matrix
Eigenvector
Eigenvalue

Av gives us:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -6\times1+3\times4 \\ 4\times1+5\times4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

λv gives us :

$$6\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

Yes they are equal! So $Av = \lambda v$ as promised.

How do we find these eigen things?

We start by finding the **eigenvalue**:

- We know this equation must be true: $Av = \lambda v$
- Now let us put in an identity matrix so we are dealing with matrix-vs-matrix:

$$Av = \lambda I v$$

- Bring all to left hand side:

$$Av - \lambda I v = 0$$

- If **v** is non-zero then we can solve for λ using just the determinant:

$$|A - \lambda I| = 0$$

Example for λ

- Start with | A − λI | = 0
- Given matrix

$$A = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

- Substitute into equation

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} -6-\lambda & 3 \\ 4 & 5-\lambda \end{bmatrix} \right| = 0$$

- Calculating the determinant

$$(-6-\lambda)(5-\lambda) - 12 = 0$$

$$(-30 + 6\lambda - 5\lambda + \lambda^2) - 12 = 0$$

$$\lambda^2 + \lambda - 42 = 0$$

- **Final Eigenvalues**

$$\lambda^2 + \lambda - 42 = 0$$

$\lambda_1 = -7$

$\lambda_2 = 6$

Find the Eigenvector for Eigenvalue λ=6

- Substitute eigenvalue into the equation:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

- Matrix multiplication gives

$$-6x + 3y = 6x$$

$$4x + 5y = 6y$$

$$-6x + 3y - 6x = 0$$

$$-12x + 3y = 0$$

$$4x + 5y - 6y = 0$$

$$4x - y = 0$$

- Solve the it

$$4x - y = 0$$

$$y = 4x$$

- Thus the eigenvector is any non-zero multiple of:

$$v = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Verify Av=λv

- Compute AV

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} (-6)(1) + (3)(4) \\ (4)(1) + (5)(4) \end{bmatrix}$$

$$= \begin{bmatrix} -6 + 12 \\ 4 + 20 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

- Compute λv

$$6 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

PCA

Principal components analysis (PCA) is a technique that can be used to simplify a dataset

- It is a linear transformation that chooses a new coordinate system for the data set such that greatest variance by any projection of the data set comes to lie on the first axis (then called the first principal component),the second greatest variance on the second axis, and so on.
- PCA can be used for reducing dimensionality by eliminating the later principal components.

Steps Involved in the PCA

Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

Step 3: Calculate the eigenvalues and eigenvectors for the covariance matrix.

Step 4: Sort eigenvalues and their corresponding eigenvectors.

Step 5: Pick k eigenvalues and form a matrix of eigenvectors.

Step 6: Transform the original matrix.

Example 1

- Assume we have the below dataset which has 2 features and a total of 4 training examples.

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

Example 1

1. Standardize the Dataset

- First, we need to standardize the dataset and for that, we need to calculate the mean and standard deviation for each feature.

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

$$\bar{X}_1 = \frac{5+3+10+2}{4} = 5$$

$$\bar{X}_2 = \frac{10+7+15+4}{4} = 9$$

$$\sigma_1 = \sqrt{\frac{(5-5)^2+(3-5)^2+(10-5)^2+(2-5)^2}{4}} = \sqrt{\frac{0+4+25+9}{4}} = \sqrt{9.5} = 3.082$$

$$\sigma_2 = \sqrt{\frac{(10-9)^2+(7-9)^2+(15-9)^2+(4-9)^2}{4}} = \sqrt{\frac{1+4+36+25}{4}} = \sqrt{16.5} = 4.062$$

Example 1

- New data

$$x_{new} = \frac{x - \mu}{\sigma}$$

Size (X_1)	Rooms (X_2)
5	10
3	7
10	15
2	4

Obs	(Z1 = (X1-5)/3.082)	(Z_2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

Example 1

Obs	(Z1 = (X1-5)/3.082)	(Z_2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

Step 2. Calculate the covariance matrix for the whole dataset

The formula to calculate the covariance matrix:

	Z1	Z2
Z1	Var(Z1)	COV(Z1,Z2)
Z2	COV(Z2,Z1)	Var(Z2)

- var(z1) = ((0-0)² + (-0.649-0)² + (1.622-0)² +(0.973–0)²)/4=3.99/4

var (z1) = 1

- cov(z1,z2) =((0)(0.246)+(−0.649)(−0.492)+(1.622)(1.478)+(−0.973)(−1.231))/3

=3.914/3

$$C = \begin{bmatrix} 1 & 1.305 \\ 1.305 & 1 \end{bmatrix}$$

- **cov(z1,z2) = 1.305**

Example 1

3. Calculate eigenvalues and eigen vectors.

$Av - \lambda Iv = 0$;

$(A - \lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A - \lambda I) = 0$

Example 1

- $(A-\lambda I)v = 0$

$$\begin{vmatrix} 1-\lambda & 1.305 \\ 1.305 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - (1.305)^2 = 0$$

$$(1-\lambda)^2 - 1.703 = 0$$

$$(1-\lambda) = \pm\sqrt{1.703} = \pm 1.305$$

- Eigen values

$$\lambda_1 = 1 + 1.305 = 2.305$$

$$\lambda_2 = 1 - 1.305 = -0.305$$

Example 1

- For λ1=2.305

$$\begin{bmatrix} -1.305 & 1.305 \\ 1.305 & -1.305 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- For λ2=−0.305

$$\begin{bmatrix} 1.305 & 1.305 \\ 1.305 & 1.305 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example 1

$$\begin{bmatrix} -1.305 & 1.305 \\ 1.305 & -1.305 \end{bmatrix} \Rightarrow v_1 =$$

$$\begin{bmatrix} 1.305 & 1.305 \\ 1.305 & 1.305 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

- A **unit vector** is a vector whose **magnitude (length) is exactly 1**.

For a vector $v = [v_1, v_2, \ldots, v_n]$:

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

If:

$$\|v\| = 1$$

then v is a **unit vector**.

For $v_1 = [1, 1]$:

$$\|v_1\| = \sqrt{2} \neq 1$$

- For normalized

$$v_1^{norm} = \frac{1}{\sqrt{2}}[1, 1]$$

Example 1

- **Step 4: Sort Eigenvalues**

Eigenvalue	Eigenvector
2.305	(v1)
-0.305	(v2)

- **Step 5: Select Top k Components**
- Choose k=1(maximum variance)

$V1 = [1/\sqrt{2}, 1/\sqrt{2}]$

Example 1

Step 6 Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

Obs	(Z1 = (X1-5)/3.082)	(Z2 = (X2-9)/4.062)
1	0	0.246
2	-0.649	-0.492
3	1.622	1.478
4	-0.973	-1.231

$$* \quad \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \quad =$$

PC1
0.174
-0.806
2.192
-1.558

Example 1 Python

Steps Involved in the PCA

Step 1: Standardize the dataset.

Step 2: Calculate the covariance matrix for the features in the dataset.

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Step 5: Pick k eigenvalues and form a matrix of eigenvectors.

Step 6: Transform the original matrix.

Example 2

- Let's Say we have marks of 6 students in 3 subjects.

Student	Math	Science	English
A	85	80	78
B	90	85	88
C	78	70	72
D	92	88	85
E	70	65	68
F	88	82	80

Example 2 –Standardize the Data

$$Z = \frac{X - \text{mean}(X)}{\text{std}(X)}$$

- Original

Student	Math	Science	English
A	85	80	78
B	90	85	88
C	78	70	72
D	92	88	85
E	70	65	68
F	88	82	80
Mean	83.83	78.33	78.5
Std	7.62	8.18	6.92

New

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

Example 2 – Covariance Matrix

	Math	Science	English
Math	VAR(M)	COV(M,S)	COV(M,E)
Science	COV(S,M)	VAR(S)	COV(S,E)
English	COV(E,M)	COV(E,S)	VAR(E)

- Since you have already standardized the features, you can consider Mean = 0 and Standard Deviation=1 for each feature.

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

Var(M)

$= ((0.153008-0)^2+(0.808755-0)^2+(-0.76504-0)^2+(1.071054-0)^2+(-1.81424-0)^2 + (0.546456-0)^2)/6$

$=6.0/5=1.2$

COV(M,S)

$= ((0.153008-0)+(0.808755-0)^2+(-0.76504-0)^2+(1.071054-0)^2+(-1.81424-0)^2 + (0.546456-0)^2)/6$

$=5.938578 /5=1.187716$

$$\text{Covariance}=\begin{vmatrix} 1.2 & 1.187716 & 1.1387 \\ 1.187716 & 1.2 & 1.1481 \\ 1.13867 & 1.148135 & 1.2 \end{vmatrix}$$

Example 2 – Eigenvalues and Eigen Vectors

Calculate eigenvalues and eigen vectors.

$Av - \lambda I v = 0$;

$(A - \lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A - \lambda I) = 0$

Example 2 Eigenvalues

1.2- λ	1.187716	1.1387
1.187716	1.2- λ	1.1481
1.13867	1.148135	1.2- λ

- When you solve the following the matrix by considering 0 on right-hand side, you can define eigen values as

λ = 3.51647796 ,0.07174828 ,0.0117737

Example 2 – Eigen Vectors

- Then, substitute each eigen value in $(A-\lambda I)v=0$ equation and solve the same for different eigen vectors

$$\begin{vmatrix} 1.2-\lambda & 1.187716 & 1.1387 \\ 1.187716 & 1.2-\lambda & 1.1481 \\ 1.13867 & 1.148135 & 1.2-\lambda \end{vmatrix} \times \begin{vmatrix} V1 \\ V2 \\ V3 \end{vmatrix} = 0$$

- For $\lambda = 3.51647796$, solving the above equation using Cramer's rule, the values for the v vector are

v1= -0.5790409, v2= -0.58058703, v3= -0.57239002

Example 2

- We can from a matrix using the **eigen vectors.**

e1	e2	e3
-0.57904	-0.66755	-0.46807
-0.58059	0.740678	-0.3381
-0.57239	-0.07598	0.816454

- Arrange them in descending order and pick up the topmost Eigen values.

Example 2 - Pick k eigenvalues and form a matrix of eigenvectors and Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

Student	Math	Science	English
A	0.153008	0.203785	-0.072232
B	0.808755	0.815139	1.372399
C	-0.765039	-1.018924	-0.939010
D	1.071054	1.181952	0.939010
E	-1.814235	-1.630278	-1.516862
F	0.546456	0.448327	0.216695

6x3

x

e1	e2
-0.57904	-0.66755
-0.58059	0.740678
-0.57239	-0.07598

3x2

=

PC1	PC2
-0.1656	-0.1995
-1.7271	0.46635
1.57204	-0.0641
-1.8439	-0.1343
2.86527	0.16194
-0.7007	-0.2304

6x2

Example 3

1. Standardize the Dataset

- Assume we have the below dataset which has 4 features and a total of 5 training examples.

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

Example 3

- First, we need to standardize the dataset and for that, we need to calculate the mean and standard deviation for each feature.

f1	f2	f3	f4
1	2	3	4
5	5	6	7
1	4	2	3
5	3	2	1
8	1	2	2

	f1	f2	f3	f4
μ =	4	3	3	3.4
σ =	3	1.58114	1.73205	2.30217

$$x_{new} = \frac{x - \mu}{\sigma}$$

After applying the formula for each feature in the dataset is transformed as below:

f1	f2	f3	f4
-1	-0.63246	0	0.26062
0.33333	1.26491	1.73205	1.56374
-1	0.63246	-0.57735	-0.17375
0.33333	0	-0.57735	-1.04249
1.33333	-1.26491	-0.57735	-0.60812

Example 3

2. Calculate the covariance matrix for the whole dataset

The formula to calculate the covariance matrix:

	f1	f2	f3	f4
f1	var(f1)	cov(f1,f2)	cov(f1,f3)	cov(f1,f4)
f2	cov(f2,f1)	var(f2)	cov(f2,f3)	cov(f2,f4)
f3	cov(f3,f1)	cov(f3,f2)	var(f3)	cov(f3,f4)
f4	cov(f4,f1)	cov(f4,f2)	cov(f4,f3)	var(f4)

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$

Since we have standardized the dataset, so the ***mean for each feature is 0*** *and the standard deviation is 1.*

Example 3

f1	f2	f3	f4
-1	-0.63246	0	0.26062
0.33333	1.26491	1.73205	1.56374
-1	0.63246	-0.57735	-0.17375
0.33333	0	-0.57735	-1.04249
1.33333	-1.26491	-0.57735	-0.60812

- var(f1) = $((-1.0-0)^2 + (0.33-0)^2 + (-1.0-0)^2 + (0.33–0)^2 + (1.33–0)^2)/5$
- **var (f1) = 0.8**
- cov(f1,f2) =((-1.0–0)*(-0.632456-0) +
  (0.33–0)*(1.264911-0) +
  (-1.0–0)* (0.632456-0)+
  (0.33–0)*(0.000000 -0)+
  (1.33–0)*(-1.264911–0))/5
- **cov(f1,f2) = -0.25298**

Example3

- In the similar way be can calculate the other covariance's and which will result in the below covariance matrix

	f1	f2	f3	f4
f1	0.8	-0.25298	0.03849	-0.14479
f2	-0.25298	0.8	0.51121	0.4945
f3	0.03849	0.51121	0.8	0.75236
f4	-0.14479	0.4945	0.75236	0.8

Example 3

3. Calculate eigenvalues and eigen vectors.

$Av - \lambda Iv = 0$;

$(A - \lambda I)v = 0$

Since we have already know v is a non- zero vector, only way this equation can be equal to zero, if $\det(A - \lambda I) = 0$

Example 3

3. Calculate eigenvalues and eigen vectors.

$$\begin{bmatrix} 0.8 & -0.25298 & 0.03849 & -0.14479 \\ -0.25298 & 0.8 & 0.51121 & 0.4945 \\ 0.03849 & 0.51121 & 0.8 & 0.75236 \\ -0.14479 & 0.4945 & 0.75236 & 0.8 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 0$$

	f1	f2	f3	f4
f1	0.8 - λ	-0.25298	0.03849	-0.14479
f2	-0.25298	0.8- λ	0.51121	0.4945
f3	0.03849	0.51121	0.8 - λ	0.75236
f4	-0.14479	0.4945	0.75236	0.8 - λ

Solving the above equation = 0

- ***λ = 2.51579324 , 1.0652885 , 0.39388704 , 0.02503121***

Example 3

- **Eigenvectors:**

Solving the (A-λI)v = 0 equation for v vector with different λ values:

$$\begin{pmatrix} 0.800000-\lambda & -(0.252982) & 0.038490 & -(0.144791) \\ -(0.252982) & 0.800000-\lambda & 0.511208 & 0.494498 \\ 0.038490 & 0.511208 & 0.800000-\lambda & 0.752355 \\ -(0.144791) & 0.494498 & 0.752355 & 0.800000-\lambda \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = 0$$

For λ = *2.51579324, solving the above equation using Cramer's rule, the values for v vector are*
v1 = 0.16195986
v2 = -0.52404813
v3 = -0.58589647
v4 = -0.59654663

Example 3

- Going by the same approach, we can calculate the eigen vectors for the other eigen values. We can from a matrix using the **eigen vectors.**

e1	e2	e3	e4
0.161960	-0.917059	-0.307071	0.196162
-0.524048	0.206922	-0.817319	0.120610
-0.585896	-0.320539	0.188250	-0.720099
-0.596547	-0.115935	0.449733	0.654547

4. **Sort eigenvalues and their corresponding eigenvectors.**

- Since eigenvalues are already sorted in this case so no need to sort them again.

Example 3

5. Pick k eigenvalues and form a matrix of eigenvectors

- If we choose the top 2 eigenvectors, the matrix will look like this:

e1	e2
0.161960	-0.917059
-0.524048	0.206922
-0.585896	-0.320539
-0.596547	-0.115935

6. Transform the original matrix.

Feature matrix * top k eigenvectors = Transformed Data

f1	f2	f3	f4
-1.000000	-0.632456	0.000000	0.260623
0.333333	1.264911	1.732051	1.563740
-1.000000	0.632456	-0.577350	-0.173749
0.333333	0.000000	-0.577350	-1.042493
1.333333	-1.264911	-0.577350	-0.608121

(5,4)

*

e1	e2
0.161960	-0.917059
-0.524048	0.206922
-0.585896	-0.320539
-0.596547	-0.115935

(4,2)

=

nf1	nf2
0.014003	0.755975
-2.556534	-0.780432
-0.051480	1.253135
1.014150	0.000239
1.579861	-1.228917

(5,2)

```python
import numpy as np
import pandas as pd
A = np.matrix([[1,2,3,4],
               [5,5,6,7],
               [1,4,2,3],
               [5,3,2,1],
               [8,1,2,2]])

df = pd.DataFrame(A,columns  = ['f1','f2','f3','f4'])
df_std  = (df - df.mean()) / (df.std())
n_components = 2
from sklearn.decomposition import PCA
pca = PCA(n_components=n_components)
principalComponents = pca.fit_transform(df_std)
principalDf = pd.DataFrame(data=principalComponents,columns=['nf'+str(i+1) for i in range(n_components)])
print(principalDf)
```

```
        nf1       nf2
0 -0.014003  0.755975
1  2.556534 -0.780432
2  0.051480  1.253135
3 -1.014150  0.000239
4 -1.579861 -1.228917
```

Solve : Apply feature reduction using PCA

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

Reference

- https://www.simplilearn.com/tutorials/machine-learning-tutorial/principal-component-analysis
- https://www.youtube.com/watch?v=YepECGlOl4Y&t=17s
- https://www.youtube.com/watch?v=H99JRtDDnvk
- https://www.youtube.com/watch?v=osgqQy9Hr8s
- https://www.slingacademy.com/article/perform-principal-component-analysis-numpy/
- https://www.statskingdom.com/pca-calculator.html
- https://www.turing.com/kb/guide-to-principal-component-analysis

Complete Linkage (Hierarchical Agglomerative Clustering)

EXAMPLE

- Find the clusters using complete link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Example complete link

- The distance matrix is

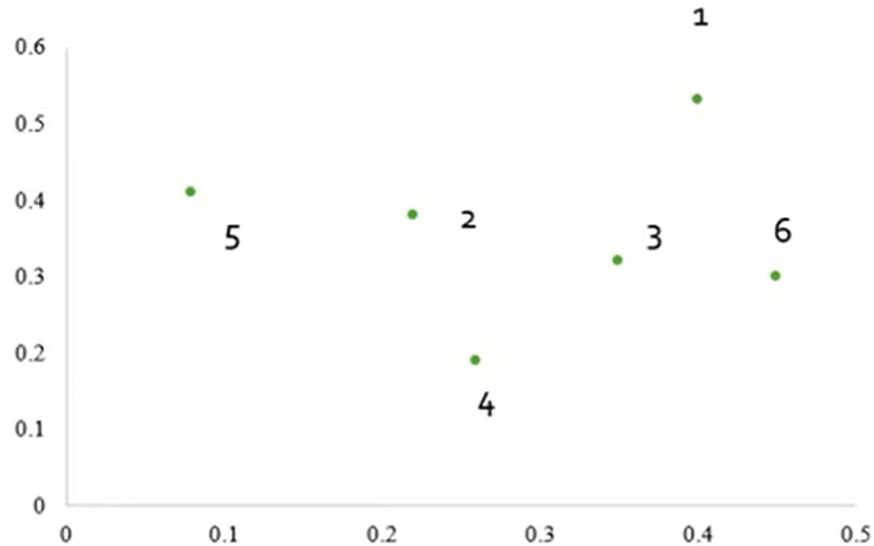
	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

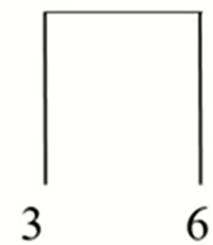
EXAMPLE COMPLETE LINK

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

Example complete link





EXAMPLE COMPLETE LINK

- The distance matrix for cluster P3,P6

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P3,P6),P1)]
- MAX(dist(P3,P1), (P6,P1))

  = MAX[(0.22,0.23)]

  = 0.23
- To update the distance matrix MAX[dist(P3,P6),P2)]
- MAX(dist(P3,P2), (P6,P2))

  = MAX[(0.15,0.25)]

  = 0.25

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P3,P6),P4)]
- MAX(dist(P3,P4), (P6,P4))

  = MAX[(0.15,0.22)]

  = 0.22
- To update the distance matrix MAX[dist(P3,P6),P5)]
- MAX(dist(P3,P5), (P6,P5))

  = MAX[(0.28,0.39)]

  = 0.39

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

Example complete link

- The updated distance matrix for cluster P3,P6

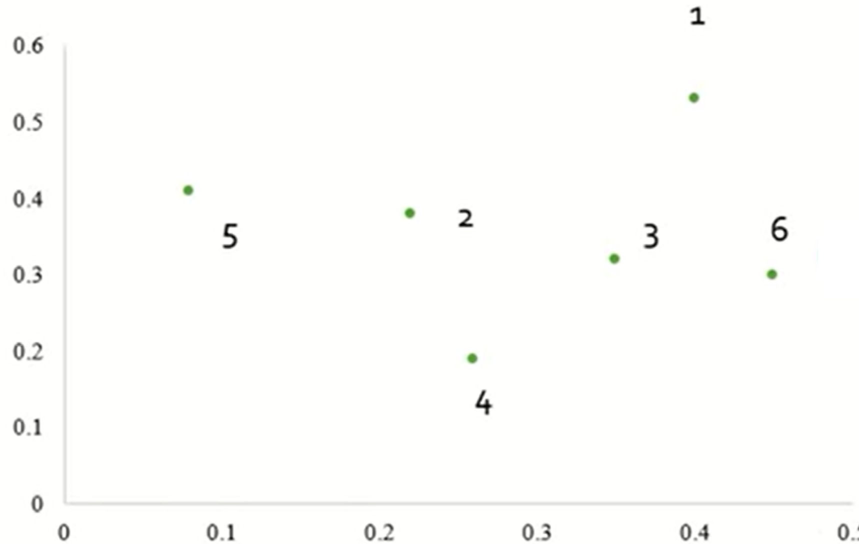
	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

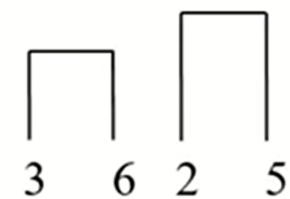
Example complete link

- The distance matrix

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK





Example complete link

- The distance matrix , cluster (P2,P5)

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK

- **T**o update the distance matrix MAX[dist(P2,P5),P1)]
- MAX[dist(P2,P1), (P5,P1)]

  = MAX[(0.23,0.34)]

  = 0.34
- **T**o update the distance matrix MAX[dist(P2,P5),(P3,P6)]
- MAX[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = MAX[(0.25,0.39)]

  = 0.39

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

EXAMPLE COMPLETE LINK

- To update the distance matrix MAX[dist(P2,P5),P4)]
- MAX[dist(P2,P4), (P5,P4)]

  = MAX[(0.20,0.29)]

  = 0.29

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

Example Complete Link

- The updated distance matrix for cluster (P2,P5)

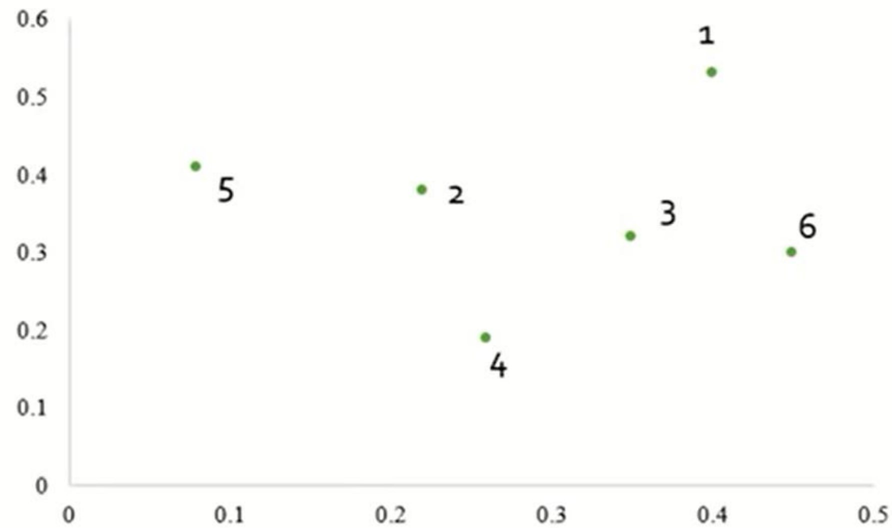
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

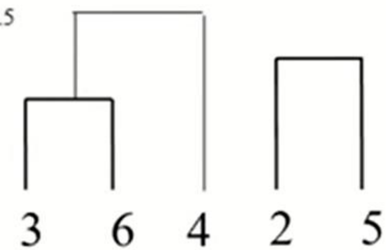
Example complete link

- The distance matrix is

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

EXAMPLE COMPLETE LINK





Example Complete Link

- To update the distance matrix[((P3,P6),P4) , P1]
- MAX[dist(P3,P6),P1), (P4,P1)]

  = MAX[(0.23,0.37)]

  = 0.37
- To update the distance matrix MAX[dist((P3,P6),P4),(P2,P5)]
- MAX[dist((P3,P6),(P2,P5)), (P4,(P2,P5))]

  = MAX[(0.39,0.29)]

  = 0.39

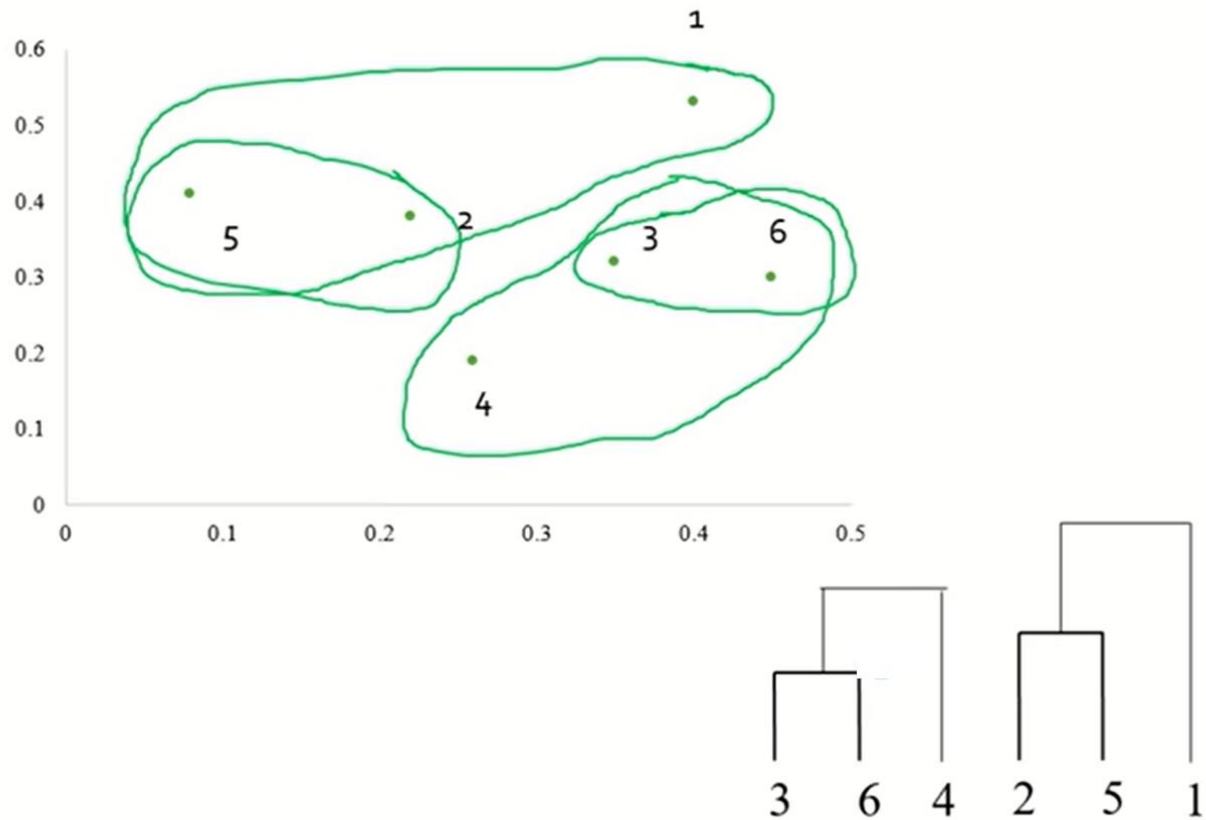
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.34	0		
P3,P6	0.23	0.39	0	
P4	0.37	0.29	0.22	0

Example complete link

- The updated distance matrix for cluster P3,P6,P4

	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.34	0	
P3,P6,P4	0.37	0.39	0

EXAMPLE COMPLETE LINK



EXAMPLE COMPLETE LINK

- **Cluster** [(P2,P5),P1]
- MAX[dist(P2,P5),(P3,P6,P4), (P1,(P3,P6,P4))]

  = MAX[(0.39,0.37)]

  = 0.39

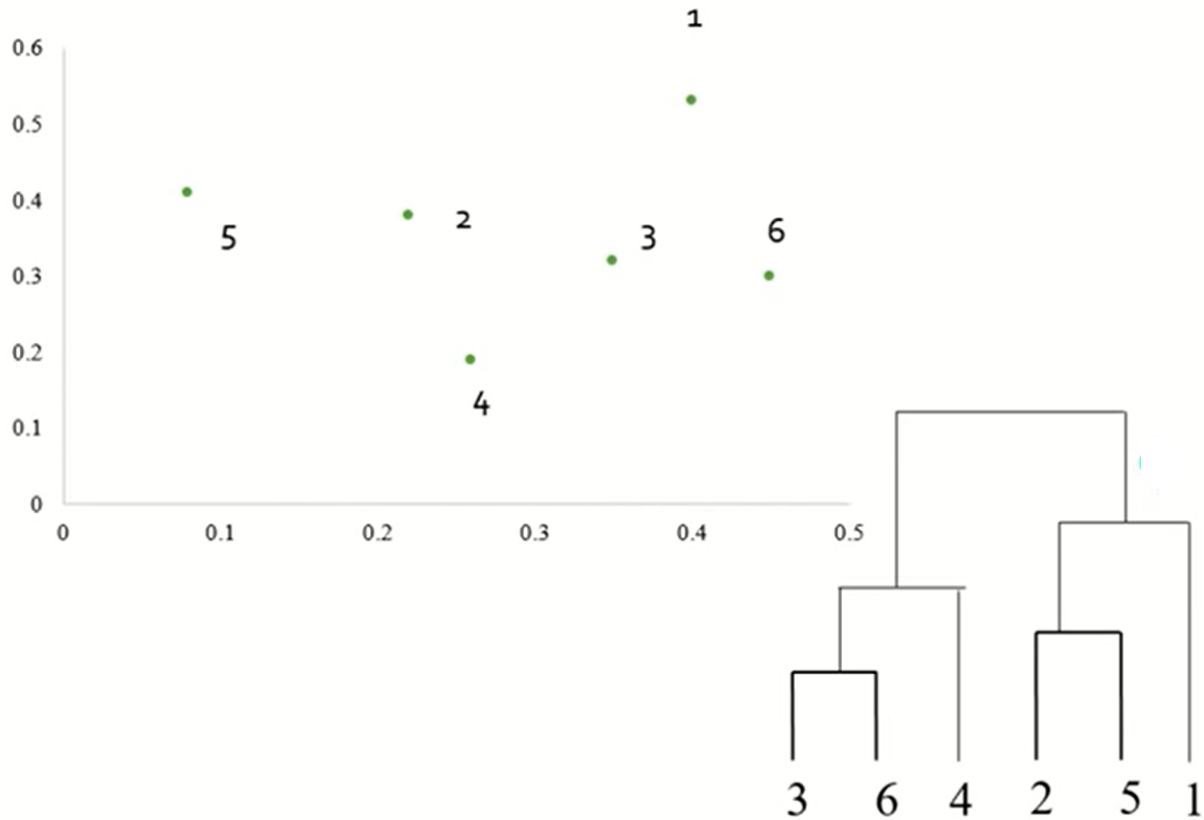
	P1	P2,P5	P3,P6,P4
P1	0		
P2,P5	0.34	0	
P3,P6,P4	0.37	0.39	0

Example complete link

- The updated distance matrix for cluster P2,P5,P1

	P2,P5,P1	P3,P6,P4
P2,P5,P1	0	
P3,P6,P4	0.39	0

EXAMPLE COMPLETE LINK



Module 3 Unsupervised Learning – Clustering

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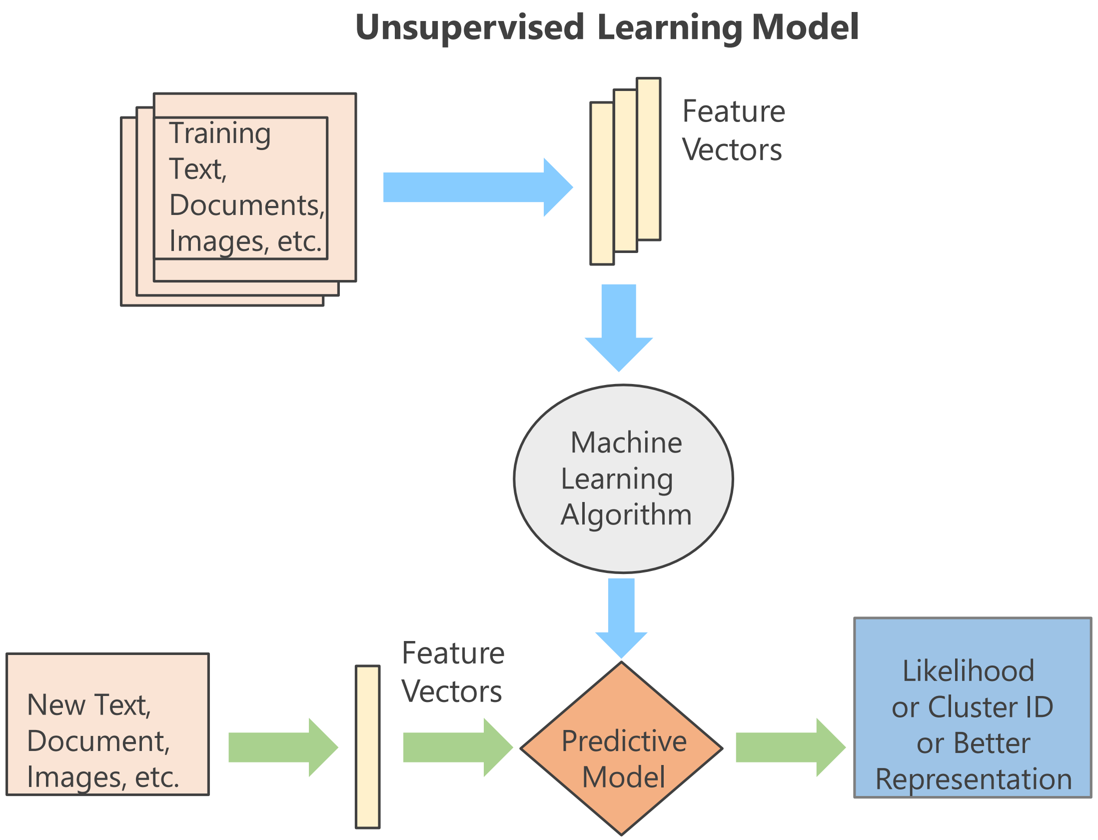
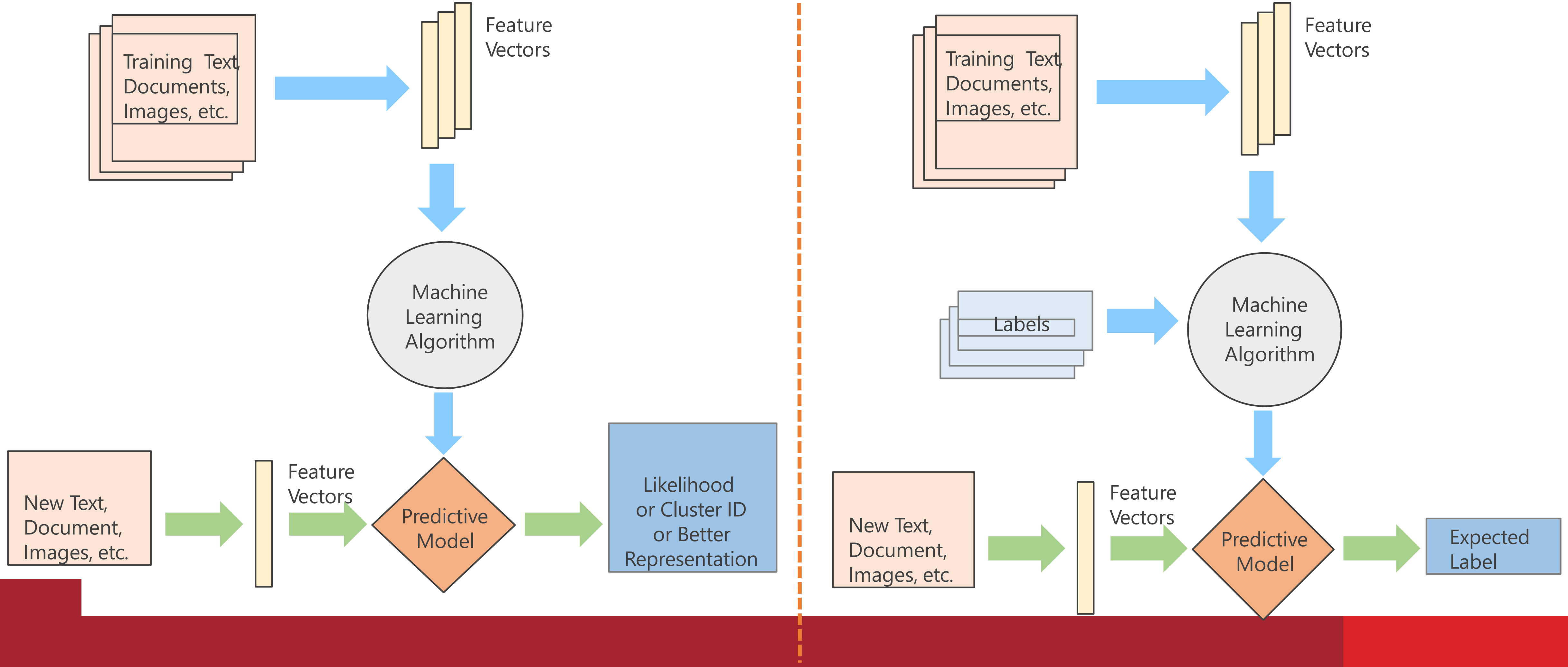
Unsupervised Learning Process Flow

The data has no labels. The machine just looks for whatever patterns it can find.



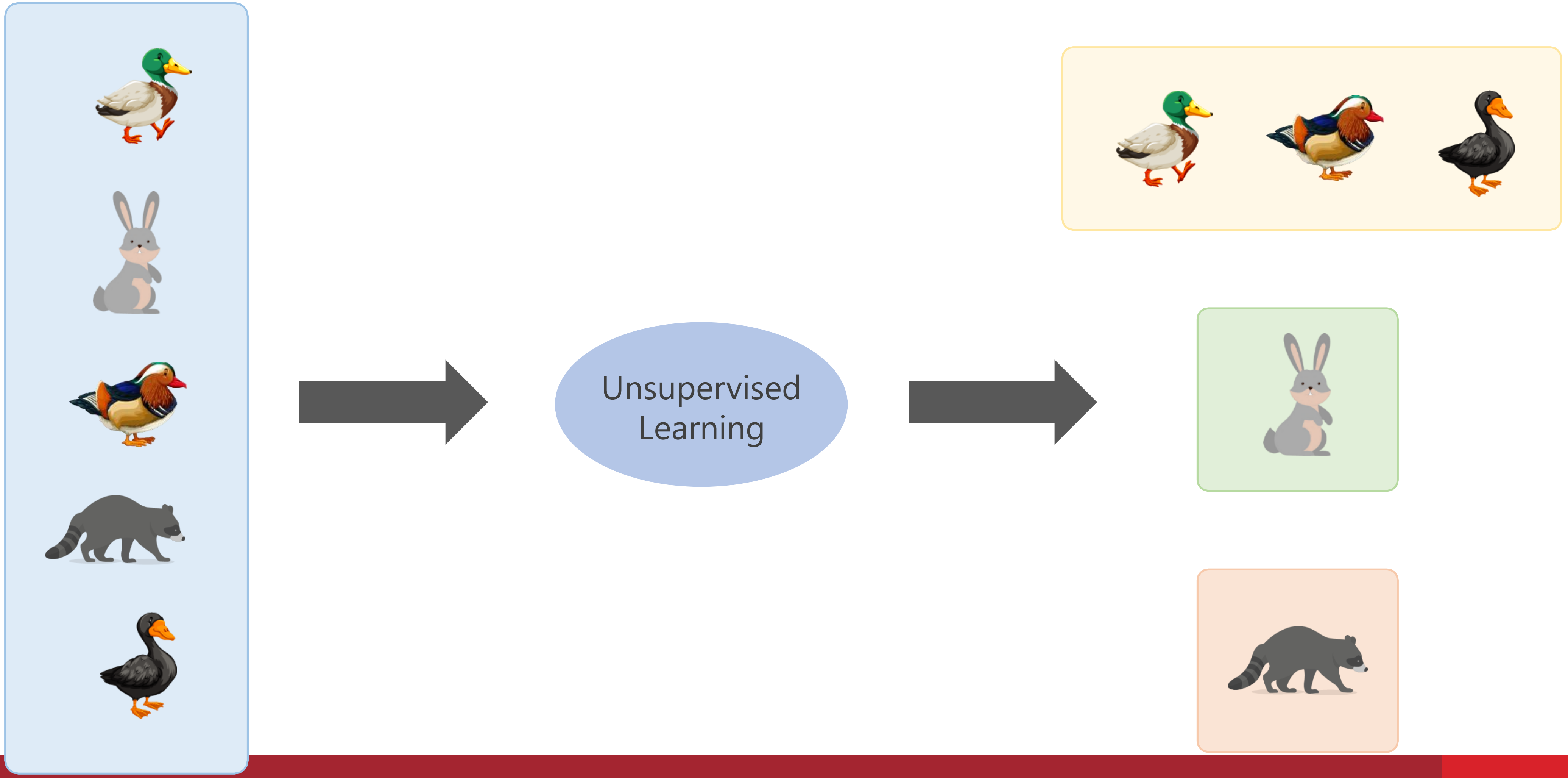
Unsupervised Learning vs. Supervised Learning

The only difference is the labels in the training data



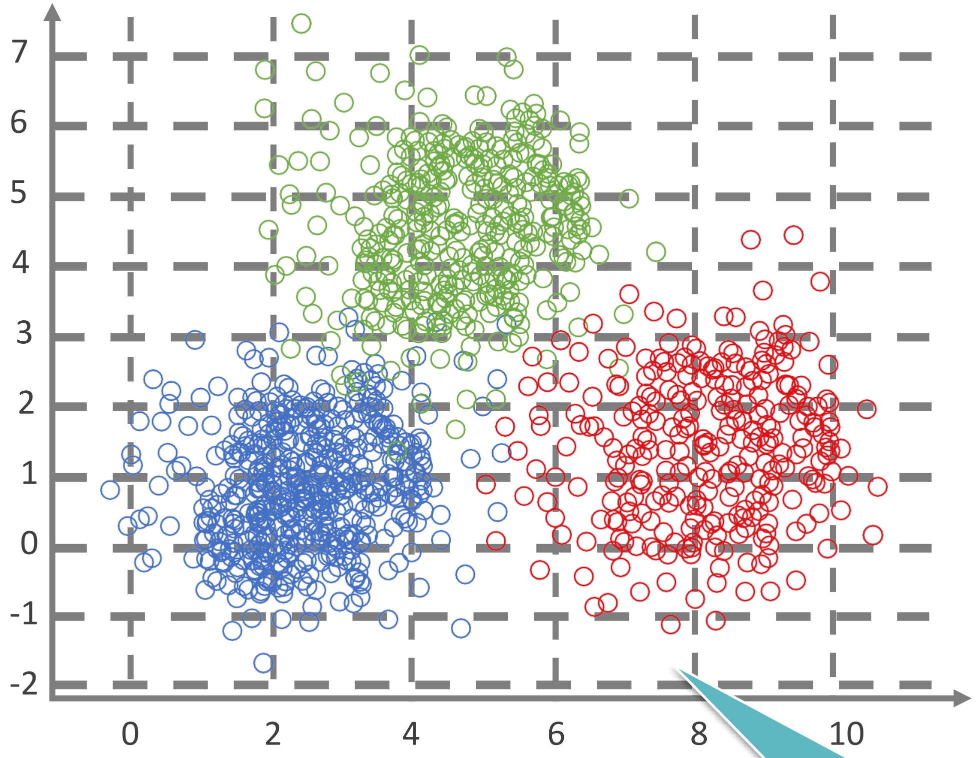
Unsupervised Learning: Example

Clustering like-looking birds/animals based on their features

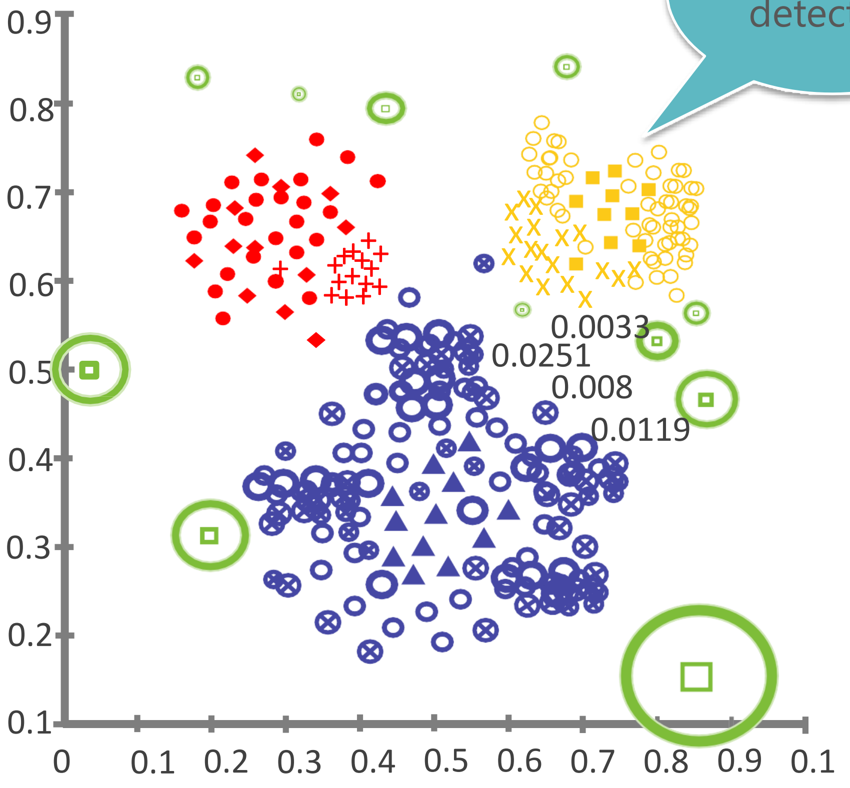


Application of Unsupervised Learning

Unsupervised learning can be used for anomaly detection as well as clustering



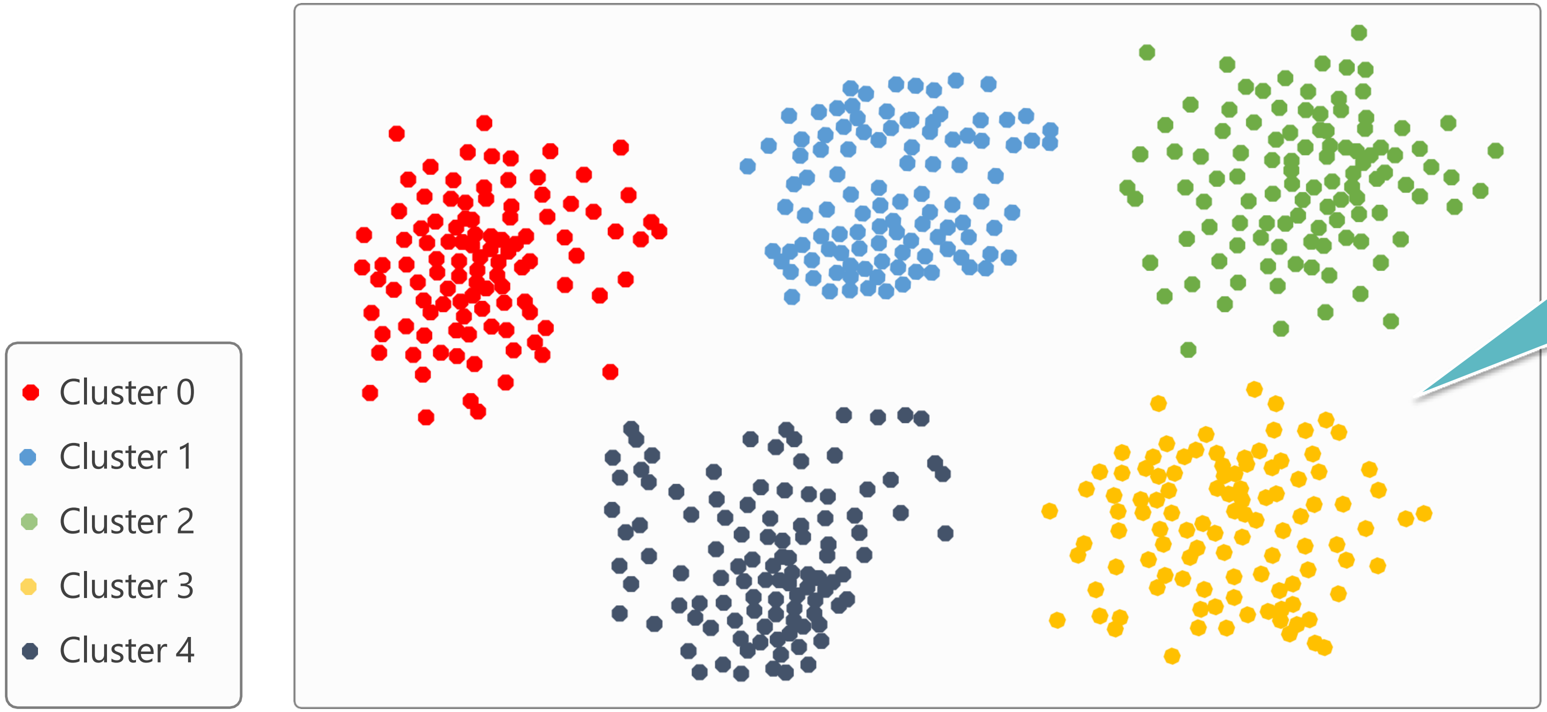
Identifying similarities in groups (Clustering)



Anomaly detection

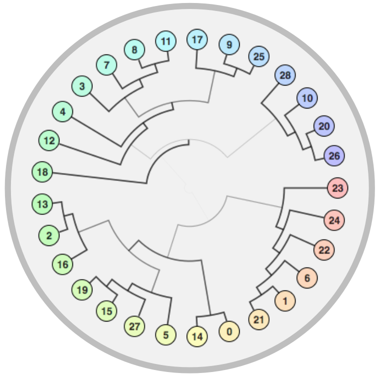
Clustering

Grouping objects based on the information found in data that describes the objects or their relationship



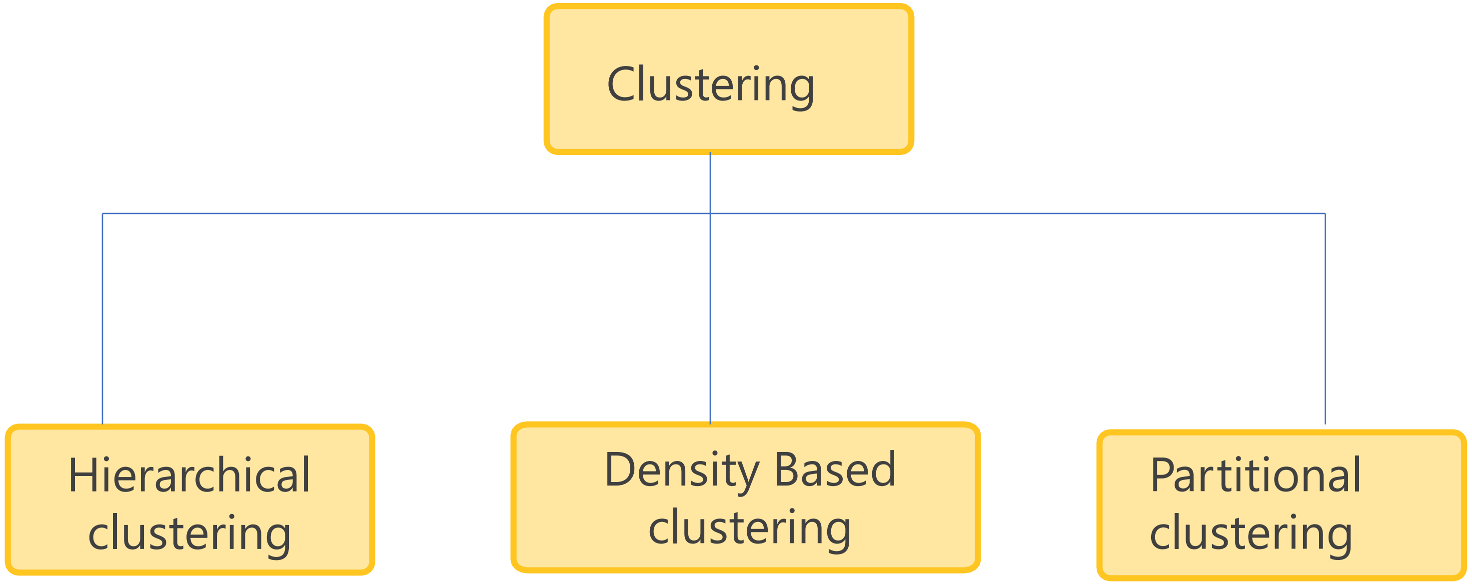
The goal is to see that similar objects are grouped into one cluster and different from objects in another cluster

Need of Clustering



- To determine the intrinsic grouping in a set of unlabeled data
- To organize data into clusters showing internal structure of the data
- To partition the data points
- To understand and extract value from large sets of structured and unstructured data

Types of Clustering



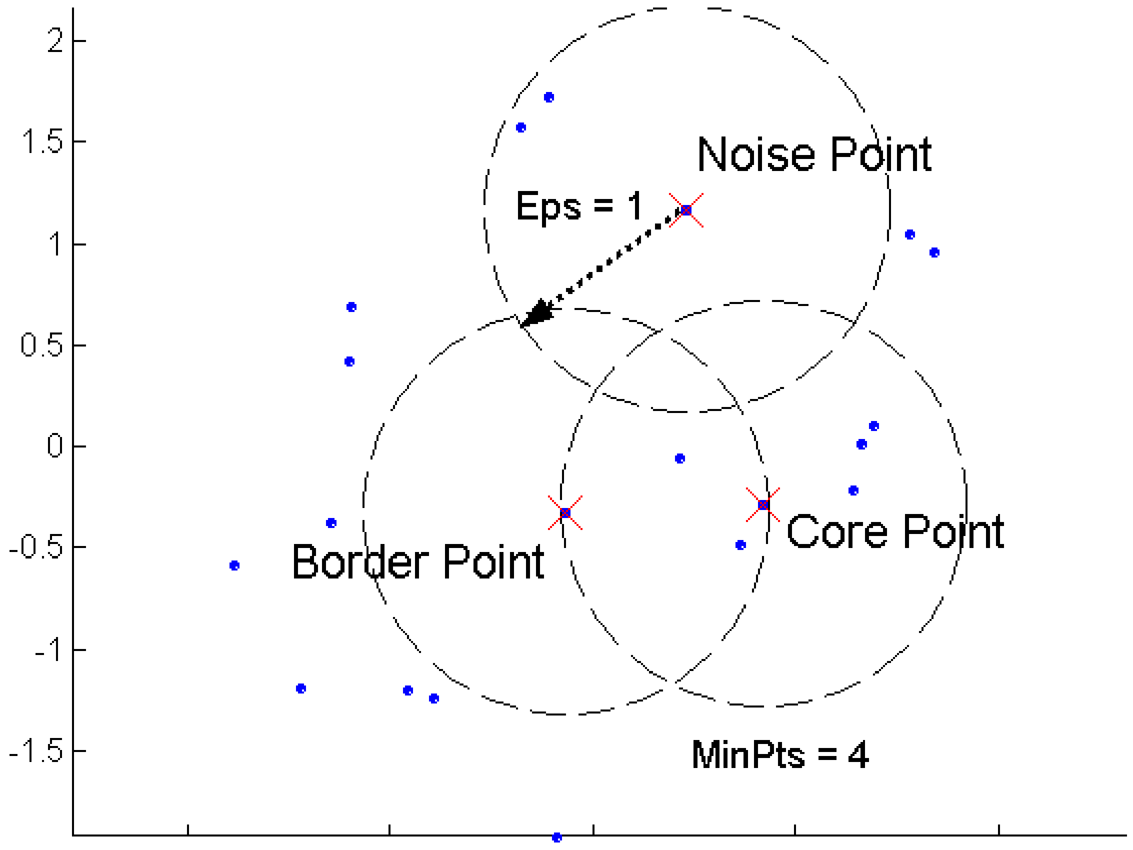
DBSCAN : Density-Based Clustering

- DBSCAN is a Density-Based Clustering algorithm
- Reminder: In density based clustering we partition points into dense regions separated by not-so-dense regions.
- Important Questions:
  - How do we measure density?
  - What is a dense region?
- DBSCAN:
  - Density at point p: number of points within a circle of radius Eps
  - Dense Region: A circle of radius Eps that contains at least MinPts points

DBSCAN

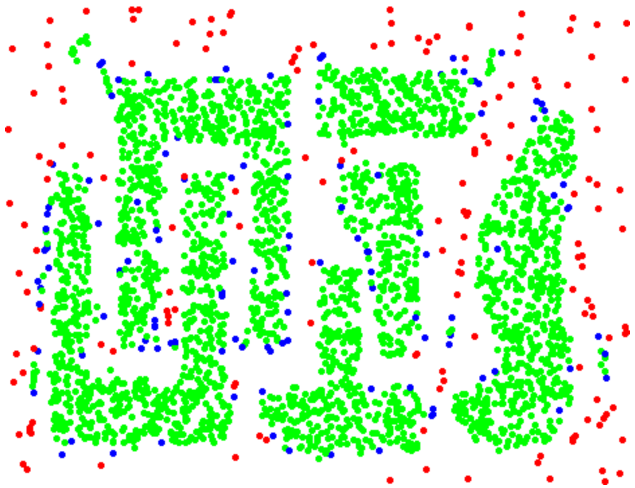
- Characterization of points
  - A point is a core point if it has more than a specified number of points (MinPts) within Eps
    - These points belong in a dense region and are at the interior of a cluster
  - A border point has fewer than MinPts within Eps, but is in the neighborhood of a core point.
  - A noise point is any point that is not a core point or a border point.

DBSCAN – Core, Border and Noise Points



DBSCAN – Core, Border and Noise Points

Original Points

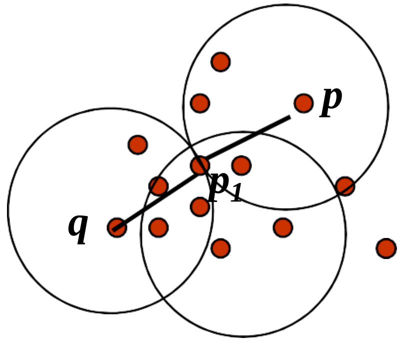


Point types: core, border and noise

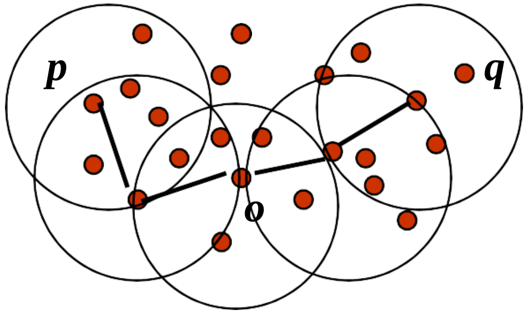
Eps = 10, MinPts = 4

Density Connected Points

- Density edge
  - We place an edge between two core points q and p if they are within distance Eps.



- Density-connected
  - A point p is density-connected to a point q if there is a path of edges from p to q



DBSCAN Algorithm

- Label points as core, border and noise
- Eliminate noise points
- For every core point p that has not been assigned to a cluster
  - Create a new cluster with the point p and all the points that are density-connected to p.
- Assign border points to the cluster of the closest core point.

Example 1

- Apply the DBSCAN algorithm to the given data points and
- Create the cluster with
- minPts=4 and
- Epsilon(ε)=1.9

Data Points:

P1: (3, 7)	P2: (4, 6)
P3: (5, 5)	P4: (6, 4)
P5: (7, 3)	P6: (6, 2)
P7: (7, 2)	P8: (8, 4)
P9: (3, 3)	P10: (2,6)
P11: (3,5)	P12: (2,4)

Example 1

P1: (3,7)
P2: (4,6)
P3: (5,5)
P4: (6,4)
P5: (7,3)
P6: (6,2)
P7: (7,2)
P8: (8,4)
P9: (3,3)
P10:(2,6)
P11:(3,5)
P12: (2,4)

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.88	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

Example 1

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1	0											
P2	1.41	0										
P3	2.83	1.41	0									
P4	4.24	2.83	1.41	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.88	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9,P11

Example 1

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9,P11

Point	Status	
P1	Noise	Border
P2	Core	
P3	Noise	Border
P4	Noise	Border
P5	Core	
P6	Noise	Border
P7	Noise	Border
P8	Noise	Border
P9	Noise	
P10	Noise	Border
P11	Core	
P12	Noise	Border

Example 1

P1: P2, P10

P2: P1, P3, P11

P3: P2, P4

P4: P3, P5

P5: P4, P6, P7, P8

P6: P5, P7

P7: P5, P6

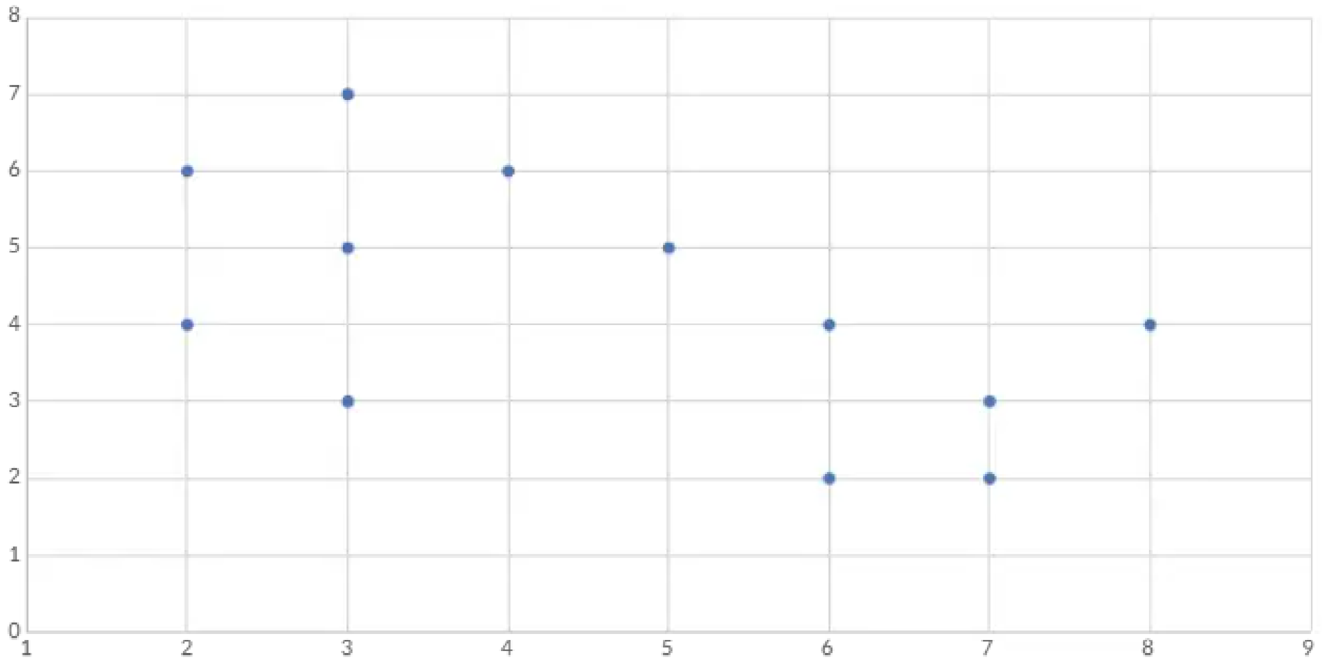
P8: P5

P9: P12

P10: P1, P11

P11: P2, P10, P12

P12: P9,P11



Evaluation Metrics

- Silhouette Score
- Adjusted Rand Index (ARI)
- Measure clustering quality

Silhouette Score

- Range: -1 to 1
- Near 1 → strong clustering
- Near 0 → overlapping clusters
- Near -1 → incorrect clustering

Silhouette Score Formula

- s(i) = (b(i) - a(i)) / max(a(i), b(i))
- **a(i)** = Average distance between point i and all other points **in the same cluster**
→ measures **cohesion (intra-cluster distance)**
- **b(i)** = Minimum average distance between point i and all points in the **nearest neighboring cluster**

Silhouette Score Interpretation

- Close to +1: well clustered
- Close to 0: overlapping clusters
- Close to -1: misclassified

ARI Formula

$$ARI = \frac{RI - Expected\ RI}{Max\ RI - Expected\ RI}$$

- 1: Perfect match
- 0: Random clustering
- <0: Worse than random

Adjusted Rand Index

- Range: 0 to 1
- >0.9 excellent
- >0.8 good clustering
- <0.5 poor clustering

Example 2

- Perform DBScan on the given problem with Eps = 2 and MinPts = 2

	X	Y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

Example 2

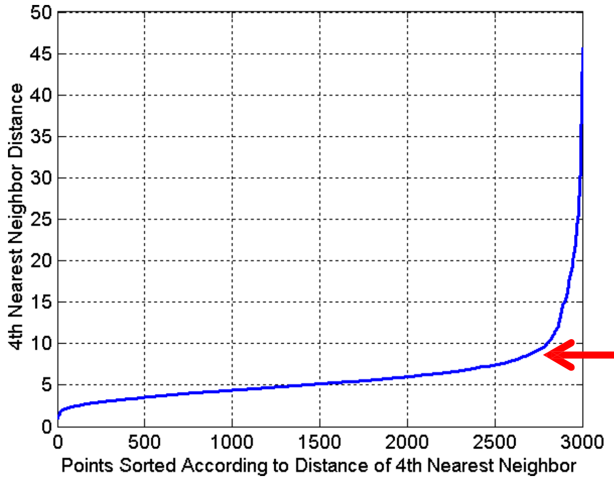
- The foundation of DBSCAN is the distance between every pair of points. Using the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$

	X	Y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0							
A2	5	0						
A3	8.49	6.08	0					
A4	3.61	4.24	5	0				
A5	7,07	5	1.41	3.61	0			
A6	7.21	4.12	2	4.12	1.41	0		
A7	8.06	3.16	7.28	7.21	6.71	5.39	0	
A8	2.24	4.47	6.4	1.41	5	5.39	7.62	0

DBSCAN – Determining Eps and MinPts

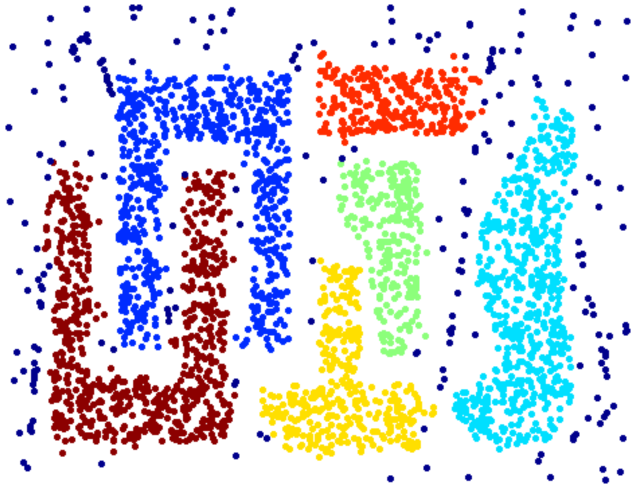
- Idea is that for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance
- Noise points have the k^{th} nearest neighbor at farther distance
- So, plot sorted distance of every point to its k^{th} nearest neighbor
- Find the distance d where there is a "knee" in the curve
  - Eps = d, MinPts = k



Eps ~ 7-10
MinPts = 4

When DBSCAN Works Well

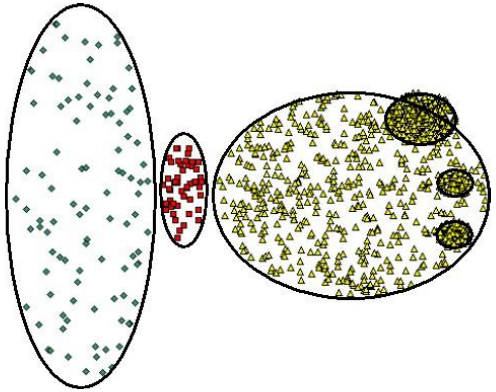
Original Points



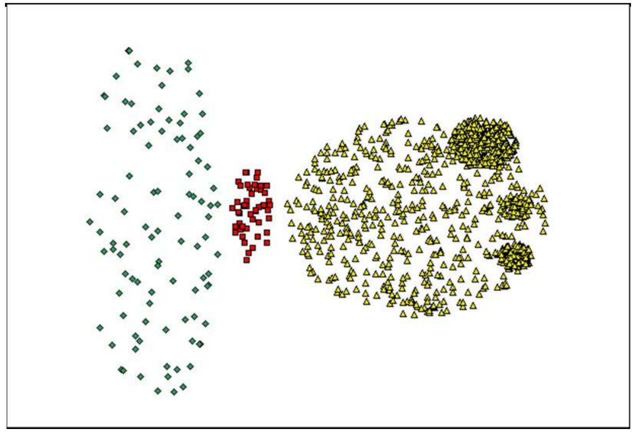
Clusters

- Resistant to Noise
- Can handle clusters of different shapes and sizes

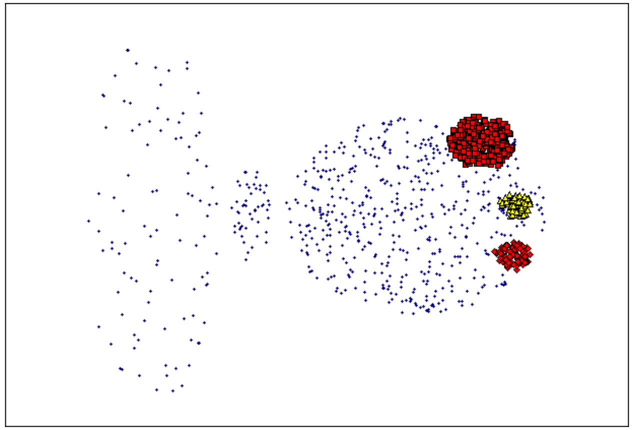
When DBSCAN not works well



Original Points



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

- Varying densities
- High-dimensional data

DBSCAN Parameter Sensitive

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

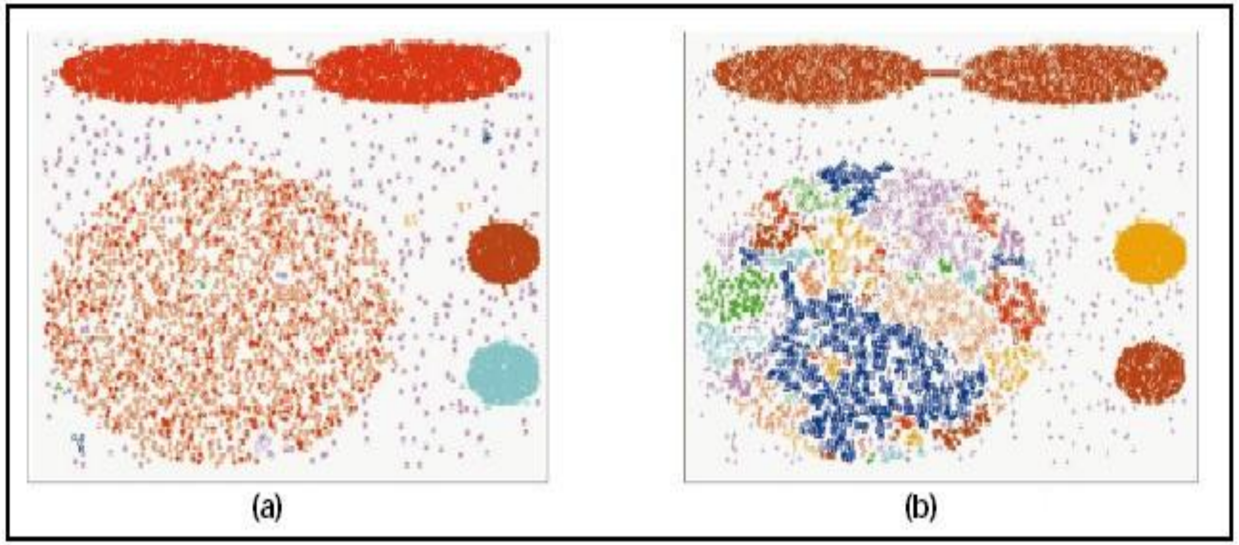
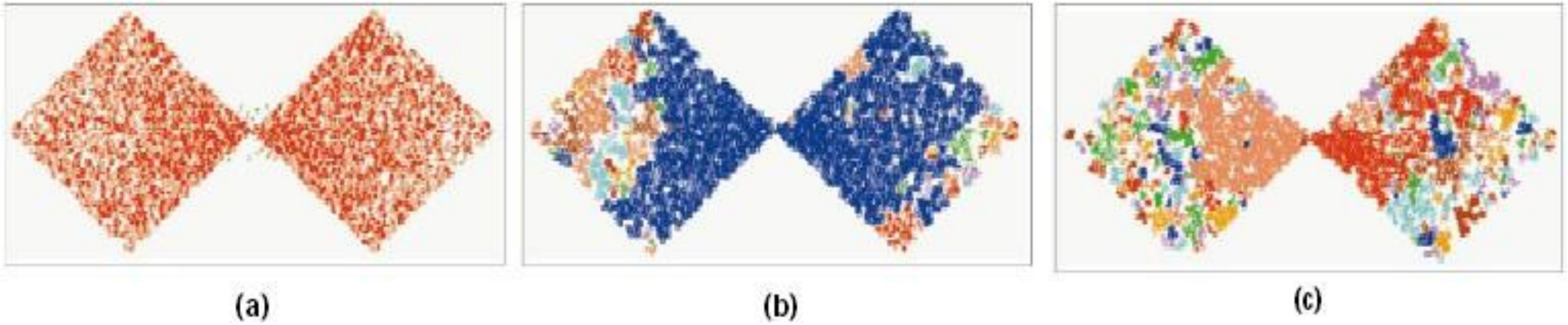


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.

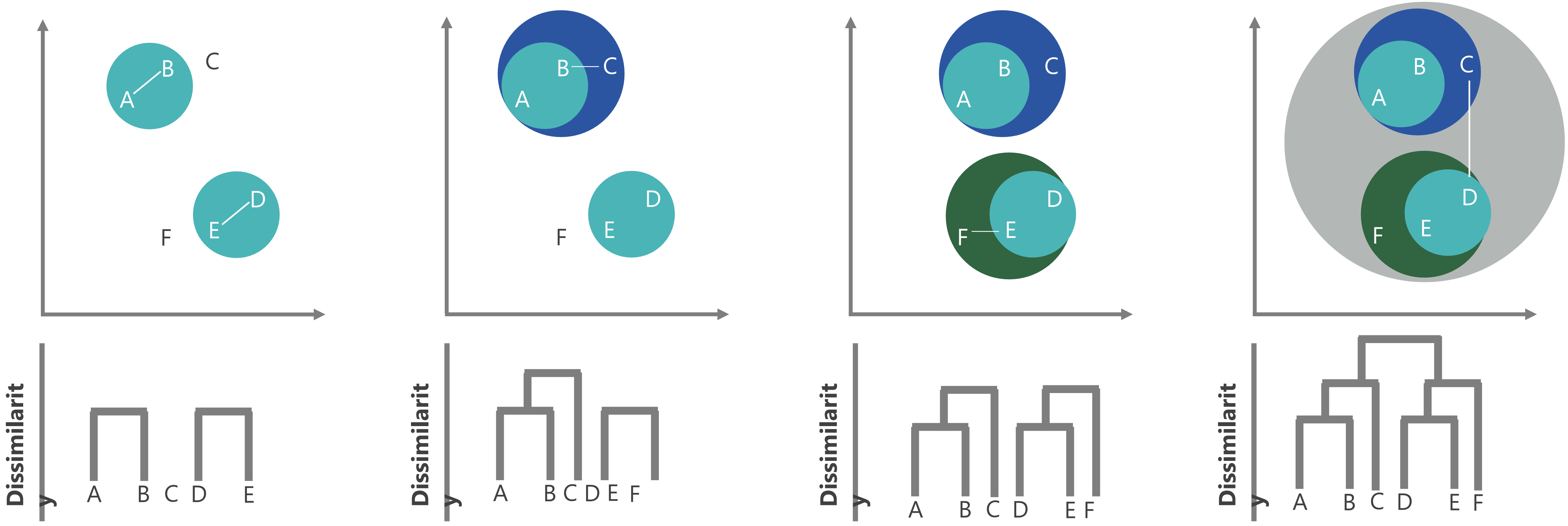


Other algorithms

- PAM, CLARANS: Solutions for the k-medoids problem
- BIRCH: Constructs a hierarchical tree that acts a summary of the data, and then clusters the leaves.
- MST: Clustering using the Minimum Spanning Tree.
- ROCK: clustering categorical data by neighbor and link analysis
- LIMBO, COOLCAT: Clustering categorical data using information theoretic tools.
- CURE: Hierarchical algorithm uses different representation of the cluster
- CHAMELEON: Hierarchical algorithm uses closeness and interconnectivity for merging

Hierarchical Clustering

Outputs a hierarchy, a structure that is more informative than the unstructured set of clusters returned by flat clustering



Combine A and B based on similarity
Combine D and E based on similarity

1

Combination of A and B is combined with C

2

Combination of D and E is combined with F

3

Final tree contains all clusters Combined into a single cluster

4

Working: Hierarchical Clustering

Step 1

Assign each item to its own cluster, such that if you have N number of items, you now have N number of clusters

Step 2

Find the closest (most similar) pair of clusters and merge them into a single cluster. Now you have one less cluster

Step 3

Compute distances (similarities) between the new cluster and every old cluster

Step 4

Repeat steps 2 and 3 until all items are clustered into a single cluster of size N

Distance Measures

Complete - Linkage clustering

- Find the maximum possible distance between points belonging to two different clusters

Single - Linkage Clustering

- Find the minimum possible distance between points belonging to two different clusters

Mean - Linkage Clustering

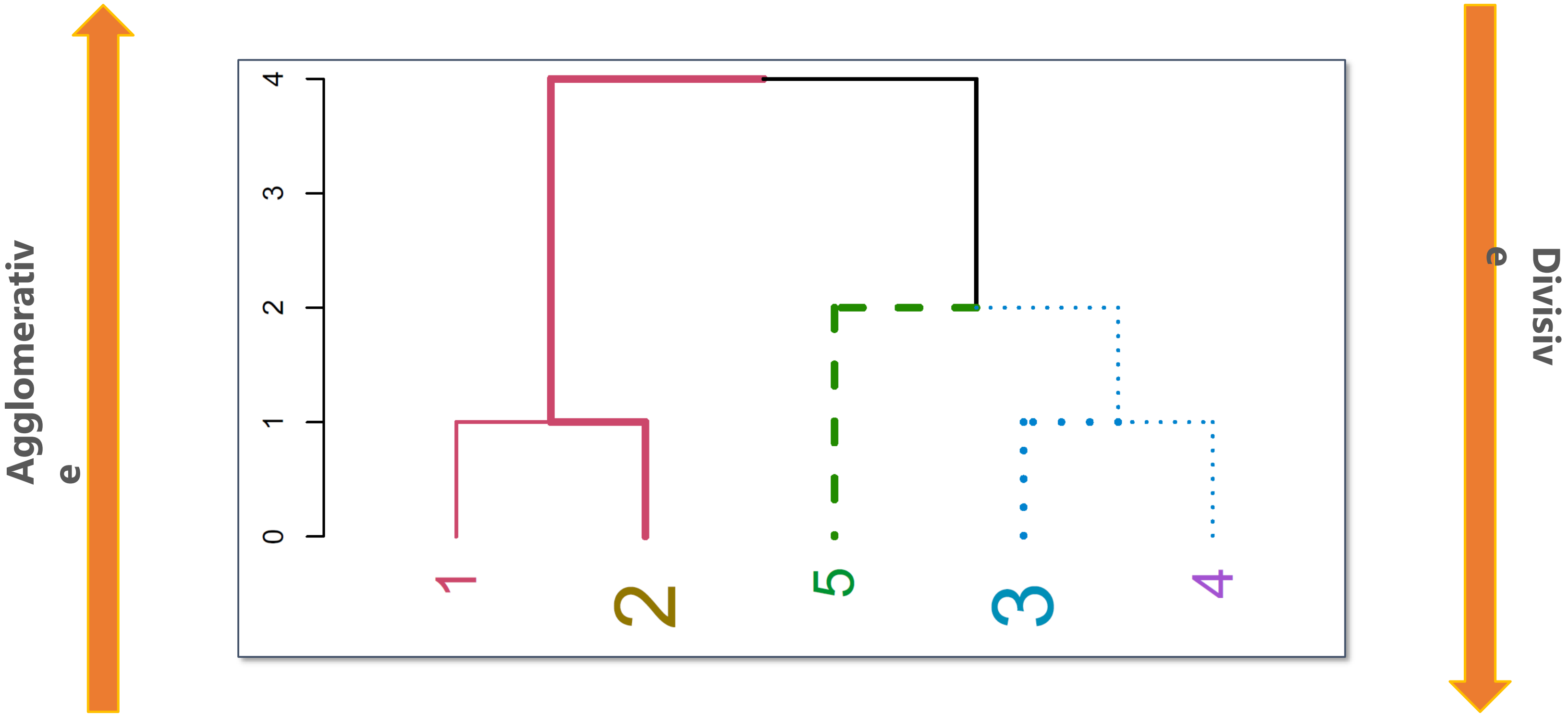
- Find all possible pair-wise distances for points belonging to two different clusters and then calculate the average

Centroid - Linkage Clustering

- Find the centroids of each cluster and calculate the distance between them

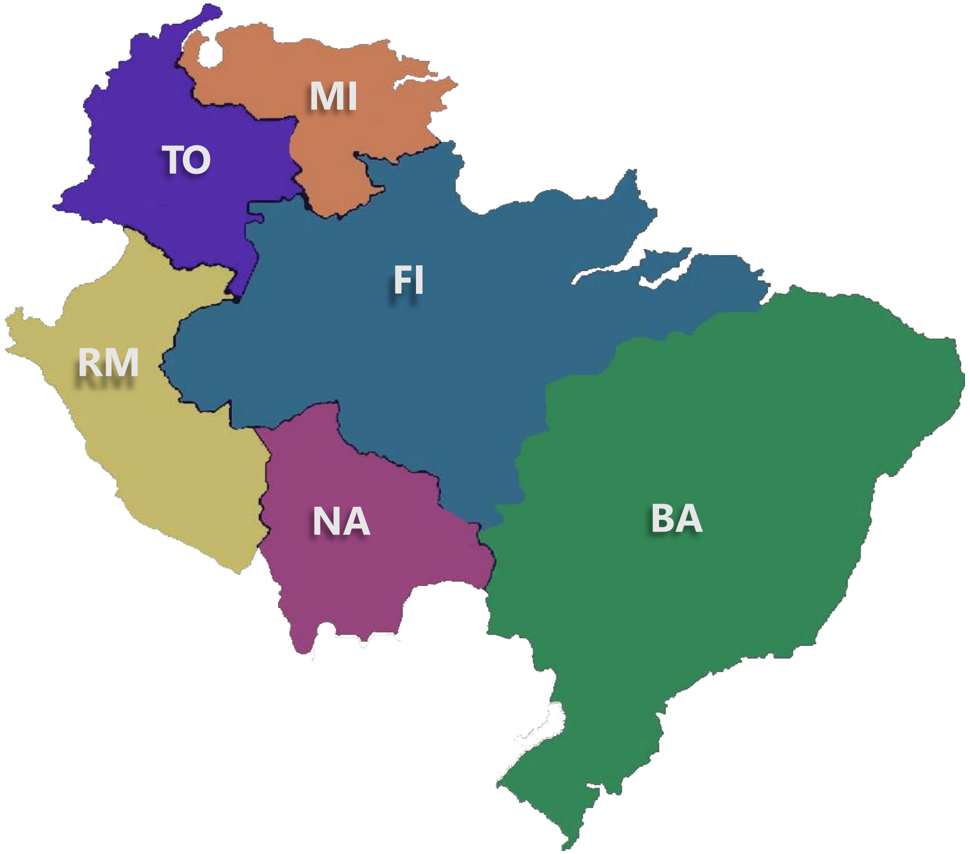
The Dendrogram

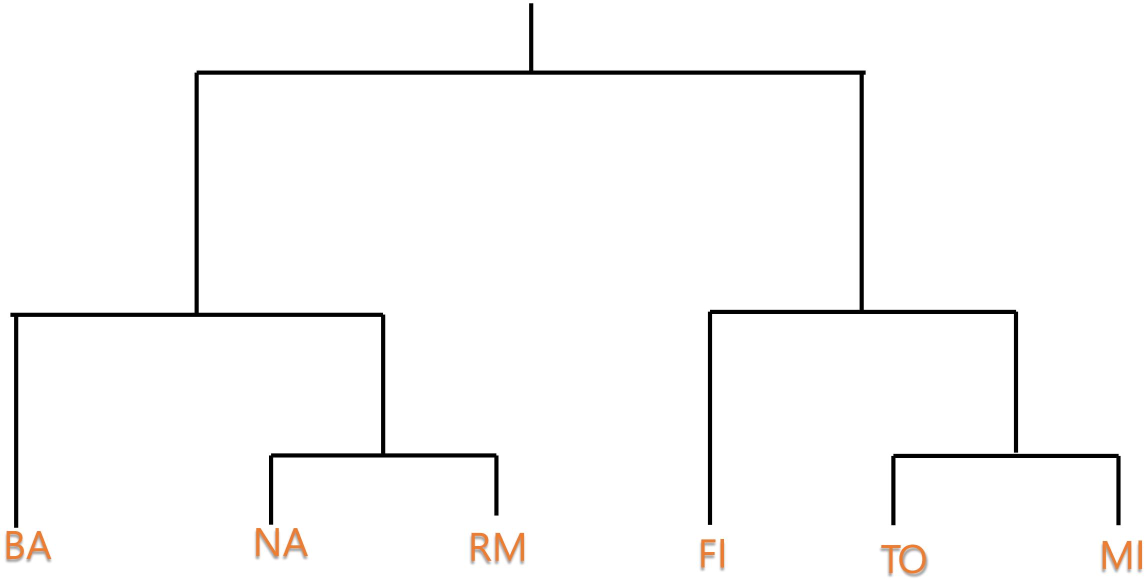
Dendrogram ((in Greek, *dendro* means tree and *gramma* means drawing) is a tree diagram frequently used to illustrate the arrangement of the clusters produced by hierarchical clustering.



Hierarchical Clustering: Example

A hierarchical clustering of distances between cities in kilometers





Hierarchical Clustering: Step 1

Create distance matrix of data

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0

Distance between TO and MI

Distance Matrix

Hierarchical Clustering: Step 2

From the distance matrix, you can see that MI has least distance with TO and they form a cluster together

	BA	FI	MI	NA	RM	TO
BA	0	662	877	255	412	996
FI	662	0	295	468	268	400
MI	877	295	0	754	564	138
NA	255	468	754	0	219	869
RM	412	268	564	219	0	669
TO	996	400	138	869	669	0

	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0

As the MI column has lower values than TO column, MI/TO consists of MI column values

Hierarchical Clustering: Step 3

Repeat clustering until a single cluster is obtained with all the members in it

	BA	FI	MI/TO	NA	RM
BA	0	662	877	255	412
FI	662	0	295	468	268
MI/TO	877	295	0	754	564
NA	255	468	754	0	219
RM	412	268	564	219	0

	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	468	564	0

Hierarchical Clustering: Step 3 (Contd.)

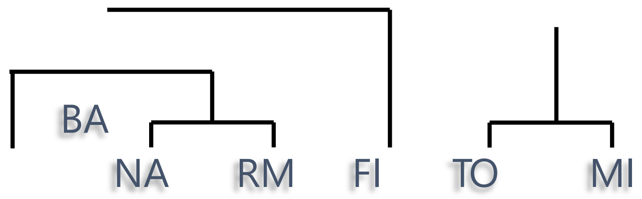
	BA	FI	MI/TO	NA/RM
BA	0	662	877	255
FI	662	0	295	268
MI/TO	877	295	0	564
NA/RM	255	268	564	0

	BA/(NA/RM)	FI	MI/TO
BA/(NA/RM)	0	268	564
FI	268	0	295
MI/TO	564	295	0

Hierarchical Clustering: Step 3 (Contd.)

	BA/(NA/RM)	FI	MI/TO
BA/(NA/RM)	0	268	564
FI	268	0	295
MI/TO	564	295	0

	BA/(NA/RM)/FI	(MI/TO)
BA/(NA/RM)/FI	0	295
(MI/TO)	295	0



Example

- Construct the dendrogram using Agglomerative Hierarchical Clustering (Single Linkage Method) for the given distance matrix.
  - Initially each data point forms a cluster.
  - Compute the distance matrix between the clusters.
  - Repeat
    - Merge the two closest clusters
    - Update the distance matrix
  - Until only a single cluster remains.

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Step 1 : {a}, {b}, {c}, {d}, {e}

Step 2 : Distance is given

Step 3: Minimum is 2 ; form c,e as cluster

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Example

Step 3: Updated Distance Matrix after forming cluster {ce}

d(a,{ce}) , d(d,{ce})

	a	b	d	ce
a	0	9	6	3
b	9	0	5	6
d	6	5	0	8
ce	3	6	8	0

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b},{c,e}, {d}

Example

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

- Step 3: Minimum distance = 3 → Merge (a, {ce}) → New cluster = {ace}
- Updated Distance Matrix after forming cluster {ace}
- d(b,{ace}) , d(d,{ace})

	ace	b	d
ace	0	6	6
b	6	0	5
d	6	5	0

	ace	b	d
ace	0		
b		0	
d			0

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b},{c,e}, {d}

{a,c,e}, {b},{d}

Example

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

- Step 3: Minimum distance = 5 → Merge (b, d) → New cluster = {bd}

Cluster will

{a}, {b}, {c}, {d}, {e}

{a}, {b},{c,e}, {d}

{a,c,e}, {b},{d}

{a,c,e}, {b,d}

Example

- Complete Linkage and Average Linkage Solved

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	6
c	3	7	0	9	2
d	6	5	9	0	8
e	11	6	2	8	0

Advantages

- Easy to understand
- No need to choose number of clusters initially
- Dendrogram gives visual representation

Disadvantages

- Computationally expensive for large datasets
- Sensitive to noise and outliers
- Once merged, clusters cannot be split

Applications

- Gene expression analysis
- Document clustering
- Customer segmentation
- Image segmentation

Applications

Refer Machine Learning book by Anuradha

Hierarchical Agglomerative Clustering [HAC]

HAC

- Starts with one cluster , individual item in its own cluster and iteratively merge clusters until all the items belong to one cluster.
- Bottom up approach is followed to merge the clusters together.
- Dendrograms are pictorially used to represent the HAC.

HAC

- Single-nearest distance or single linkage.
- Complete-farthest distance or complete linkage.
- Average-average distance or average linkage.

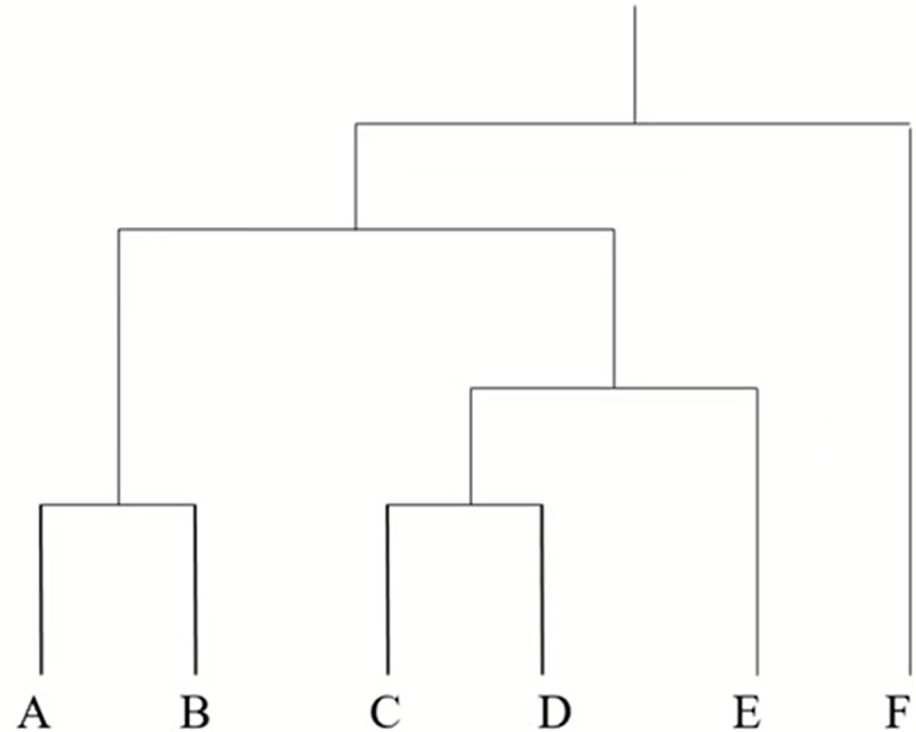
HAC

Single Linkage	This is the distance between the closest members of the two clusters.
Complete Linkage	This is the distance between the members that are farthest apart.
Average Linkage	This method involves looking at the distances between all pairs and averages all of these distances. This is also called Unweighted Pair Group Mean Averaging.

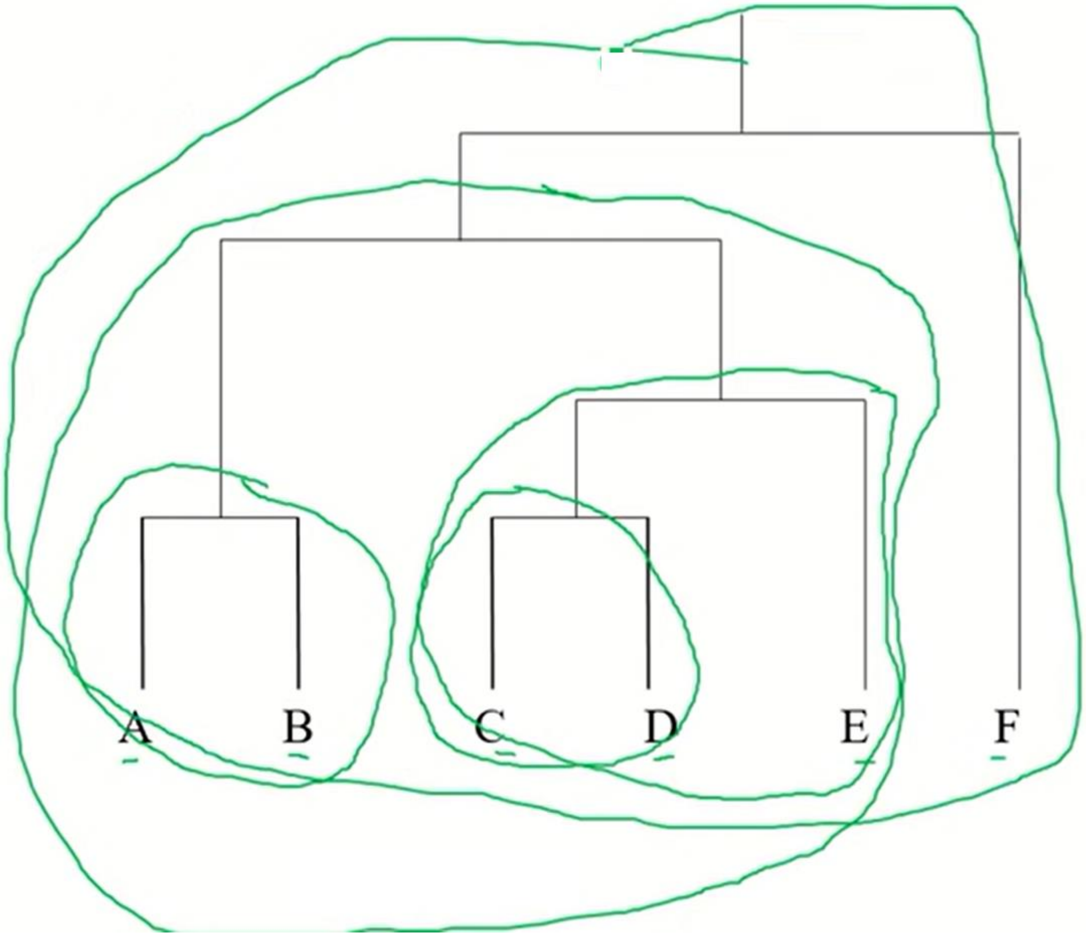
DENDROGRAM

- **A** tree like structure which represents hierarchical technique.
  - ✓ Leaf- Individual.
  - ✓ Root – One cluster.
- A cluster at level 1, is the merger of its child cluster at level i + 1.

DENDROGRAM



DENDROGRAM



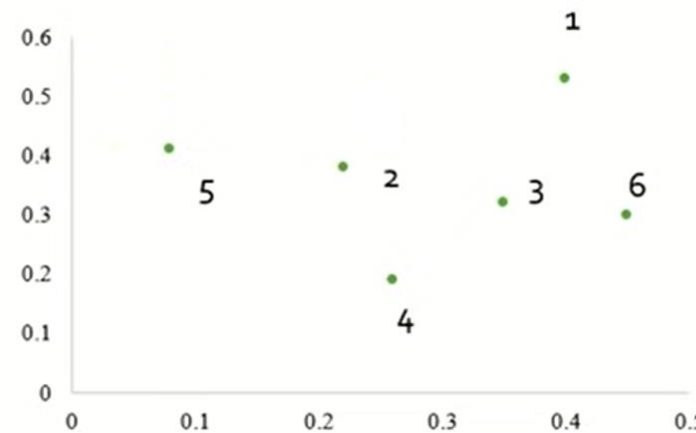
EXAMPLE

- Find the clusters using single link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

EXAMPLE SINGLE LINK

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30



EXAMPLE SINGLE LINK

- Calculate Euclidean distance, create the distance matrix.

Distance [(x,y), (a,b)] = $\sqrt{(x-a)^2+(x-b)^2}$

Distance (P1,P2) $= \sqrt{(0.40-0.22)^2+(0.53-0.38)^2}$

(0.40,0.53), (0.22,0.38) $= \sqrt{(0.18)^2+(0.15)^2}$

$= \sqrt{0.0324+0.0225}$

$= \sqrt{0.0549}$

$= 0.23$

Example

Single Link

- The distance matrix is

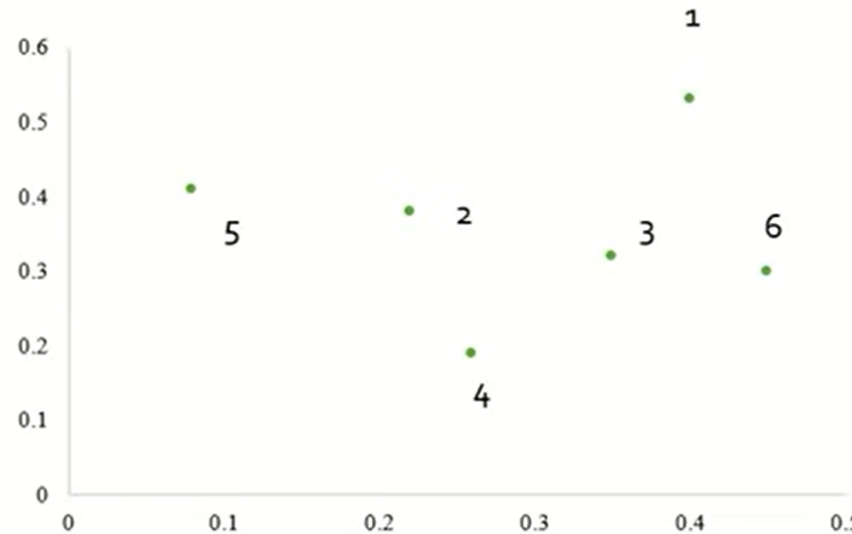
	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

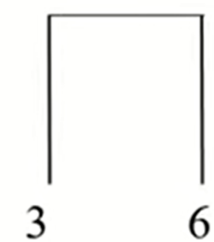
EXAMPLE SINGLE LINK

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.24	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE SINGLE LINK





EXAMPLE SINGLE LINK

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist(P3,P6),P1)]
- MIN(dist(P3,P1), (P6,P1))

  = min[(0.22,0.23)]

  = 0.22
- To update the distance matrix MIN[dist(P3,P6),P2)]
- MIN(dist(P3,P2), (P6,P2))

  = min[(0.15,0.25)]

  = 0.15

EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist(P3,P6),P4)]
- MIN(dist(P3,P4), (P6,P4))

  = min[(0.15,0.22)]

  = 0.15
- To update the distance matrix MIN[dist(P3,P6),P5)]
- MIN(dist(P3,P5), (P6,P5))

  = min[(0.28,0.39)]

  = 0.28

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P3, P6

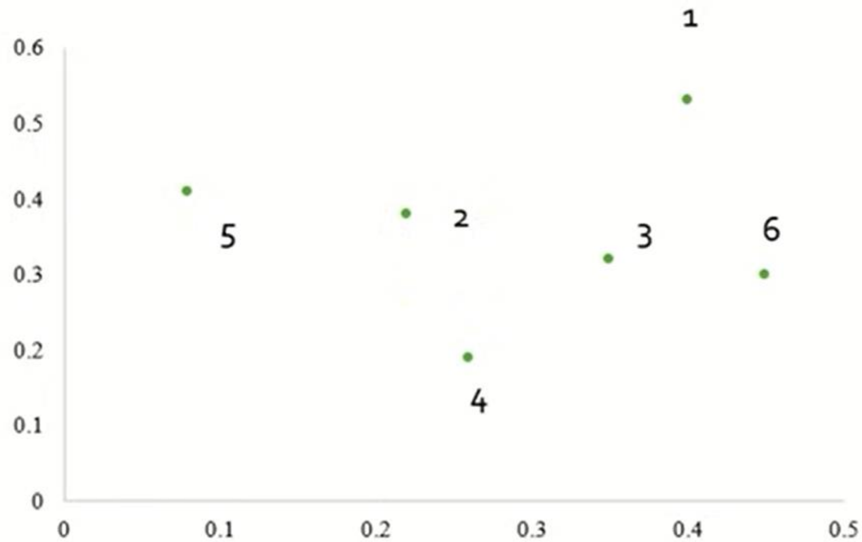
	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

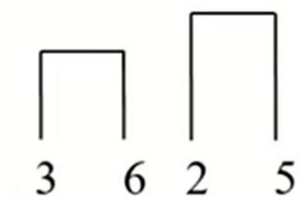
EXAMPLE SINGLE LINK

- The distance matrix fro cluster P2, P5

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

Example single link





EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist(P2,P5),P1)]
- MIN[dist(P2,P1), (P5,P1)]

  = min[(0.23,0.34)]

  = 0.23
- To update the distance matrix MIN[dist(P2,P5),(P3,P6)]
- MIN[dist(P2,(P3,P6)), (P5,(P3,P6))]

  = min[(0.15,0.28)]

  = 0.15

EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist(P2,P5),P4)]
- MIN[dist(P2,P4), (P5,P4)]

  = min[(0.20,0.29)]

  = 0.20

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

Example Single Link

- The distance matrix is

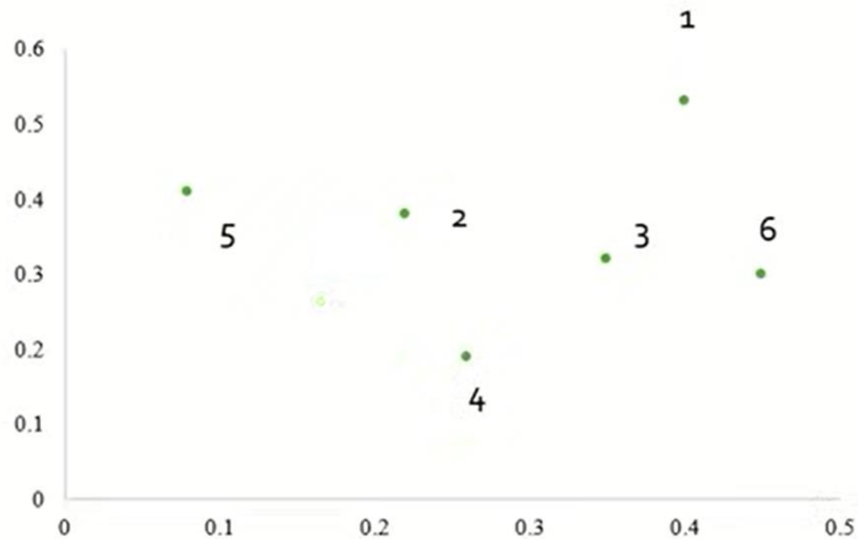
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

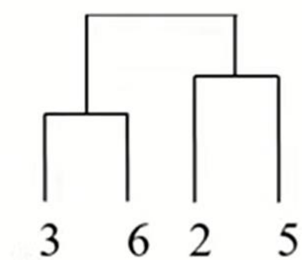
EXAMPLE SINGLE LINK

- The distance matrix for cluster (P3,P6), (P2,P5)

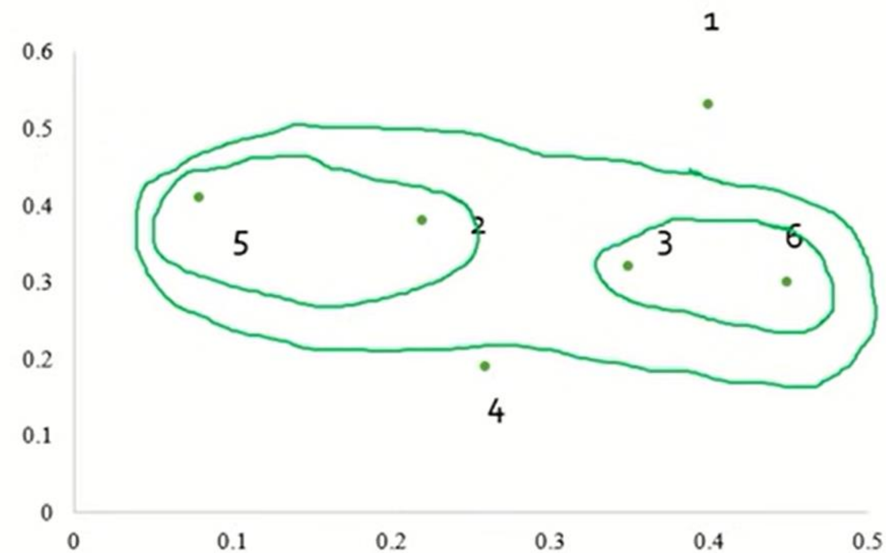
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

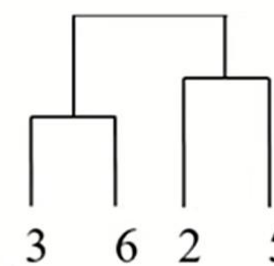
EXAMPLE SINGLE LINK





EXAMPLE SINGLE LINK





EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P1]
- MIN[dist((P2,P5),P1), ((P3,P6),P1)]

    = min[(0.23,0.22)]

    = 0.22

EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist((P2,P5),(P3,P6)),P4]
- MIN[dist((P2,P5),P4), ((P3,P6),P4)]

  = min[(0.20,0.15)]

  = 0.15

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5,P3,P6

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

Example Single Link

- The distance matrix is

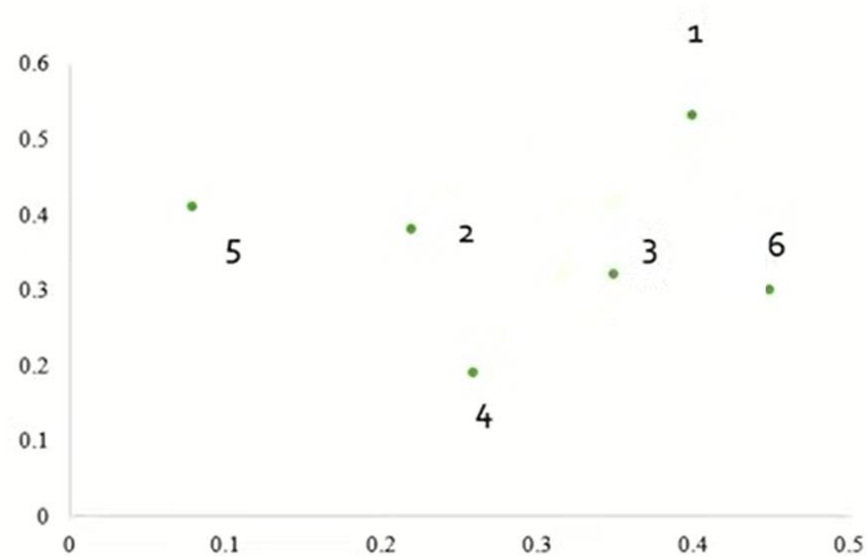
	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

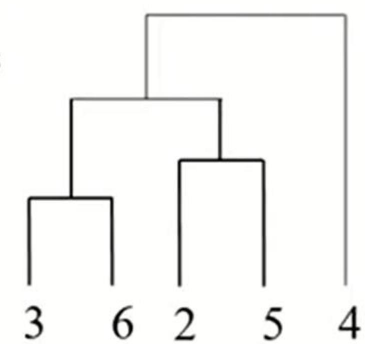
EXAMPLE SINGLE LINK

- The distance matrix is

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

EXAMPLE SINGLE LINK





EXAMPLE SINGLE LINK

- To update the distance matrix MIN[dist(P2,P5,P3,P6),P4]
- MIN[dist((P2,P5,P3,P6),P1), (P4,P1)]

  = min[(0.22,0.37)]

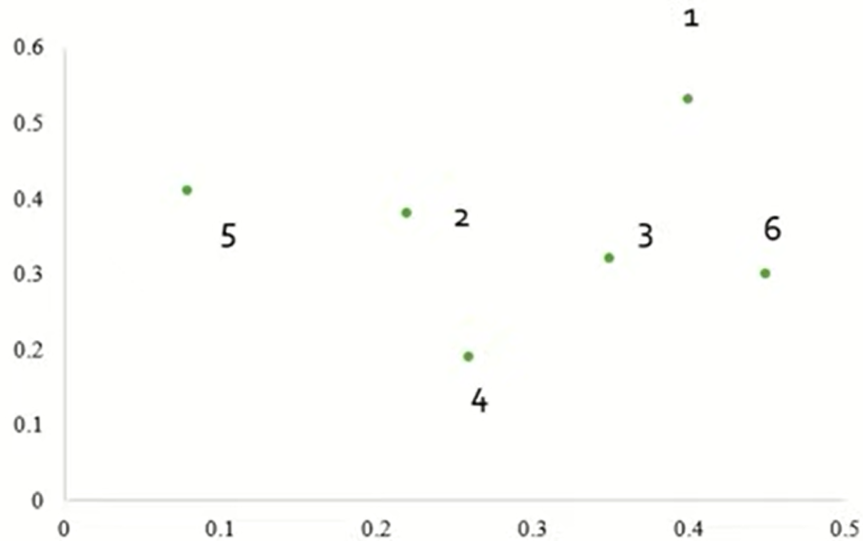
  = 0.22

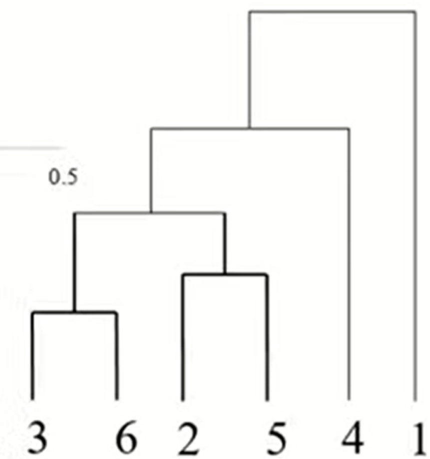
EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5,P3,P6,P4

	P1	P2,P5,P3,P6,P4
P1	0	
P2,P5,P3,P6,P4	0.22	0

EXAMPLE SINGLE LINK





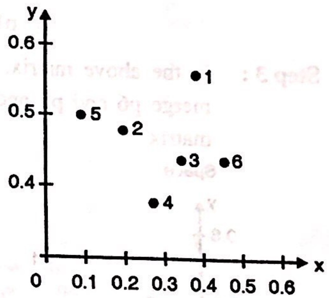
Thank you

Ex. 9.5.1 : Assume that the database D is given by the table below. Follow single link technique to find clusters in D. Use Euclidean distance measure.

		X	Y
	p1	0.40	0.53
	p2	0.22	0.38
D =	p3	0.35	0.32
	p4	0.26	0.19
	p5	0.08	0.41
	p6	0.45	0.30

Soln. :

Step 1 : Plot the objects in n-dimensional space (where n is the number of attributes). In our case we have 2 attributes x and y, so we plot the objects p1, p2, … p6 in 2-dimensional space :



Step 2 : Calculate the distance from each object (point) to all other points, using Euclidean distance measure, and place the numbers in a distance matrix.

The formula for Euclidean distance between two points i and j is :

$$d(i, j) = \sqrt{|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j1}|^2 + \ldots + |x_{ip} - x_{jp}|^2}$$

where x_{i1} is the value of attribute 1 for i and x_{j1} is the value of attribute 1 for j and so on, as many attributes we have … shown up to p i.e. x_{ip} in the formula.

In our case, we only have 2 attributes. So, the Euclidean distance between our points p1 and p2, which have attributes x and y would be calculated as follows:

$$d(p1, p2) = \sqrt{|xp_1 - xp_1|^2 + |yp_1 - yp_2|^2}$$

$$= \sqrt{|0.40 - 0.22|^2 + |0.53 - 0.38|^2}$$

$$= \sqrt{|0.18|^2 + |0.15|^2}$$

$$= \sqrt{0.0324 + 0.0225}$$

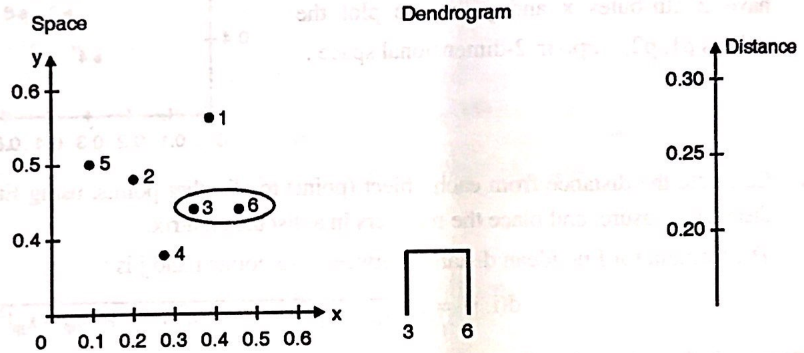
$$= \sqrt{0.0549} = 0.2343$$

Analogically, we calculate the distance to the remaining points and we will receive the following values :

Distance matrix :

	p1	p2	p3	p4	p5	p6
p1	0					
p2	0.24	0				
p3	0.22	0.15	0			
p4	0.37	0.20	0.15	0		
p5	0.34	0.14	0.28	0.29	0	
p6	0.23	0.25	**0.11**	0.22	0.39	0

Step 3 : In the above matrix, p6 and p3 are two clusters with shortest distance 0.11, so merge p6 and p3 and make single cluster (p3,p6). Now re-compute the distance matrix.



Distance matrix :

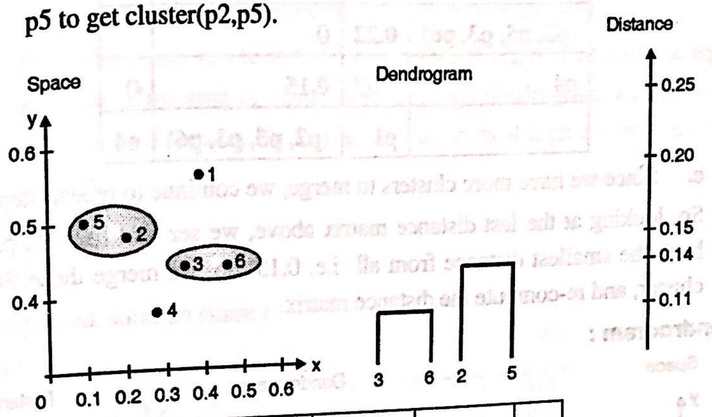
	p1	p2	(p3, p6)	p4	p5
p1	0				
p2	0.24	0			
(p3, p6)	**0.22**	**0.15**	0		
p4	0.37	0.20	**0.15**	0	
p5	0.34	0.14	**0.28**	0.29	0

To calculate the distance of p1 from (p3,p6) :

$$\text{dist}((p3, p6), p1) = \text{MIN}(\text{dist}(p3, p1), \text{dist}(p6, p1))$$
$$= \text{MIN}(0.22, 0.23) \quad \text{//from original matrix}$$
$$= 0.22$$

Step 4 : Repeat Step 3 until one single cluster is formed i.e. merge all the clusters.

a. Now p2 and p5 have the smallest distance from above matrix, so merge p2 and p5 to get cluster(p2,p5).



p1	0			
(p2, p5)	**0.24**	0		
(p3, p6)	0.22	**0.15**	0	
p4	0.37	**0.20**	0.15	0
	p1	**(p2, p5)**	(p3, p6)	p4

The distance between (p3, p6) and (p2, p5) is calculated as given below :

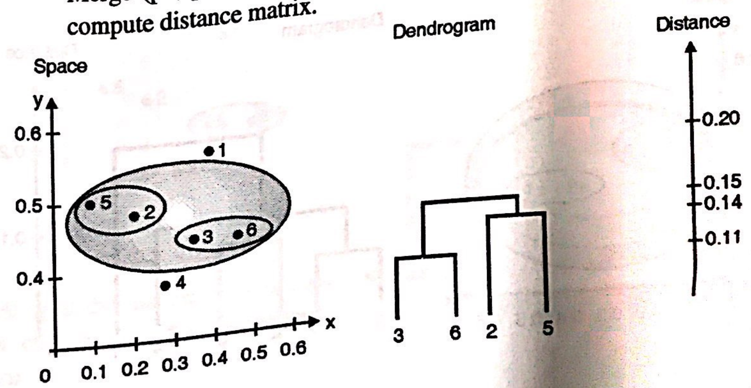
dist((p3, p6), (p2, p5)) = MIN (dist(p3, p2) , dist(p6, p2), dist(p3, p5), dist(p6, p5))

= MIN (0.15, 0.25, 0.28, 0.39) //from original matrix

= 0.15

b. Repeat Step 3.

Merge (p2, p5) and (p3, p6) as having minimum distance i.e.0.15 and again compute distance matrix.



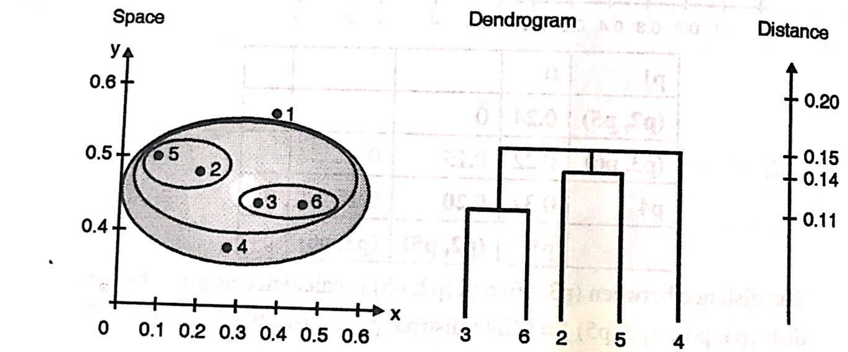
Distance matrix :

p1	0		
(p2, p5, p3, p6)	0.22	0	
p4	0.37	0.15	0
	p1	(p2, p5, p3, p6)	p4

c. Since we have more clusters to merge, we continue to repeat Step 3.

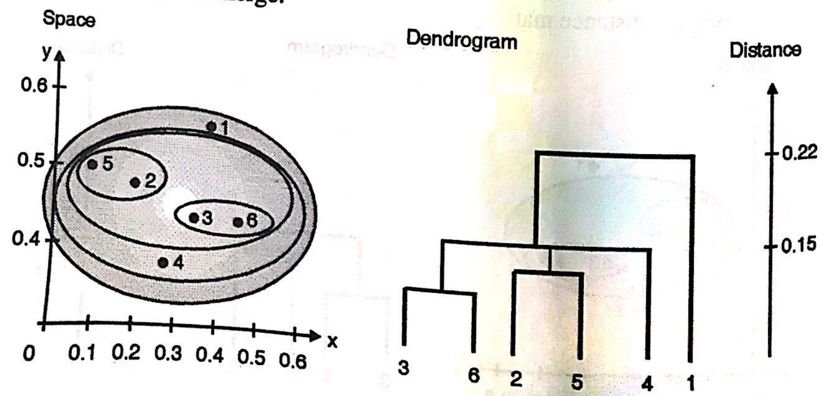
So, looking at the last distance matrix above, we see that (p2, p5, p3, p6) and p4 have the smallest distance from all i.e. 0.15. So, we merge those two in a single cluster, and re-compute the distance matrix.

Space dendrogram :



d. Since we have more clusters to merge, we continue to repeat Step 3.

So, looking at the last distance matrix above, we see that (p2, p5, p3, p6, p4) and p1 have the smallest distance - 0.22 (the only one left). So, we merge those two in a single cluster. There is no need to re-compute the distance matrix, as there are no more clusters to merge.



Stopping condition :

We indicated that "each object is placed in a separate cluster, and at each step we merge the closest pair of clusters, *until certain termination conditions are satisfied*".

To stop clustering either user has to specify the number of clusters he wants or algorithm has to make decision to stop clustering at which level. Through dendrogram, we can notice the merging of clusters at various distances. If merging of clusters is at high distance then we can stop clustering at that level.

Complete link :

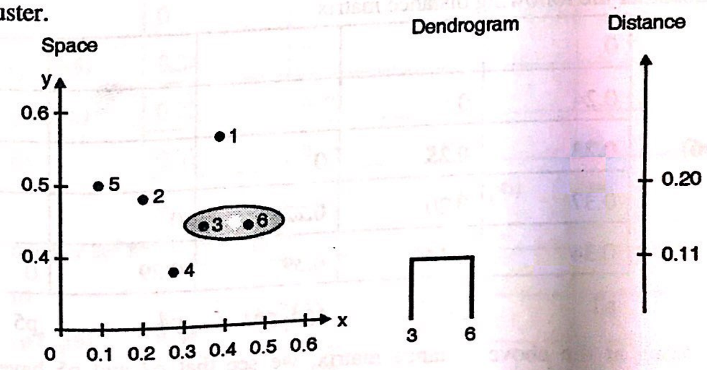
Step 1 and step 2 :

Refer the single link solution (same)

Distance matrix :

	p1	p2	p3	p4	p5	p6
p1	0					
p2	0.24	0				
p3	0.22	0.15	0			
p4	0.37	0.20	0.15	0		
p5	0.34	0.14	0.28	0.29	0	
p6	0.23	0.25	0.11	0.22	0.39	0

Step 3 : Identify the two clusters with the shortest distance in the matrix, and merge them together. Re-compute the distance matrix, as those two clusters are now in a single cluster.



By looking at the distance matrix above, we see that p3 and p6 have the *smallest* distance from all i.e. 0.11 So, we merge those two in a single cluster and re-compute the distance matrix.

dist((p3, p6), p1) = MAX (*dist*(p3, p1) , *dist*(p6, p1))
= MAX (0.22, 0.23) //from original matrix
= 0.23

dist((p3, p6), p2) = MAX (*dist*(p3, p2) , *dist*(p6, p2))
= MAX (0.15, 0.25) //from original matrix
= 0.25

dist((p3, p6), p4) = MAX (*dist*(p3, p4) , *dist*(p6, p4))
= MAX (0.15, 0.22) //from original matrix
= 0.22

dist((p3, p6), p5) = MAX (*dist*(p3, p5) , *dist*(p6, p5))
= MAX (0.28, 0.39) //from original matrix
= 0.39

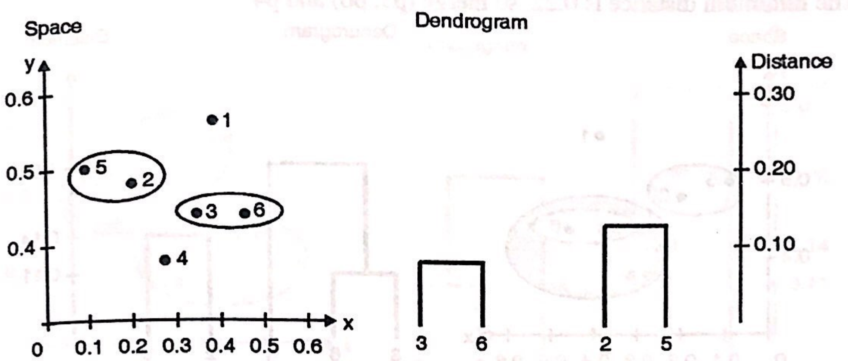
New distance matrix :

p1	0				
p2	0.24	0			
(p3, p6)	**0.23**	**0.25**	0		
p4	0.37	0.20	**0.22**	0	
p5	0.34	0.14	**0.39**	0.29	0
	p1	p2	**(p3, p6)**	p4	p5

Step 4 : Consider the following distance matrix

p1	0				
p2	0.24	0			
(p3, p6)	**0.23**	**0.25**	0		
p4	0.37	0.20	**0.22**	0	
p5	0.34	0.14	**0.39**	0.29	0
	p1	p2	**(p3, p6)**	p4	p5

So, looking at the above distance matrix, we see that p2 and p5 have the smallest distance from all – 0.14. So, we merge those two in a single cluster, and re-compute the distance matrix using the following calculations.



$dist$((p2, p5), p1) = MAX ($dist$(p2, p1) , $dist$(p5, p1))
= MAX (0.24, 0.34) //from original matrix
= 0.34

$dist$((p2, p5), (p3,p6)) = MAX ($dist$(p2, p3) , $dist$(p2, p6), $dist$(p5, p3), $dist$(p5, p6))
= MAX (0.15, 0.25, 0.28, 0.39)
//from original matrix
= 0.39

$dist$((p2, p5), p4) = MAX ($dist$(p2, p4) , $dist$(p5, p4))
= MAX (0.20, 0.29) //from original matrix
= 0.29

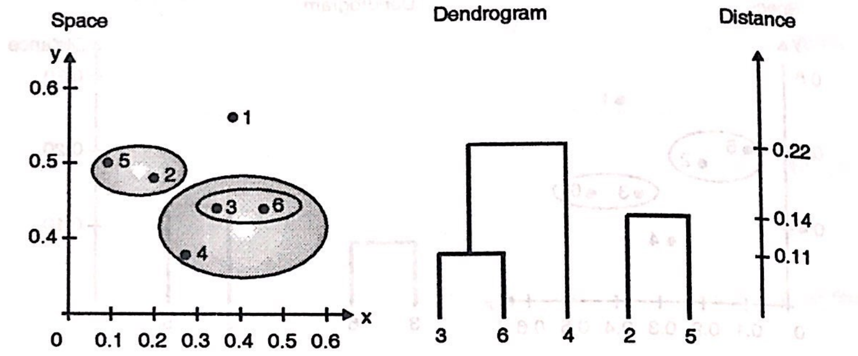
Therefore new distance matrix is :

p1	0			
(p2, p5)	**0.34**	0		
(p3, p6)	0.23	**0.39**	0	
p4	0.37	**0.29**	0.22	0
	p1	**(p2, p5)**	(p3, p6)	p4

Step 5 : Consider the following matrix

p1	0			
(p2, p5)	**0.34**	0		
(p3, p6)	0.23	**0.39**	0	
p4	0.37	**0.29**	0.22	0
	p1	**(p2, p5)**	(p3, p6)	p4

The minimum distance is 0.22, so merge (p3, p6) and p4



dist((p3, p6, p4) , p1) = MAX (*dist*(p3, p1) , *dist*(p6, p1), dist(p4, p1))

= MAX (0.22, 0.23, 0.37) //from original matrix

= 0.37

dist((p3, p6, p4) , (p2,p5)) = MAX (*dist*(p3, p2) , *dist*(p3, p5),

dist(p6, p2),dist(p6, p5),

dist(p4, p2) , dist(p4, p5))

= MAX (0.15, 0.28, 0.25, 0.39, 0.20, 0.29)

= 0.39

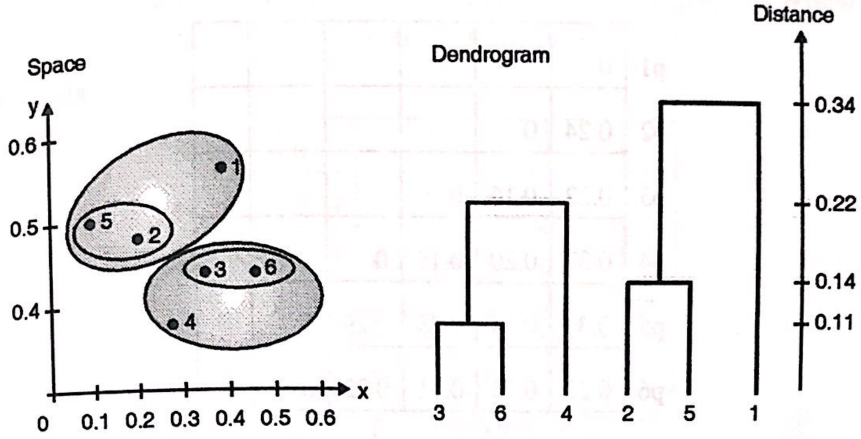
Therefore new distance matrix

p1	0		
(p2, p5)	**0.34**	0	
(p3, p6, p4)	0.37	**0.39**	0
	p1	**(p2, p5)**	(p3, p6, P4)

Step 6 : Now consider the following distance matrix

p1	0		
(p2, p5)	**0.34**	0	
(p3, p6, p4)	0.37	**0.39**	0
	p1	**(p2, p5)**	(p3, p6, P4)

Since the minimum distance is 0.34, merge (p2, p5) with p1.

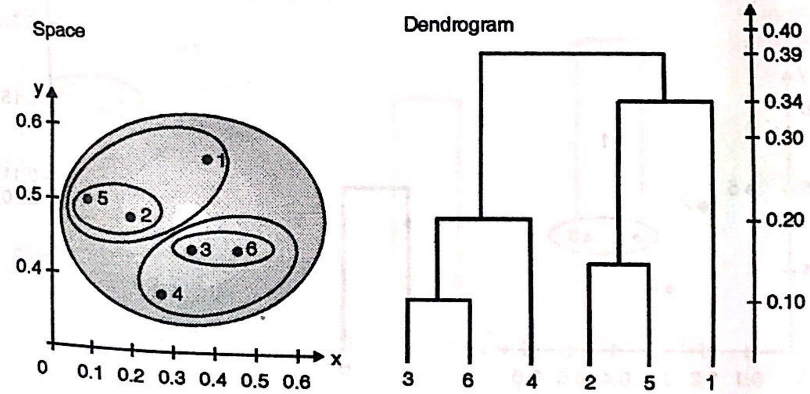


dist ((p2, p5, p1), (p3, p6, p4) = 0.39

Therefore new distance matrix

	(p2, p5,p1)	(p3, p6, P4)
(p2, p5,p1)	0	
(p3, p6, p4)	0.39	0

Finally, merge the cluster (p2, p5,p1) and (p3,p6,p4)



Average link :

Step 1 and Step 2 :

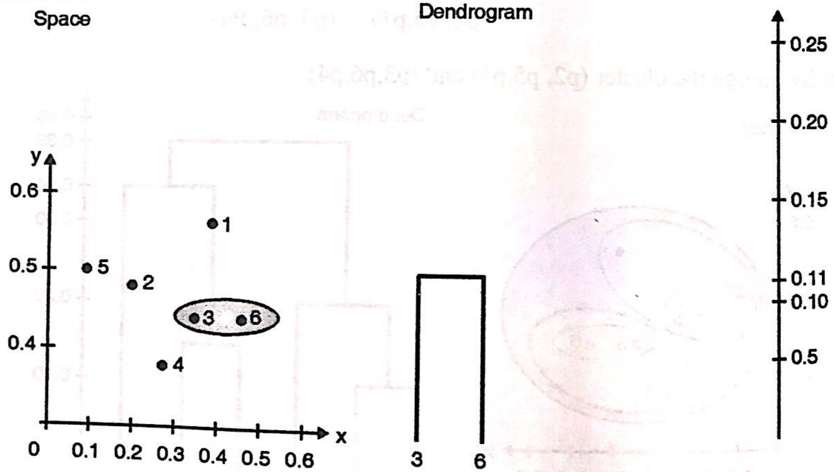
Refer the single link solution

Distance matrix :

	p1	p2	p3	p4	p5	p6
p1	0					
p2	0.24	0				
p3	0.22	0.15	0			
p4	0.37	0.20	0.15	0		
p5	0.34	0.14	0.28	0.29	0	
p6	0.23	0.25	0.11	0.22	0.39	0

Step 3 : Identify the two clusters with the shortest distance in the matrix, and merge them together. Re-compute the distance matrix, as those two clusters are now in a single cluster.

By looking at the distance matrix above, we see that p3 and p6 have the smallest distance from all – 0.11 So, we merge those two in a single cluster, and re-compute the distance matrix.



$$\text{dist}((p3\ p6),\ p1) = 1/2\,(\text{dist}(p3,p1) + \text{dist}(p6,p1))$$
$$= 0.5 \times (0.22 + 0.23) = 0.23$$
$$\text{dist}((p3\ p6),\ p2) = 1/2\,(\text{dist}(p3,p2) + \text{dist}(p6,p2))$$
$$= 0.5 \times (0.15 + 0.25) = 0.2$$
$$\text{dist}((p3\ p6),\ p4) = 1/2\,(\text{dist}(p3,p4) + \text{dist}(p6,p4))$$
$$= 0.5 \times (0.15 + 0.22) = 0.19$$

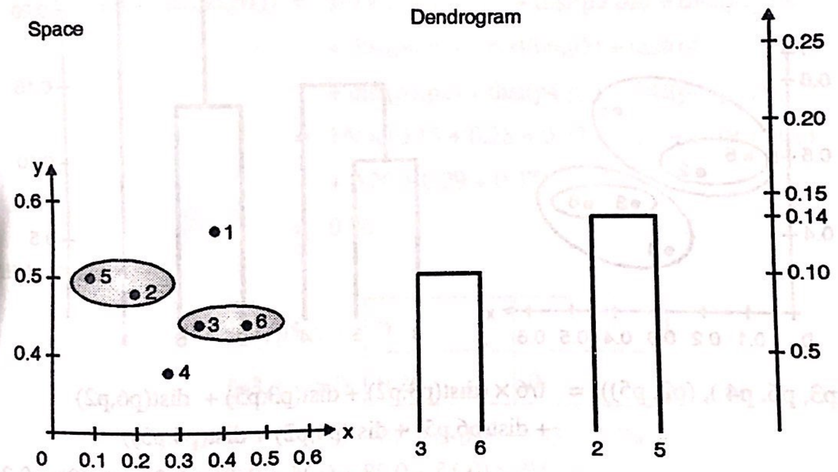
$$\text{dist}((p3\ p6),\ p5) = 1/2\ (\text{dist}(p3,p5) + \text{dist}(p6,p5))$$
$$= 0.5 \times (0.28 + 0.39) = 0.34$$

Distance matrix :

	p1	p2	(p3, p6)	p4	p5
p1	0				
p2	0.24	0			
(p3, p6)	**0.23**	**0.2**	0		
p4	0.37	0.20	**0.19**	0	
p5	0.34	0.14	**0.34**	0.29	0

Step 4 :

So, looking at the above distance matrix above, we see that p2 and p5 have the smallest distance from all – 0.14. So, we merge those two in a single cluster, and re-compute the distance matrix.



$$\text{dist}((p2,\ p5),\ p1) = 1/2\ (\text{dist}(p2,p1) + \text{dist}(p5,p1))$$
$$= 0.5 \times (0.24 + 0.34) = 0.29$$

$$\text{dist}((p2,\ p5),\ (p3,p6)) = 1/4\ (\text{dist}(p2,p3) + \text{dist}(p2,p6) + \text{dist}(p5,p3) + \text{dist}(p5,p6))$$
$$= 1/4 \times (0.15 + 0.25 + 0.28 + 0.39) = 0.27$$

$$\text{dist}((p2,\ p5),\ p4) = 1/2\ (\text{dist}(p2,p4) + \text{dist}(p5,p4))$$
$$= 0.5 \times (0.14 + 0.29) = 0.22$$

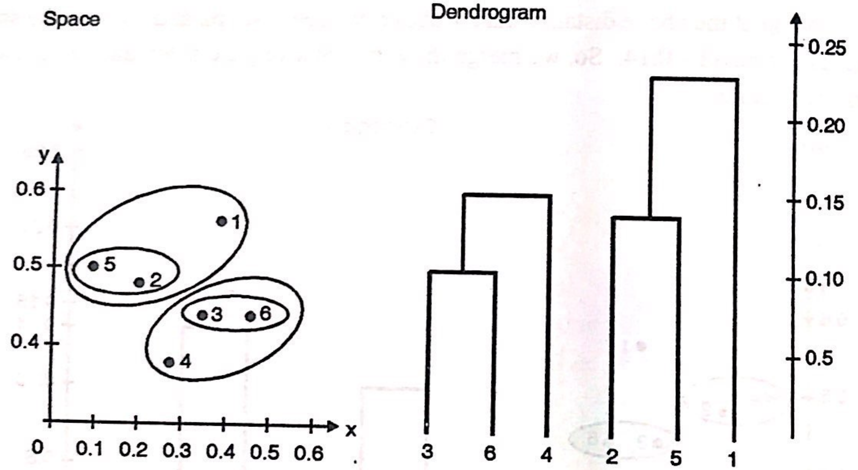
Distance matrix :

	p1	**(p2, p5)**	(p3, p6)	p4
p1	0			
(p2, p5)	**0.29**	0		
(p3, p6)	0.22	**0.27**	0	
p4	0.37	**0.22**	0.15	0

Since, we have merged (p2, p5) together in a cluster, we now have one entry for (p2, p5) in the table, and no longer have p2 or p5 separately.

Step 5 :

Now the closest clusters are merged, where distance is the smallest measure by looking at the maximum distance between any two points.



$$\text{dist}((p3, p6, p4), (p2, p5)) = 1/6 \times (\text{dist}(p3,p2) + \text{dist}(p3,p5) + \text{dist}(p6,p2) + \text{dist}(p6,p5) + \text{dist}(p4,p2) + \text{dist}(p4,p5))$$

$$= 1/6 \times (0.15 + 0.28 + 0.25 + 0.39 + 0.20 + 0.29) = 0.26$$

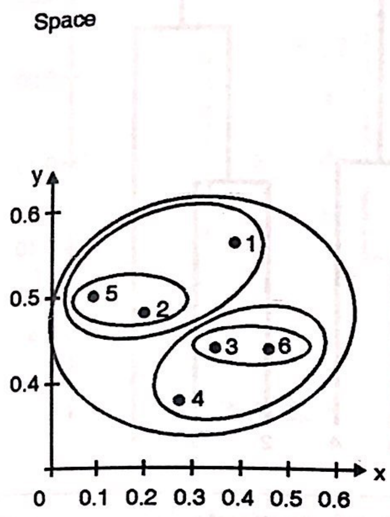
$$\text{dist}((p3, p6, p4), p1) = 1/3 \times (\text{dist}(p3,p1) + \text{dist}(p6,p1) + \text{dist}(p4,p1))$$

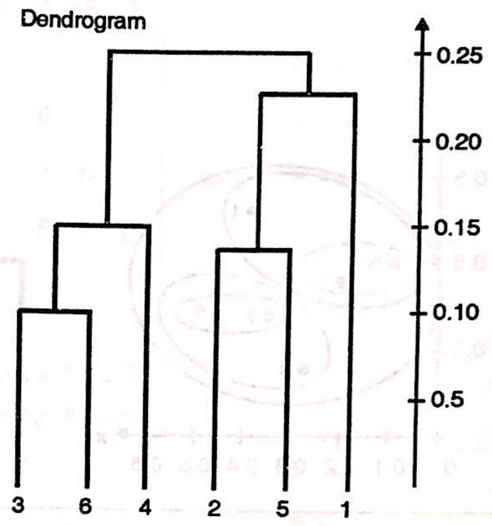
$$= 1/3 \times (0.22 + 0.23 + 0.37) = 0.27$$

Distance matrix :

	p1	**(p2, p5)**	(p3, p6, P4)
p1	0		
(p2, p5)	**0.24**	0	
(p3, p6, p4)	0.27	**0.26**	0

Step 6 : So merge the cluster (p2,p5) and p1.





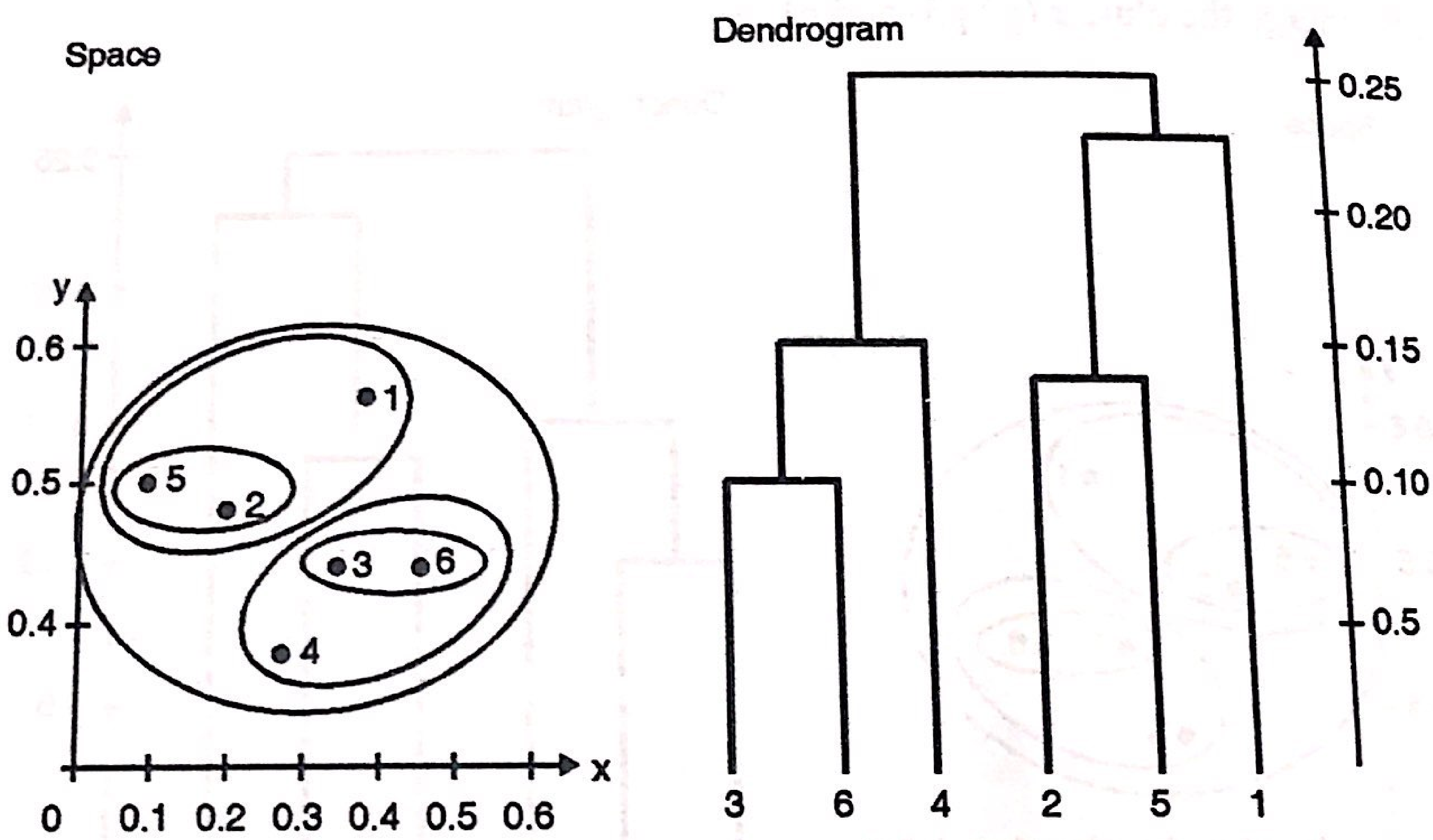
dist((p3 p6 p4), (p2,p5,p1)) = 1/9 × (dist(p3,p2) + dist(p3,p5) + dist(p3,p1)
+ dist(p6,p2) + dist(p6,p5) + dist(p6,p1)
+ dist(p4,p2) + dist(p4,p5) + dist(p4+p1))
= 1/9 × (0.15 + 0.28 + 0.22 + 0.25 + 0.39 + 0.23
+ 0.20 + 0.29 + 0.37)
= 0.26

Distance matrix :

(p2, p5,p1)	0	
(p3, p6, p4)	0.26	0
	(p2, p5,p1)	(p3, p6, P4)

We need to re-compute the distance from all other points / clusters to our new cluster - (p2, p5,p1) to (p3,p6,p4) and enter the maximum distance in the above matrix (use original distance matrix).

Finally merge the cluster (p2, p5,p1) and (p3,p6,p4).


Space
y
0.6
0.5
0.4
x
0
0.1
0.2
0.3
0.4
0.5
0.6
1
5
2
3
6
4
Dendrogram
0.25
0.20
0.15
0.10
0.5
3
6
4
2
5
1

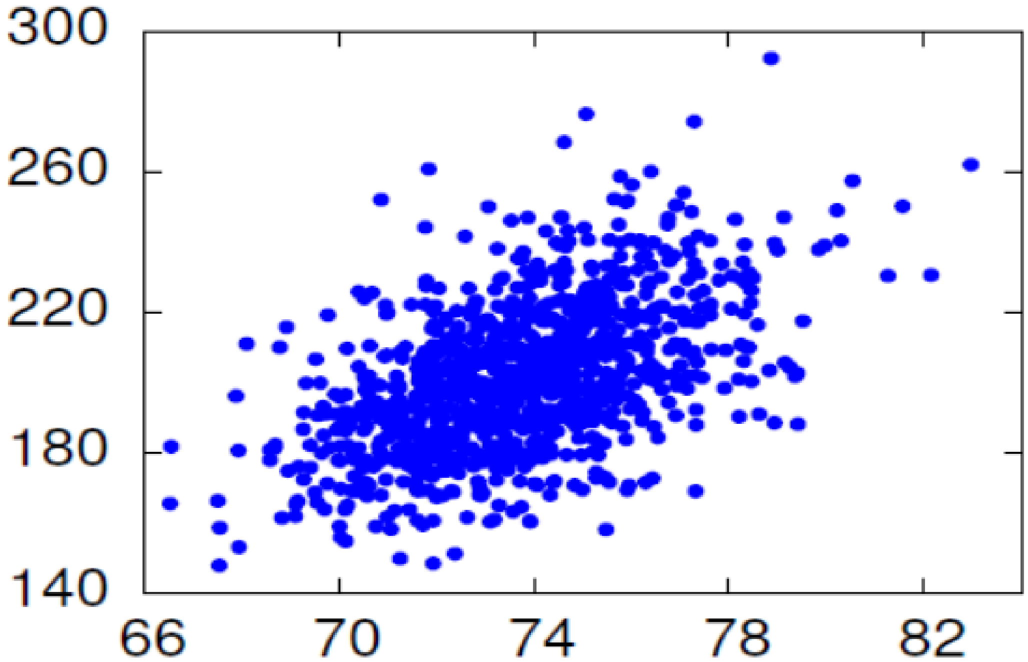
Gaussian Mixture Model

Gauss

Johann Carl Friedrich Gauss (1777-1855)
"Greatest Mathematician since Antiquity"

Estimating Gaussians

- Give training data, how to choose parameters μ, Σ?
- Find parameters so that resulting distribution . . .
  - "Matches" data as well as possible.
  - Sample data: height, weight of baseball players



Why Maximum-Likelihood Estimation ?

- Assume we have “correct” model form.
- Then, as the number of training samples increases . . .
  - ML estimates approach “true” parameter values (consistent)
  - ML estimators are the best! (efficient)
- ML estimation is easy for many types of models.
  - Count and normalize!

What is ML Estimate for Gaussians ?

- Much easier to work with log likelihood $L = \ln \mathcal{L}$:

$$L(x_1^N|\mu,\sigma) = -\frac{N}{2}\ln 2\pi\sigma^2 - \frac{1}{2}\sum_{i=1}^{N}\frac{(x_i-\mu)^2}{\sigma^2}$$

- Take partial derivatives w.r.t. μ, σ:

$$\frac{\partial L(x_1^N|\mu,\sigma)}{\partial\mu} = \sum_{i=1}^{N}\frac{(x_i-\mu)}{\sigma^2}$$

$$\frac{\partial L(x_1^N|\mu,\sigma)}{\partial\sigma^2} = -\frac{N}{2\sigma^2} + \sum_{i=1}^{N}\frac{(x_i-\mu)^2}{\sigma^4}$$

- Set equal to zero; solve for μ, σ^2.

$$\mu = \frac{1}{N}\sum_{i=1}^{N}x_i \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2$$

What is ML Estimate for Gaussians?

- Multivariate case.

$$\mu = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$$

$$\mathbf{\Sigma} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \mu)^T(\mathbf{x}_i - \mu)$$

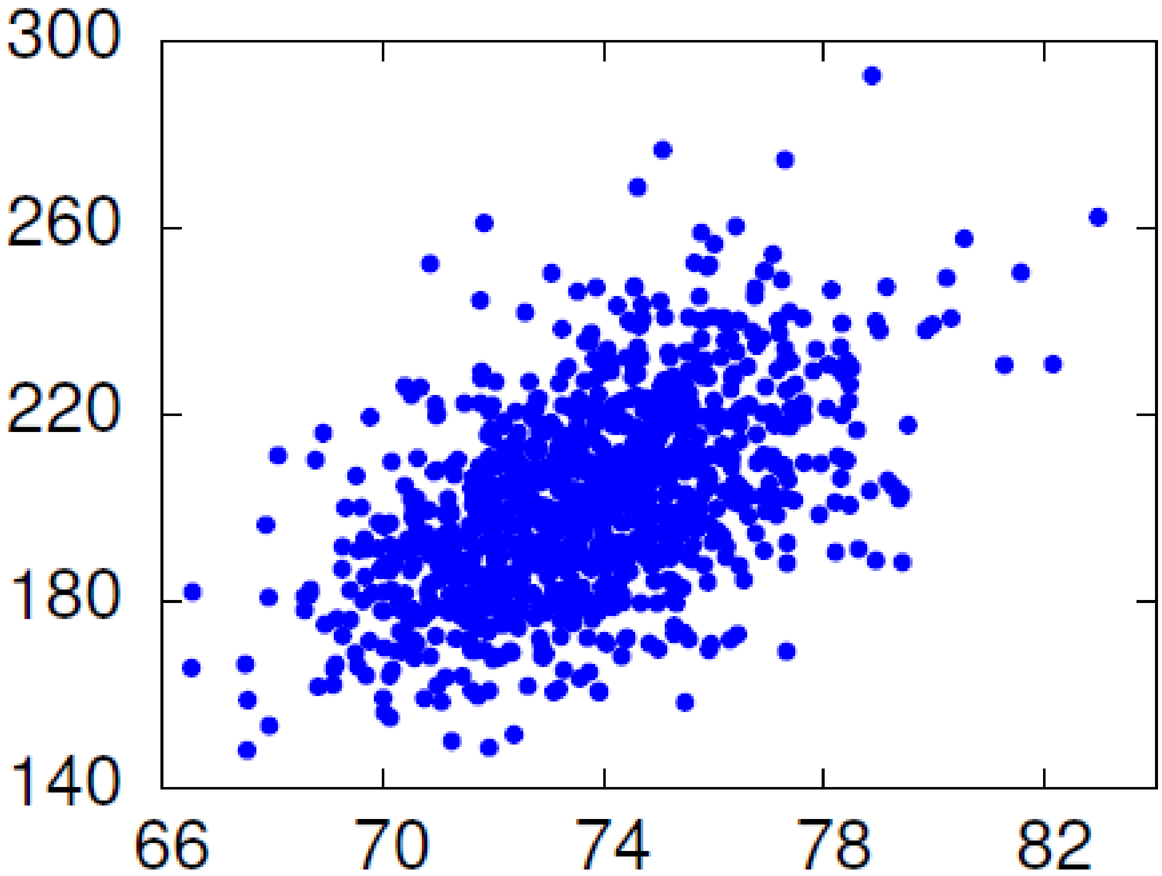
- What if diagonal covariance?
  - Estimate params for each dimension independently.

Example: ML Estimation

- Heights and weights of 1033 pro baseball players. Noise added to hide discretization effects.

height	weight
74.34	181.29
73.92	213.79
72.01	209.52
72.28	209.02
72.98	188.42
69.41	176.02
68.78	210.28
...	...
...	...

Example: ML Estimation



Example: Diagonal Covariance

$$\mu_1 = \frac{1}{1033}(74.34 + 73.92 + 72.01 + \cdots) = 73.71$$

$$\mu_2 = \frac{1}{1033}(181.29 + 213.79 + 209.52 + \cdots) = 201.69$$

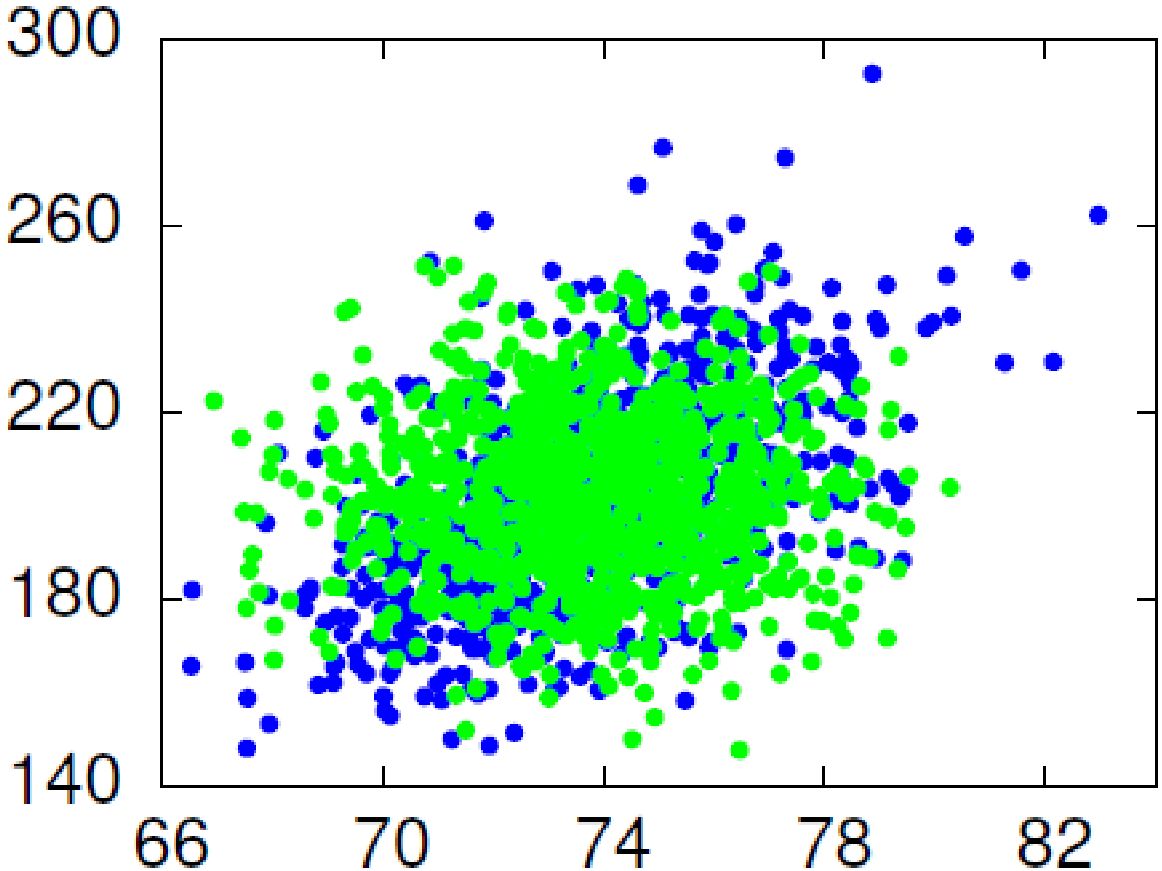
$$\sigma_1^2 = \frac{1}{1033}\left[(74.34 - 73.71)^2 + (73.92 - 73.71)^2 + \cdots)\right]$$

$$= 5.43$$

$$\sigma_2^2 = \frac{1}{1033}\left[(181.29 - 201.69)^2 + (213.79 - 201.69)^2 + \cdots)\right]$$

$$= 440.62$$

Example: Diagonal Covariance



Example: Full Covariance

- Mean; diagonal elements of covariance matrix the same.

$$\mathbf{\Sigma}_{12} = \mathbf{\Sigma}_{21}$$

$$= \frac{1}{1033}[(74.34 - 73.71) \times (181.29 - 201.69) +$$

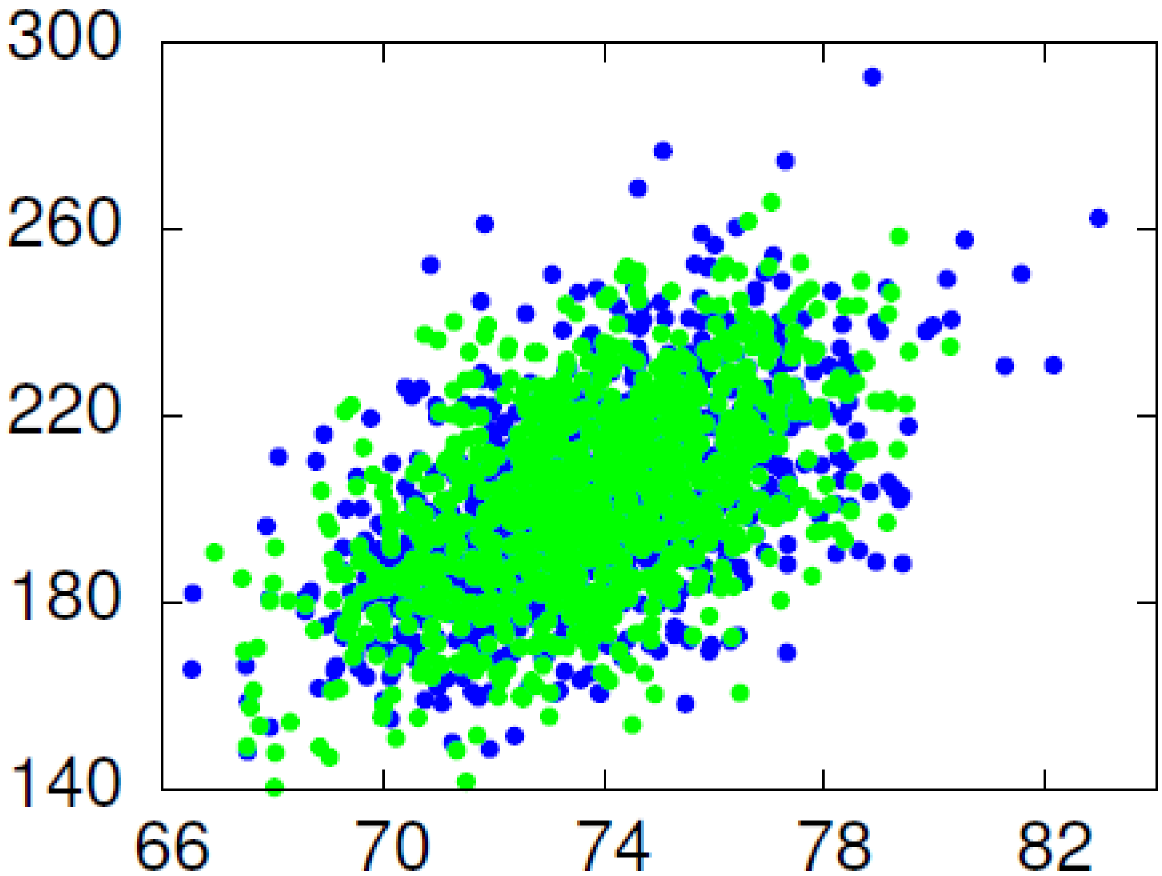
$$(73.92 - 73.71) \times (213.79 - 201.69) + \cdots)]$$

$$= 25.43$$

$$\boldsymbol{\mu} = [\ 73.71 \quad 201.69\ ]$$

$$\mathbf{\Sigma} = \begin{vmatrix} 5.43 & 25.43 \\ 25.43 & 440.62 \end{vmatrix}$$

Example: Full Covariance



Part II

Gaussian Mixture Models

Gaussian Distribution

- Gaussian or Normal Distribution, 1D

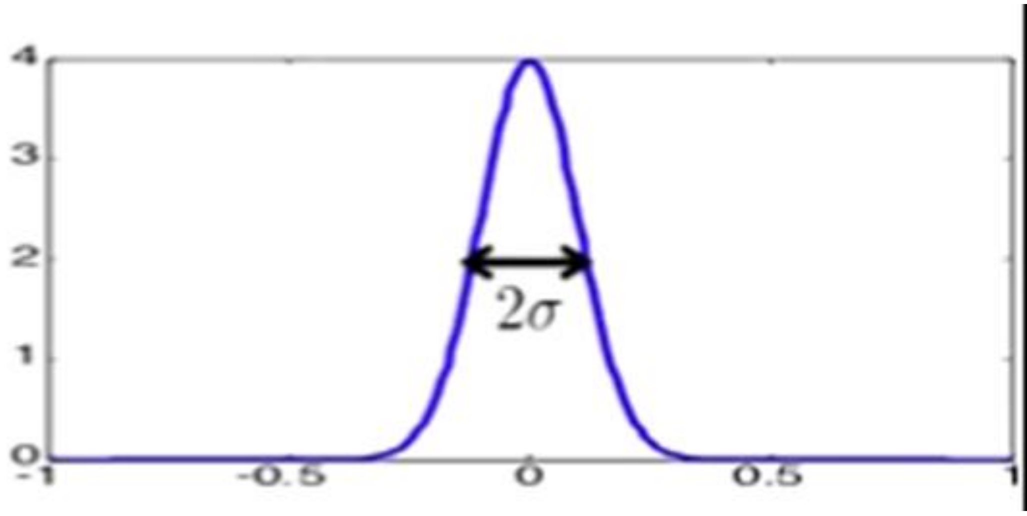
$$N(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\frac{1}{2}(x - \mu)^2}{\sigma^2}\right]$$

Parameters: mean μ, variance σ^2
(standard deviation)

Maximum Likelihood estimates

$$\mu = \frac{1}{N}\sum x^{(i)}$$

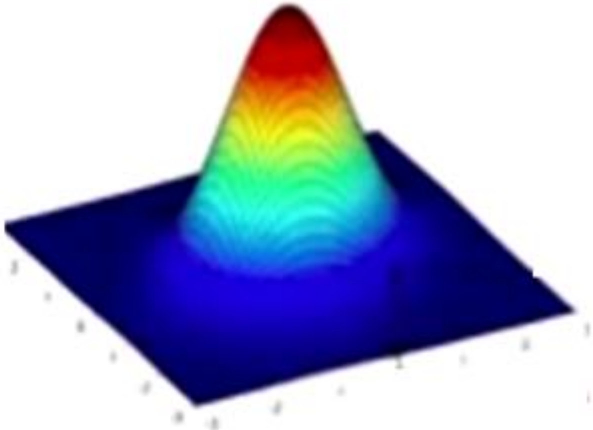
$$\sigma^2 = \frac{1}{N}\sum\left(x^{(i)} - \mu\right)^2$$



Multivariate Gaussian

- Similar to Univariate Case

$$N(x: \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{d}{2}}} |\Sigma|^{-\frac{1}{2}} exp\left\{-\frac{1}{2}(x - \mu)\Sigma^{-1}(x - \mu)^T\right\}$$



μ = length – d row vector

∑ d x d matrix

|∑| = matrix determinant

Multivariate Gaussians

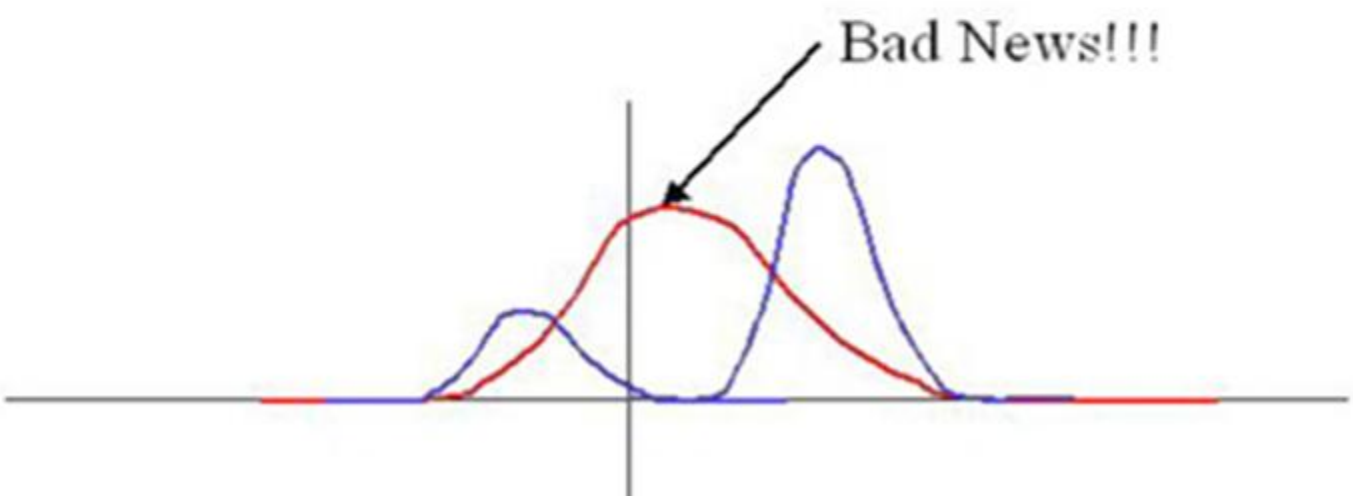
- Defining μ and ∑

$$\mu = E(x)$$
$$\Sigma = E\left[(x - \mu)(x - \mu)^T\right]$$

- So the i-jth element of ∑ is:

$$\sigma_{ij}^2 = E\left[(x_i - \mu_i)(x_i - \mu_j)\right]$$

- Single Gaussian may do a bad job of modeling distribution in any dimension

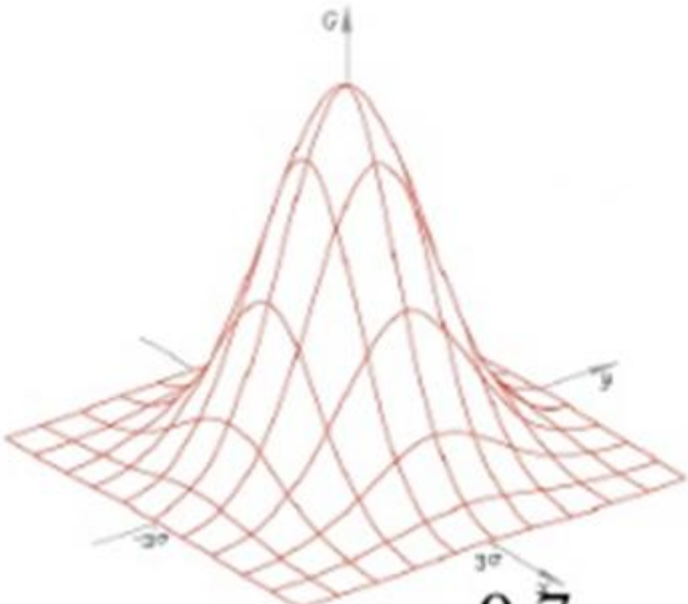


What's Gaussian Mixture?

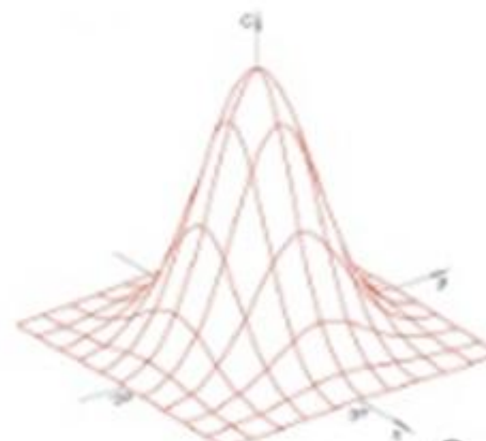
$$p(X) = \pi_1 N(X \mid \mu_1, \Sigma_1) + \pi_2 N(X \mid \mu_2, \Sigma_2)$$

Mixing Coefficient

$$\sum_{k=1}^{k} \pi_k = 1$$



$\pi_1 = 0.7$



$\pi_2 = 0.3$

Gaussian Mixture Models

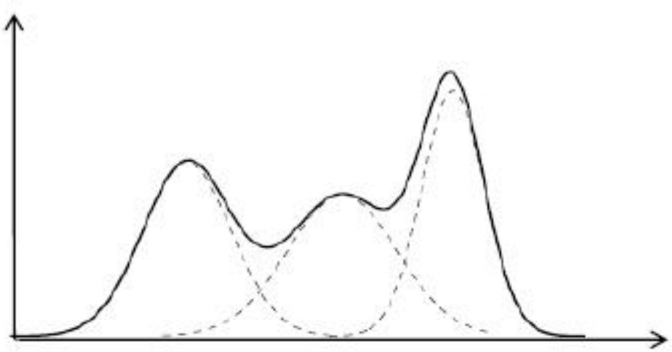
- GMM: the weighted sum of a number of Gaussian where the weights are determined by distribution, π

$$p(X) = \pi_0 N(x \mid \mu_0, \Sigma_0) + \pi_1 N(x \mid \mu_1, \Sigma_1) + \ldots + \pi_k N(x \mid \mu_K, \Sigma_K)$$

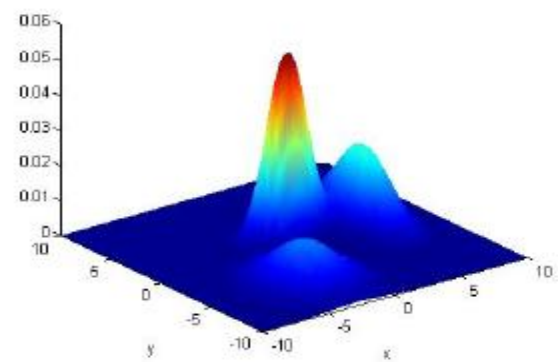
Where $\sum_{i=0}^{k} \pi_i = 1$ $$p(x) = \sum_{i=0}^{k} \pi_i N(x \mid \mu_k, \Sigma_k)$$

Examples:

d=1:



d=2:



What is a Gaussian Mixture Model

- **Problem**

  Given a set of data $X = \{x_1, x_2, \dots \dots \dots \dots x_N\}$ drawn from an unknown distribution (probably a GMM) estimates the parameters θ of the GMM model that fits the data

- **Solution**

  Maximize the likelihood $p(X\,|\theta)$ of the data with regard to the model parameters?

$$\theta^* = arg \max_{\theta} p(X\,|\theta) = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i|\theta)$$

Maximum Likelihood Estimation

- **Consider log of Gaussian Distribution**

$$\ln p(x \mid \mu, \Sigma) = -\frac{1}{2}\, in(2\pi) - \frac{1}{2}\, ln|\Sigma| - \frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)$$

- **Take the derivative and equate it to zero**

$$\frac{\partial \ln p(x \mid \mu, \Sigma)}{\partial \mu} = 0 \qquad \frac{\partial \ln p(x \mid \mu, \Sigma)}{\partial \Sigma} = 0$$

$$\mu_{ML} = \frac{1}{N} \sum_{a=1}^{N} X_n \qquad \Sigma_{ML} = \frac{1}{N} \sum_{a=1}^{N} (x_n - \mu_{ML})(x_n - \mu_{ML})^T$$

Maximum Likelihood Estimation

$$p(x) = \sum_{k=1}^{k} \pi_k N(x \,|\mu_k, \textstyle\sum_K)$$

$$N(x|\textstyle\sum_k) = \dfrac{1}{(2\pi)^{\frac{d}{2}} \|\textstyle\sum\|^{\frac{1}{2}}} \exp\left\{-\dfrac{1}{2}(x - \mu)^T \textstyle\sum^{-1}(x - \mu)\right\}$$

Which is called a Mixture of Gaussian or Gaussian Mixture Model

Each Gaussian density is called component of the mixtures and has its own mean μ_k and covariance $\sum_k$.

The parameters are called mixing coefficients ($\sum_k \pi_k = k$)

- The log likelihood function is given by

$$\ln p(X|\pi, \mu, \textstyle\sum) = \sum_{n=1}^{N} \ln \left\{ \sum_{K=1}^{K} \pi_k N(X_n|\mu_k, \textstyle\sum_k) \right\}$$

- Goal : Find parameter which maximize log likelihood
- Problem: Hard to compute maximum likelihood
- Solution: Use EM algorithm

The Expectation-Maximization algorithm

One of the most popular approaches to maximize the likelihood is to use the Expectation-Maximization (EM) algorithm

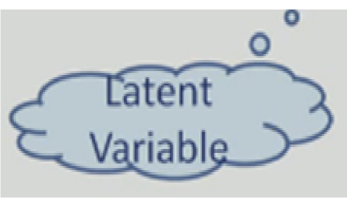
- Basic Ideas of EM algorithm :
  - Introduce the hidden variable such that its knowledge would simply the maximization of the likelihood

  - At each iteration :
    - E-Step Estimate the distribution of the hidden variable given the data and the current value of parameters.
    - M-Step Maximize the joint distribution of the data and the hidden variable

Latent Variable: posterior prob.

- We can think of the mixing coefficients as prior probabilities for the components
- For a given values of 'X', we can evaluate the corresponding posterior probabilities called responsibilities.
- From Baye's Rule

$$\gamma_k(x) = p(k|x) = \frac{p(k)p(x|k)}{p(x)}$$



$$= \frac{\pi_k N(x|\mu_k, \Sigma_k)}{\sum_j^k \pi_j N(x|\mu_j, \Sigma_j)} \quad where\ \pi_k = \frac{N_k}{N}$$

Interpret N_k as the effective number of points assigned to cluster K

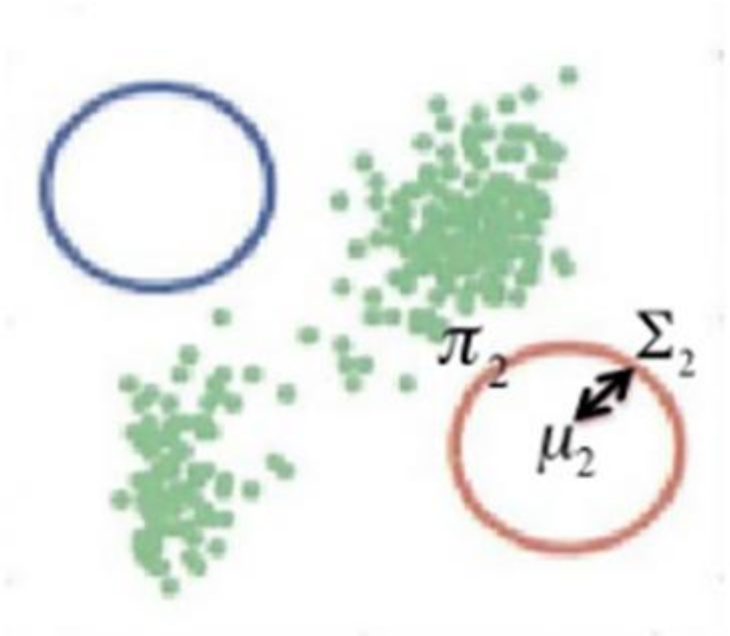
EM for Gaussian Mixtures

- Initialize μ_k, $\sum_k$, π_k one for each Gaussian K
- Tip! Use K-means result to initialize :

$\mu_k \longleftarrow \mu_k$

$\sum_k \longleftarrow$ conv(cluster (k))

$\Pi_k \longleftarrow \dfrac{Number\ of\ points\ in\ k}{Total\ number\ of\ points}$

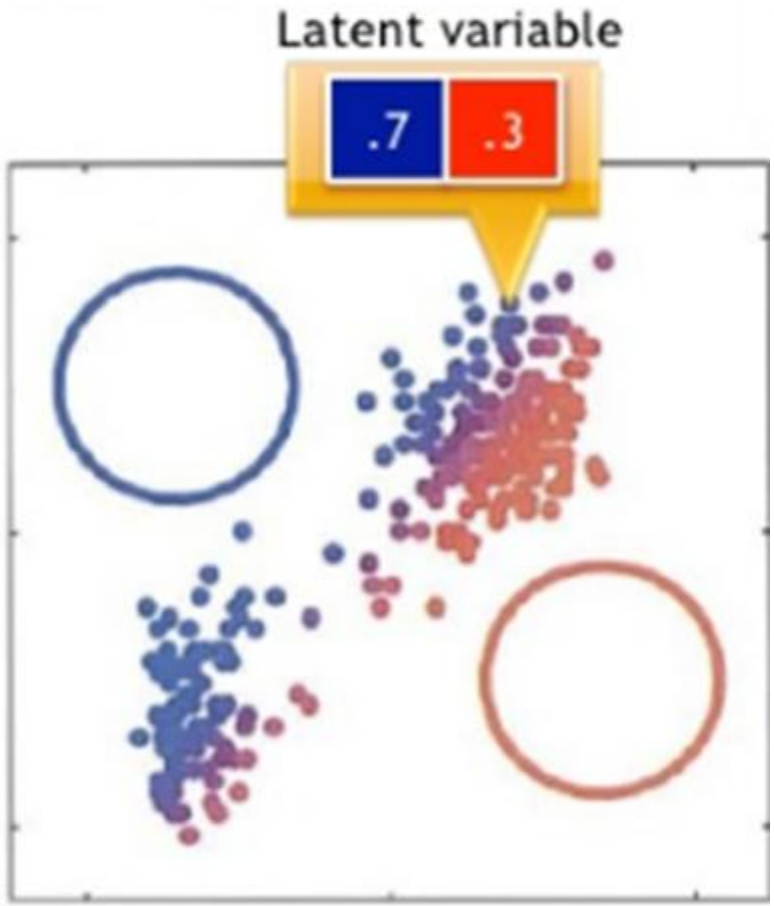


EM for Gaussian Mixtures

- **E Step**: For each point X_n, determine its assignment score to each Gaussian k:

$$\gamma(Z_{nk}) = \frac{\pi_k N(X_n|\mu_k, \Sigma_k)}{\sum_{j=1}^{k} \pi_j N(X_n|\mu_j, \Sigma_j)}$$

$\gamma(Z_{nk})$ is called a "responsibility": how much is its Gaussian k responsible for this point X_n ?



EM for Gaussian Mixtures

- **M step** : For each Gaussian k, update parameters using new $\gamma(Z_{nk})$

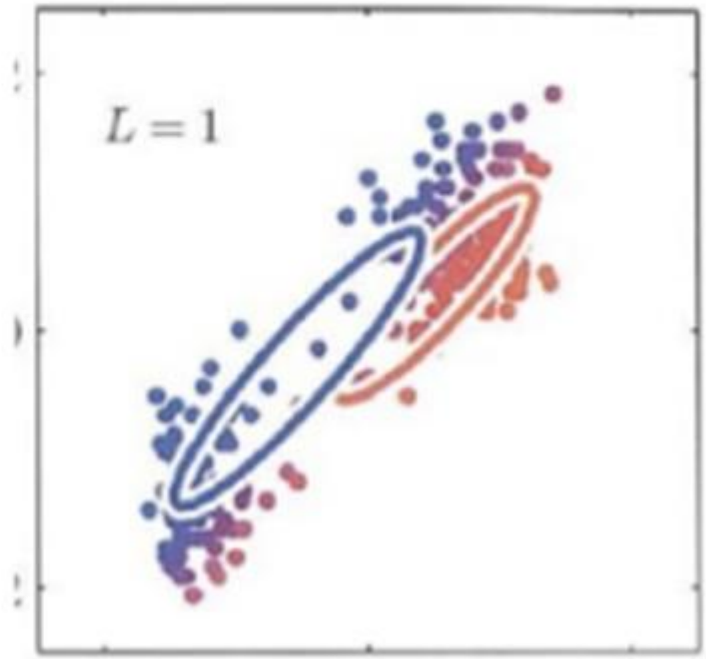
Responsibility for this Xn

- **Mean of Gaussian k**

$$\mu_k^{new} = \frac{1}{N_k} \sum_{n-1}^{N} \gamma(Z_{nk})\, X_n$$

$$N_k = \sum_{n=1}^{N} \gamma(Z_{nk})$$

Find mean that ‘Fits’ the assignment score best

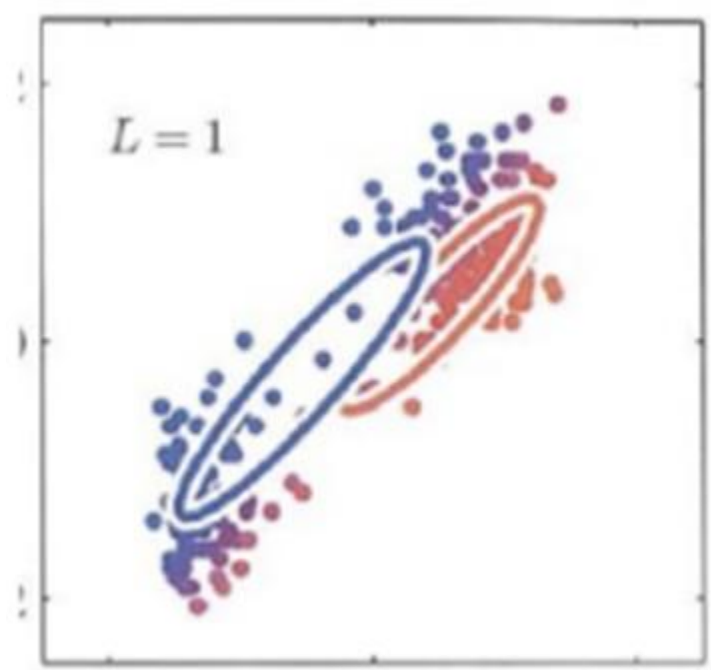


EM for Gaussian Mixture

- M step: For each Gaussian k, update parameters using new $\gamma(Z_{nk})$
- Covariance matrix of Gaussian k

$$\Sigma_k^{new} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(Z_{nk}) * (X_n - \mu_k^{new})(X_n - \mu_k^{new})^T$$

Just calculated this!



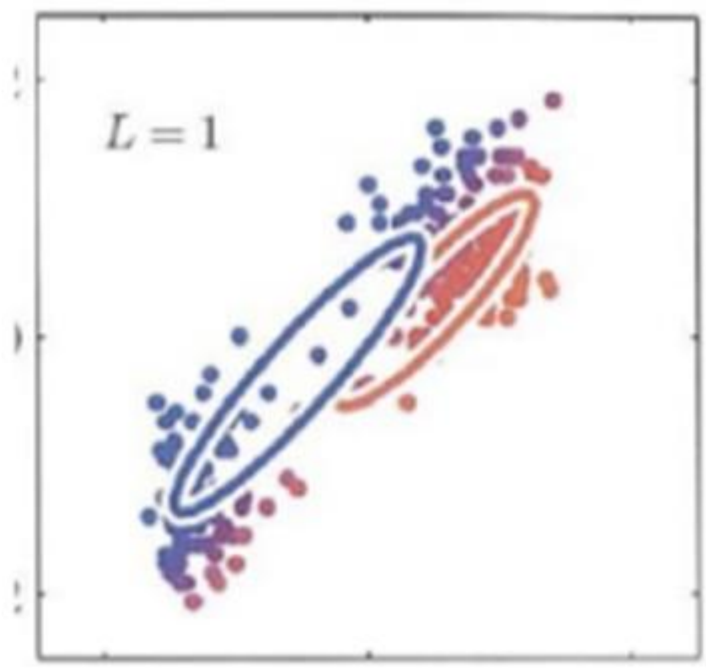
EM for Gaussian Mixture

- **M step:** For each Gaussian k, update parameters using new $\gamma(Z_{nk})$

- Mixing Coefficients for Gaussian k

$$N_k = \sum_{n=1}^{N} \gamma(Z_{nk})$$

$$\pi_k^{new} = \frac{N_k}{N}$$



Total # of points

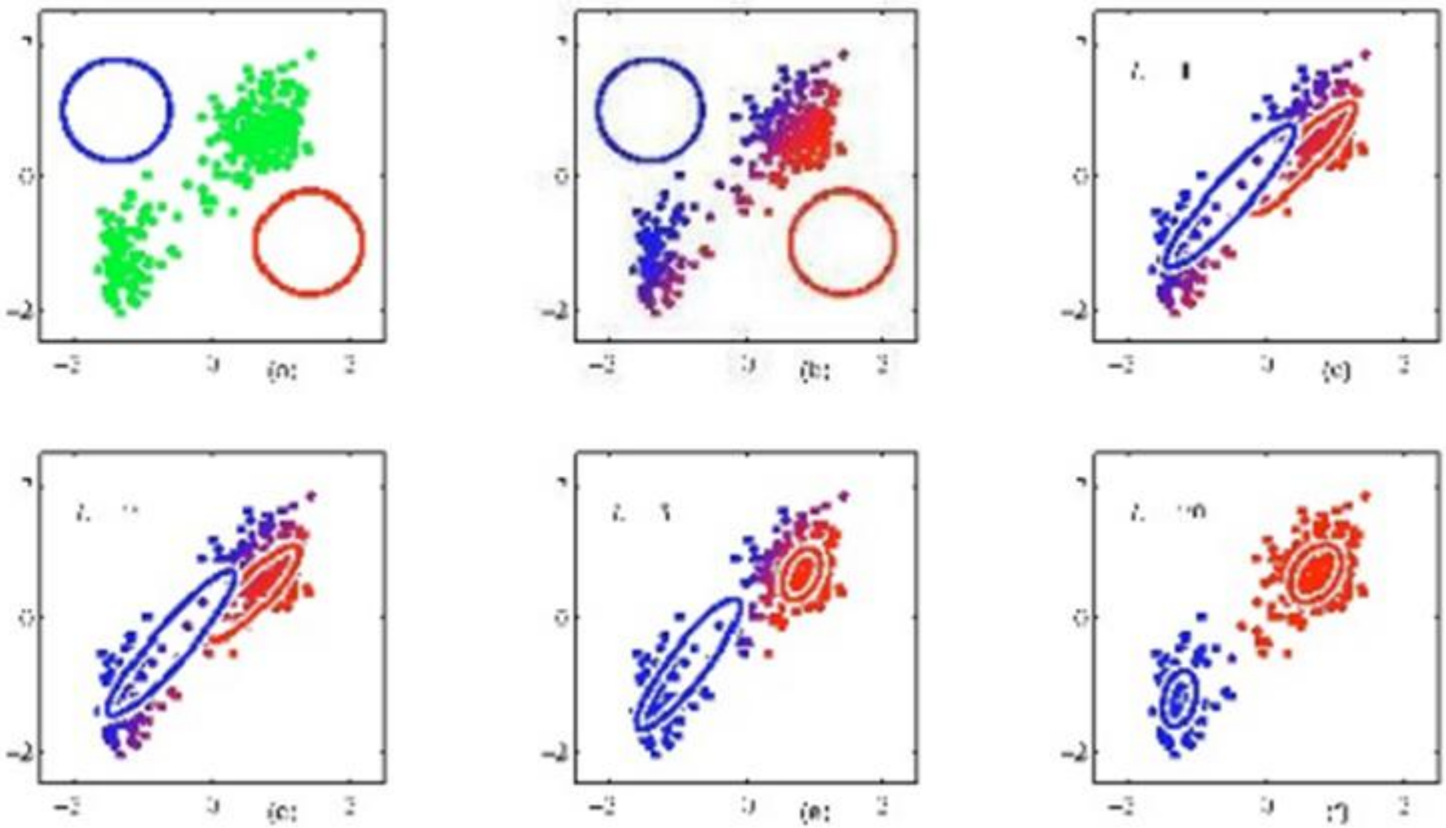
EM for Gaussian Mixtures

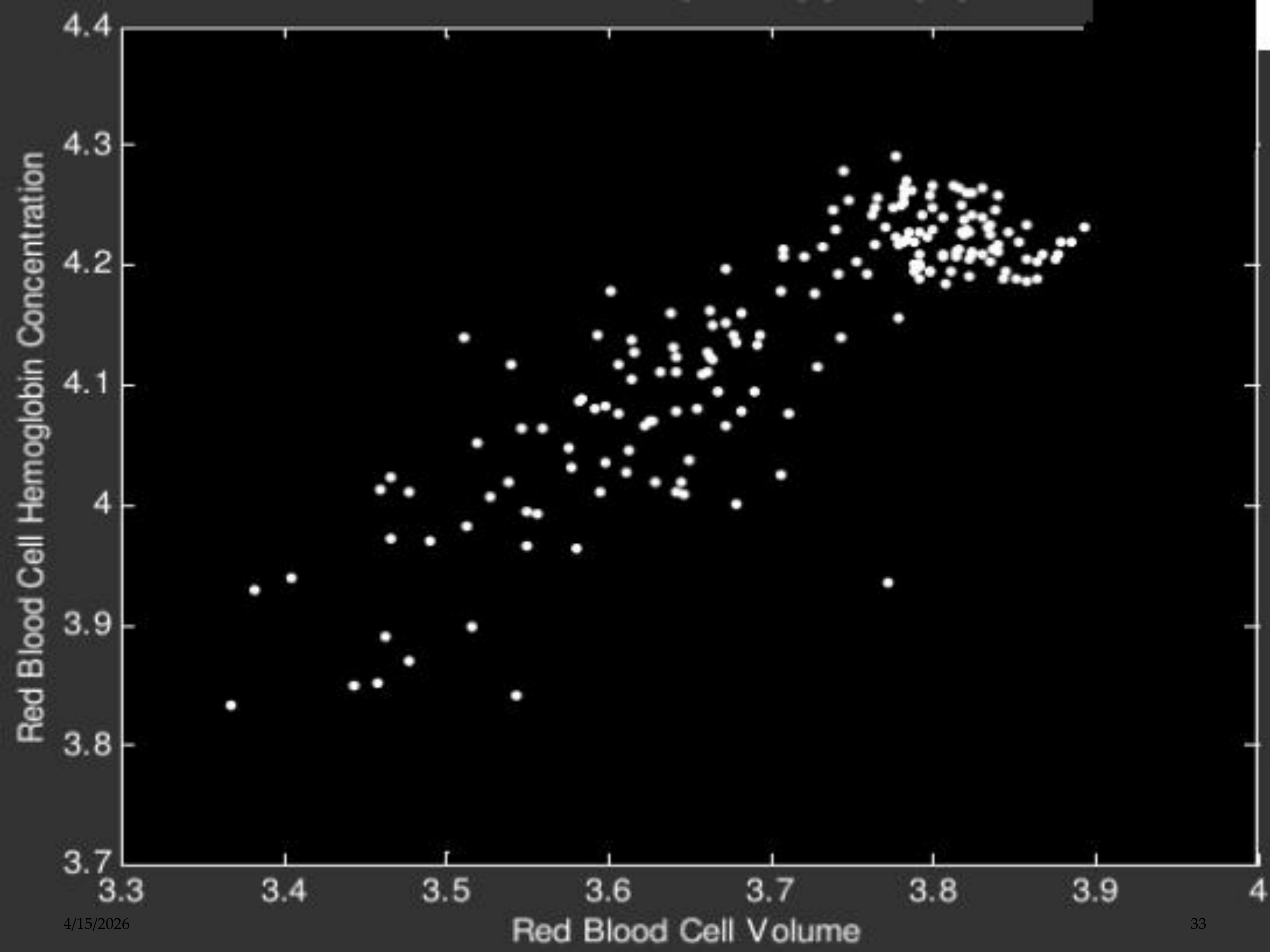
- Evaluate log likelihood. If likelihood or parameters coverage, stop. Else go to Step 2 (E step).

$$\ln p(X\,|\mu,\Sigma,\pi) = \sum_{n=1}^{N} \ln\left\{\sum_{k=1}^{K} \pi_k N(X_n|\mu_k,\Sigma_k)\right\}$$

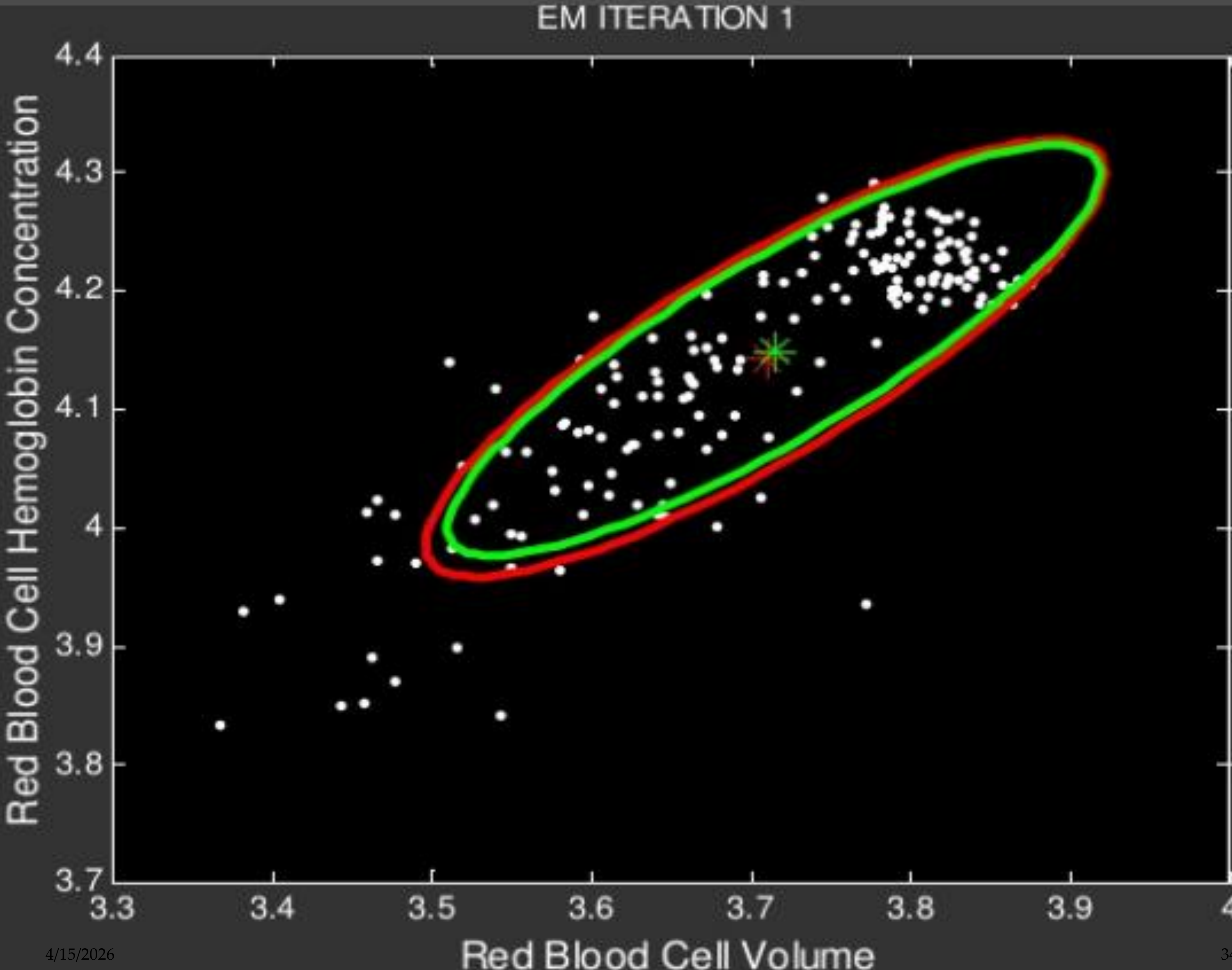
- Likelihood is the probability that the data X was generated by the parameters you found i.e. Correctness!

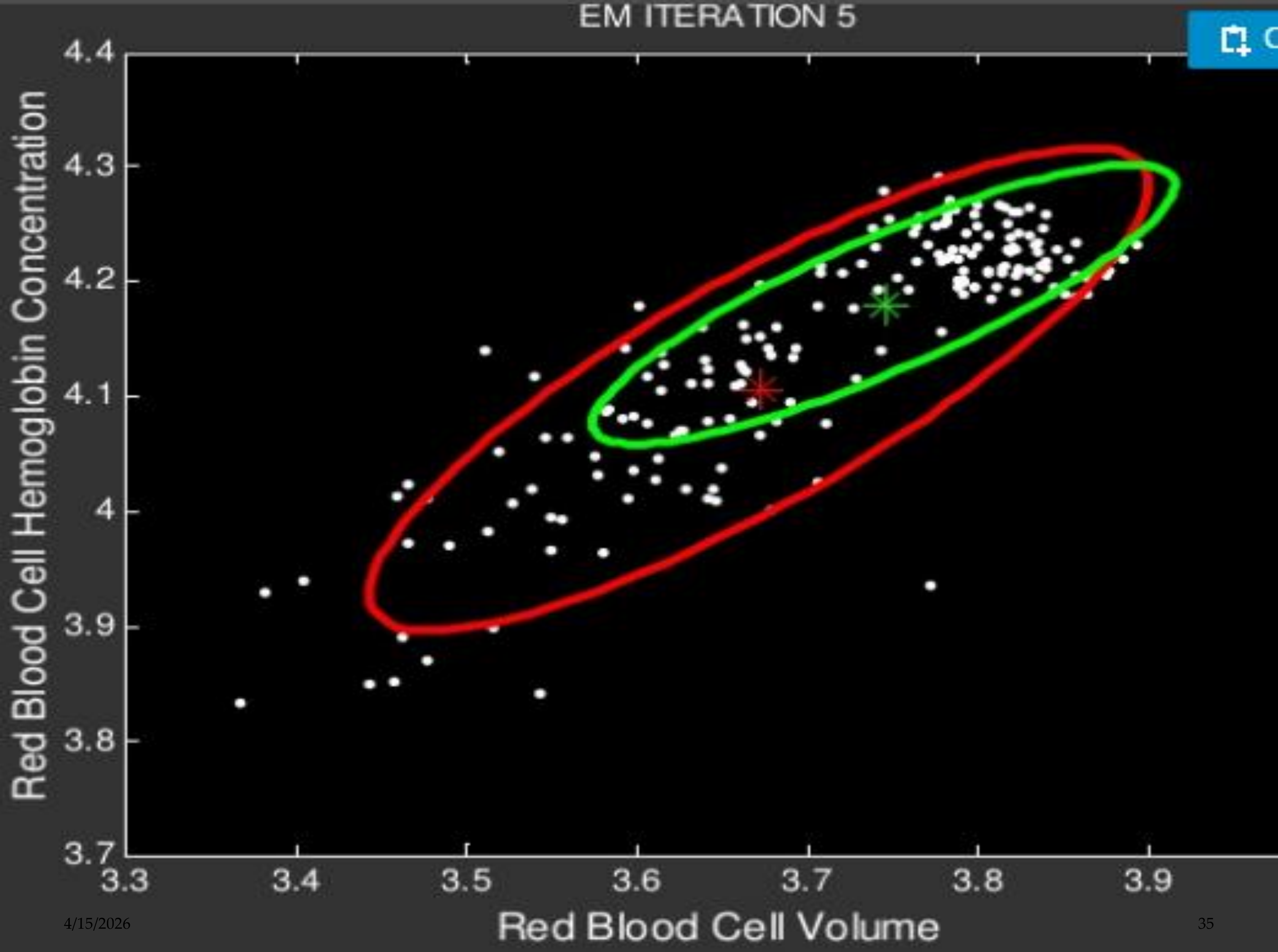
Visual Example of EM

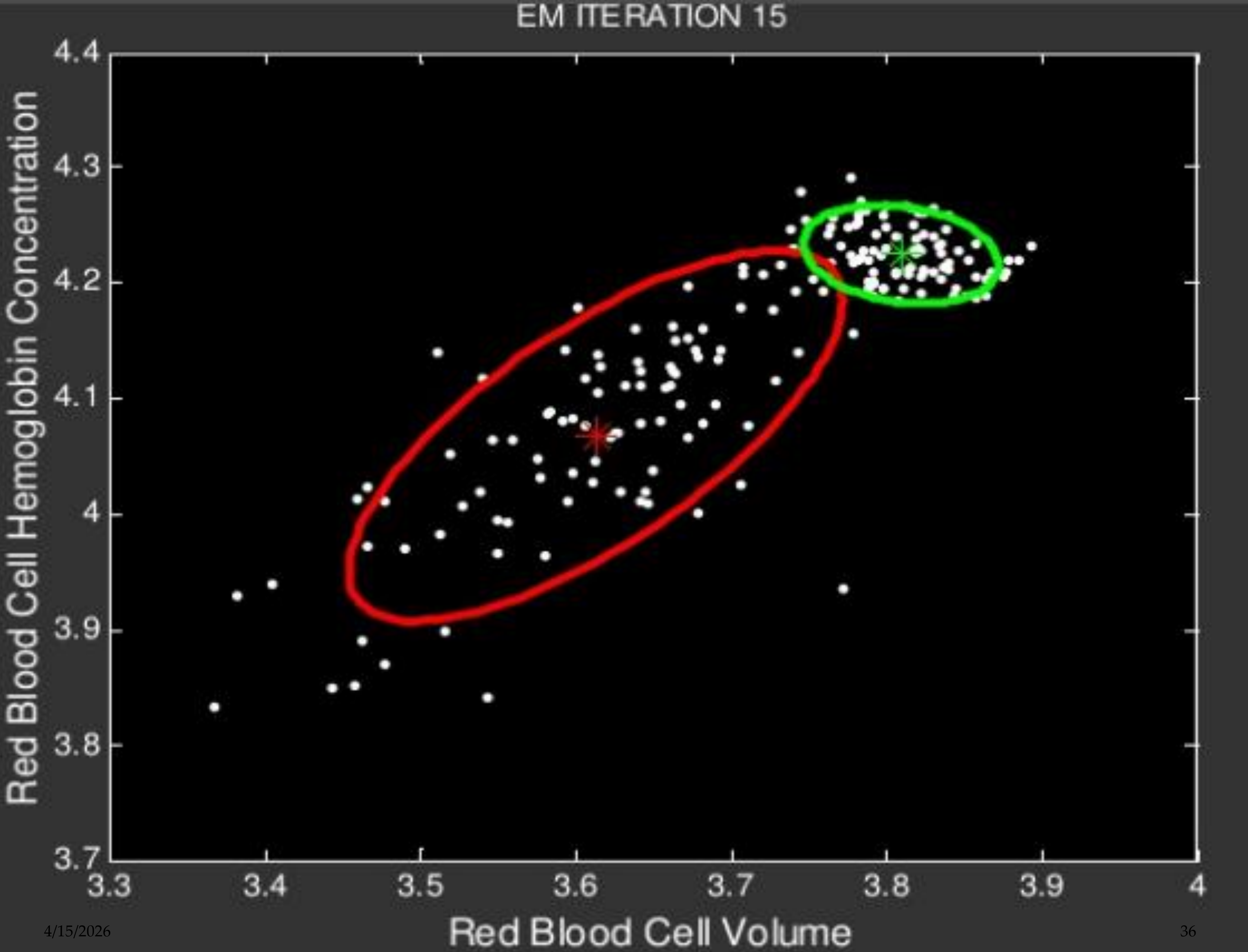


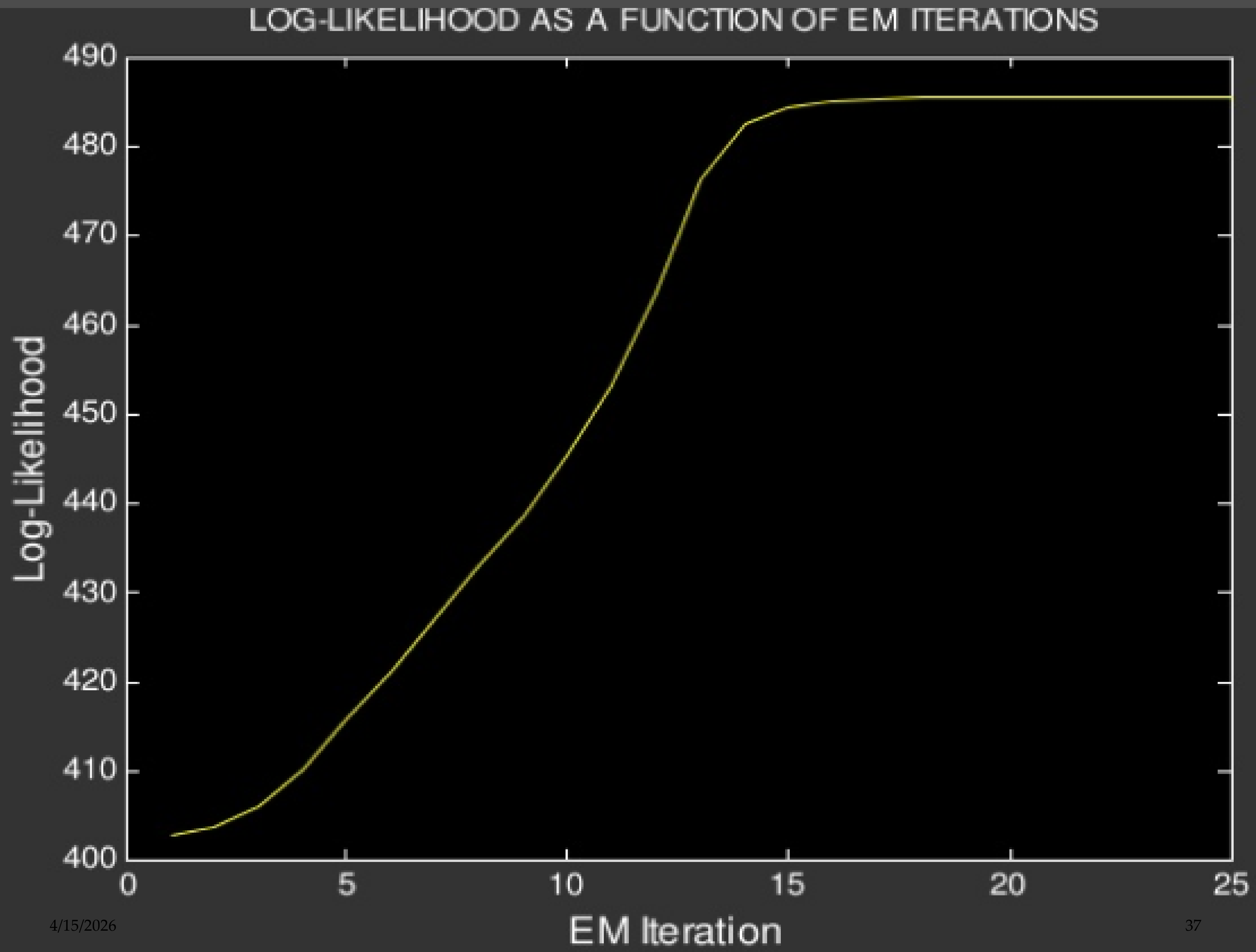


Red Blood Cell Hemoglobin Concentration
4.4
4.3
4.2
4.1
4
3.9
3.8
3.7
3.3
3.4
3.5
3.6
3.7
3.8
3.9
4
Red Blood Cell Volume


EM ITERATION 1
Red Blood Cell Hemoglobin Concentration
4.4
4.3
4.2
4.1
4
3.9
3.8
3.7
3.3
3.4
3.5
3.6
3.7
3.8
3.9
4
Red Blood Cell Volume


EM ITERATION 5
Red Blood Cell Hemoglobin Concentration
Red Blood Cell Volume
4.4
4.3
4.2
4.1
4
3.9
3.8
3.7
3.3
3.4
3.5
3.6
3.7
3.8
3.9


EM ITERATION 15
Red Blood Cell Hemoglobin Concentration
4.4
4.3
4.2
4.1
4
3.9
3.8
3.7
3.3
3.4
3.5
3.6
3.7
3.8
3.9
4
Red Blood Cell Volume


LOG-LIKELIHOOD AS A FUNCTION OF EM ITERATIONS
490
480
470
460
450
440
430
420
410
400
Log-Likelihood
0
5
10
15
20
25
EM Iteration

Choice of Covariance Matrix

- Nodal Covariance

  One covariance matrix per Gaussian Component

- Grand Covariance

  one covariance matrix for all Gaussian component

- Global Covariance

  single covariance matrix shared by all speaker component

Potential Problems

- Incorrect number of Mixtures Components

- Singularities

  - EM is an iterative algorithm which is very sensitive to initial conditions

  .

  - Usually, we use k means to get good initialization

ML Estimation For GMM's

- Given training data, how to estimate parameters . . . i.e., the μj , Σj , and mixture weights pj . . .
- To maximize likelihood of data?
  - No closed-form solution. Can't just count and normalize.
- Instead, must use an optimization technique . . .
- To find good local optimum in likelihood.
  - Gradient search
  - Newton's method Tool of choice:
  - The Expectation-Maximization Algorithm

The Basic Idea, using more Formal Terminology

- Initialize parameter values somehow.
- For each iteration . . .
- Expectation step: compute posterior (count) of h for each xi .

$$\tilde{P}(h|x_i) = \frac{P(h, x_i)}{\sum_h P(h, x_i)}$$

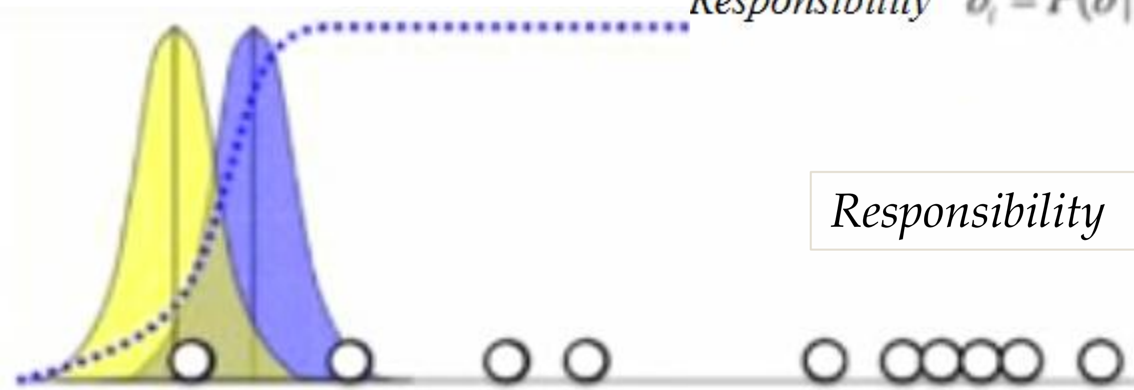
- Maximization step: update parameters.
- Instead of data xi with hidden h, pretend . . .
- Non-hidden data where . . .
- (Fractional) count of each (h, xi) is P˜(h|xi).

E-M Example

- 1 Dimensional example (10 Samples, 2 Gaussian, $\{\mu_1, \sigma^2 \mu_2\ \sigma_2^2\}$

$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right)$$

Responsibility $b_i = P(b \mid x_i) = \dfrac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$



Responsibility $p\,(a/x_i) = a_i = 1 - b_i$

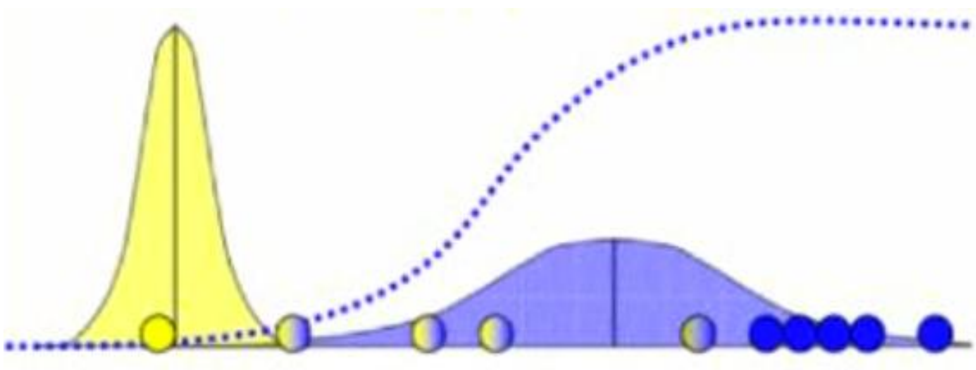
$$\mu_b = \frac{b_1x_1 + b_2x_2 + \ldots + b_nx_n}{b_1 + b_2 + \ldots + b_n}$$

$$\sigma_b^2 = \frac{b_1(x_1 - \mu_1)^2 + \ldots + b_n(x_n - \mu_n)^2}{b_1 + b_2 + \ldots + b_n}$$

E M Example

$$\mu_a = \frac{a_1 x_1 + a_2 x_2 + \ldots + a_n x_n}{a_1 + a_2 + \ldots + a_n}$$

$$\sigma_a^2 = \frac{a_1(x_1 - \mu_1)^2 + \ldots + a_n(x_n - \mu_n)^2}{a_1 + a_2 + \ldots + a_n}$$



$$\mu_k^{new} = \frac{1}{N_k} \sum_{n-1}^{N} \gamma(Z_{nk})\, X_n$$

$$\Sigma_k^{new} = \frac{1}{N_k} \Sigma_{n=1}^{N} \gamma(Z_{nk}) * (X_n - \mu_k^{new})(X_n - \mu_k^{new})^T$$

$$N_k = \sum_{n=1}^{N} \gamma(Z_{nk})$$

Example: Training a 2-component GMM

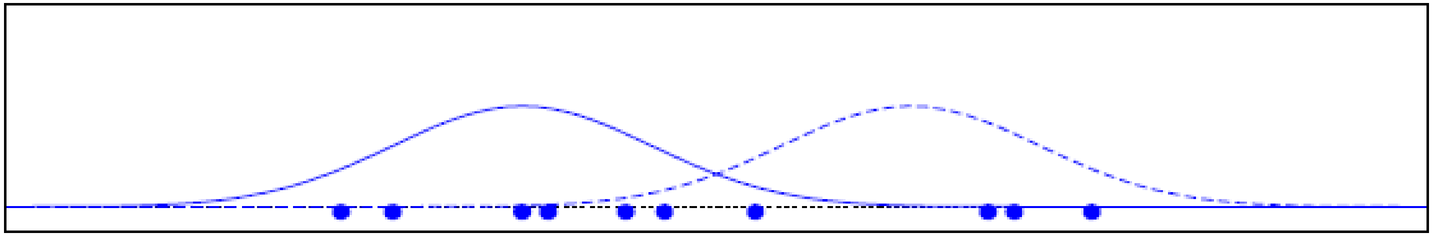
- Two-component univariate GMM; 10 data points.
- The data: $x_1, \ldots, x_{10}$

$$8.4, 7.6, 4.2, 2.6, 5.1, 4.0, 7.8, 3.0, 4.8, 5.8$$

- Initial parameter values:

p_1	μ_1	σ_1^2	p_2	μ_2	σ_2^2
0.5	4	1	0.5	7	1

- Training data; densities of initial Gaussians.



The E Step

x_i	$p_1 \cdot \mathcal{N}_1$	$p_2 \cdot \mathcal{N}_2$	$P(x_i)$	$\tilde{P}(1\|x_i)$	$\tilde{P}(2\|x_i)$
8.4	0.0000	0.0749	0.0749	0.000	1.000
7.6	0.0003	0.1666	0.1669	0.002	0.998
4.2	0.1955	0.0040	0.1995	0.980	0.020
2.6	0.0749	0.0000	0.0749	1.000	0.000
5.1	0.1089	0.0328	0.1417	0.769	0.231
4.0	0.1995	0.0022	0.2017	0.989	0.011
7.8	0.0001	0.1448	0.1450	0.001	0.999
3.0	0.1210	0.0001	0.1211	0.999	0.001
4.8	0.1448	0.0177	0.1626	0.891	0.109
5.8	0.0395	0.0971	0.1366	0.289	0.711

$$\tilde{P}(h|x_i) = \frac{P(h, x_i)}{\sum_h P(h, x_i)} = \frac{p_h \cdot \mathcal{N}_h}{P(x_i)} \qquad h \in \{1, 2\}$$

The M Step

- View: have *non-hidden* corpus for each component GMM.
  - For hth component, have $\tilde{P}(h|x_i)$ counts for event x_i.
- Estimating μ: fractional events.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \quad \Rightarrow \quad \mu_h = \frac{1}{\sum_i \tilde{P}(h|x_i)} \sum_{i=1}^{N} \tilde{P}(h|x_i) x_i$$

$$\mu_1 = \frac{1}{0.000 + 0.002 + 0.980 + \cdots} \times (0.000 \times 8.4 + 0.002 \times 7.6 + 0.980 \times 4.2 + \cdots)$$

$$= 3.98$$

- Similarly, can estimate σ_h^2 with fractional events.

The M Step (cont'd)

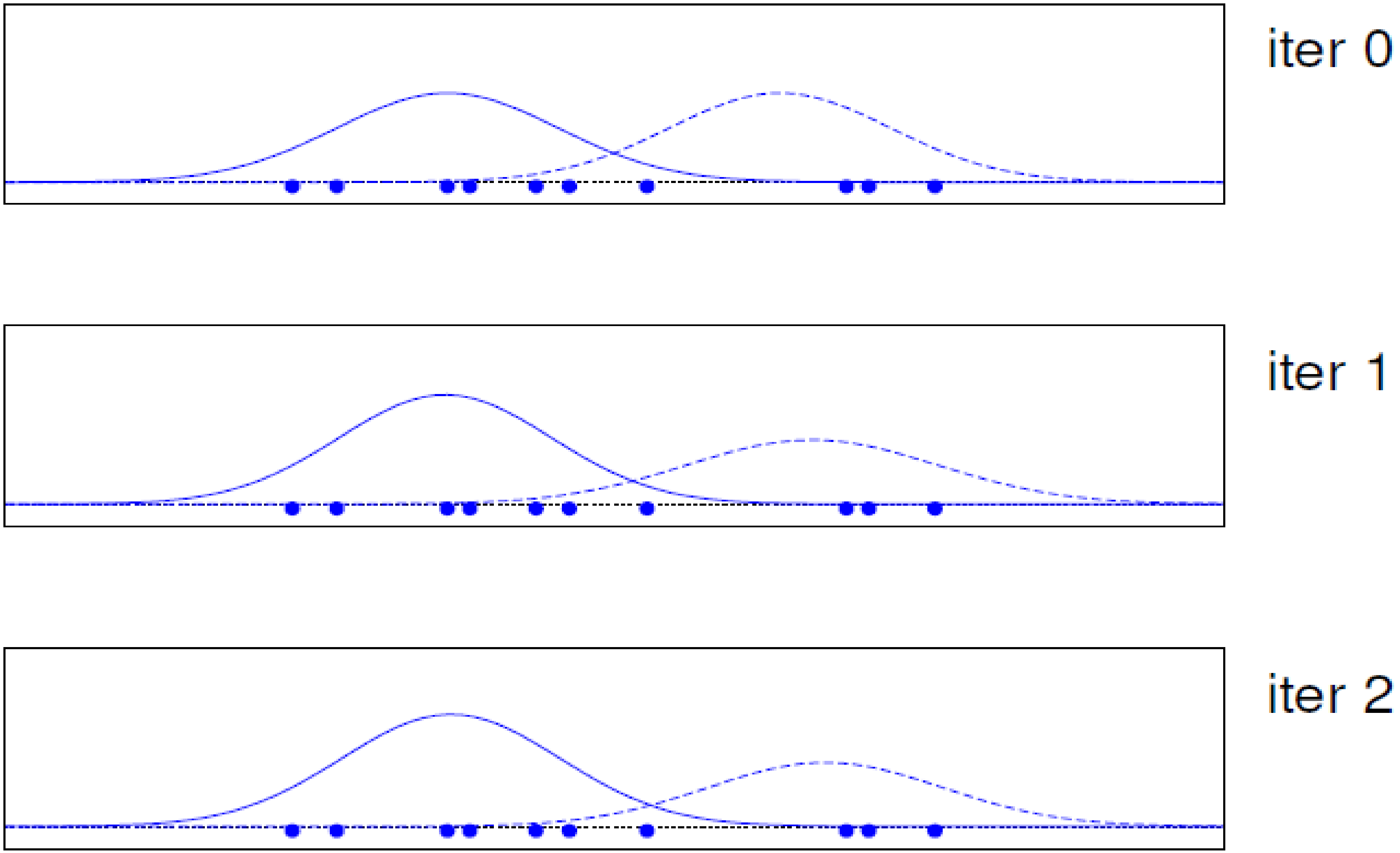
- What about the mixture weights p_h?
  - To find MLE, count and normalize!

$$p_1 = \frac{0.000 + 0.002 + 0.980 + \cdots}{10} = 0.59$$

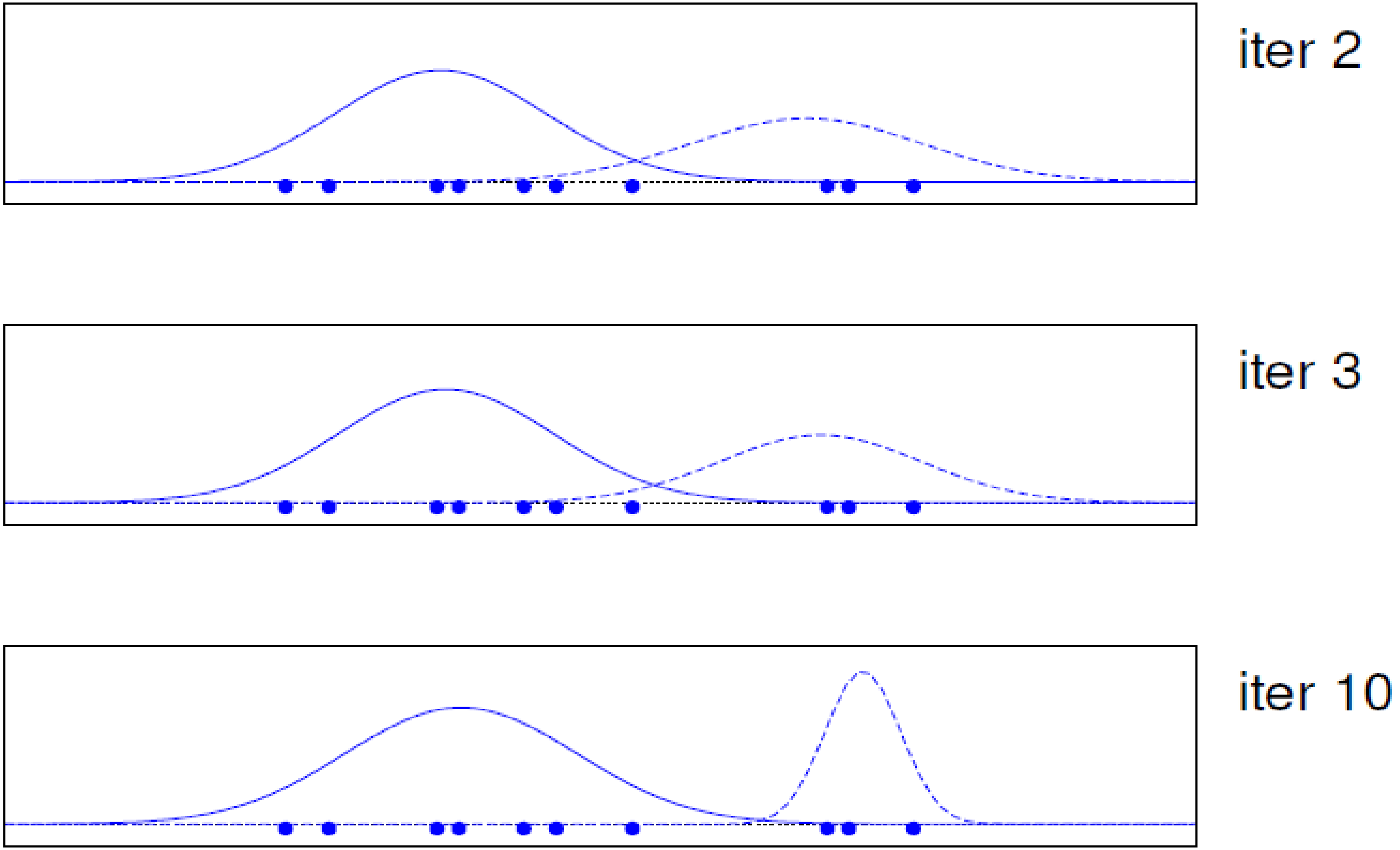
The End Result

iter	p_1	μ_1	σ_1^2	p_2	μ_2	σ_2^2
0	0.50	4.00	1.00	0.50	7.00	1.00
1	0.59	3.98	0.92	0.41	7.29	1.29
2	0.62	4.03	0.97	0.38	7.41	1.12
3	0.64	4.08	1.00	0.36	7.54	0.88
10	0.70	4.22	1.13	0.30	7.93	0.12

First Few Iterations of EM



Later Iterations of EM



Advantages and Disadvantages of Mixture Models

- Strength
  - Mixture models are more general than partitioning and fuzzy clustering
  - Clusters can be characterized by a small number of parameters
  - The results may satisfy the statistical assumptions of the generative models
- Weakness
  - Converge to local optimal (overcome: run multi-times w. random initialization)
  - Computationally expensive if the number of distributions is large, or the data set contains very few observed data points
  - Need large data sets
  - Hard to estimate the number of clusters